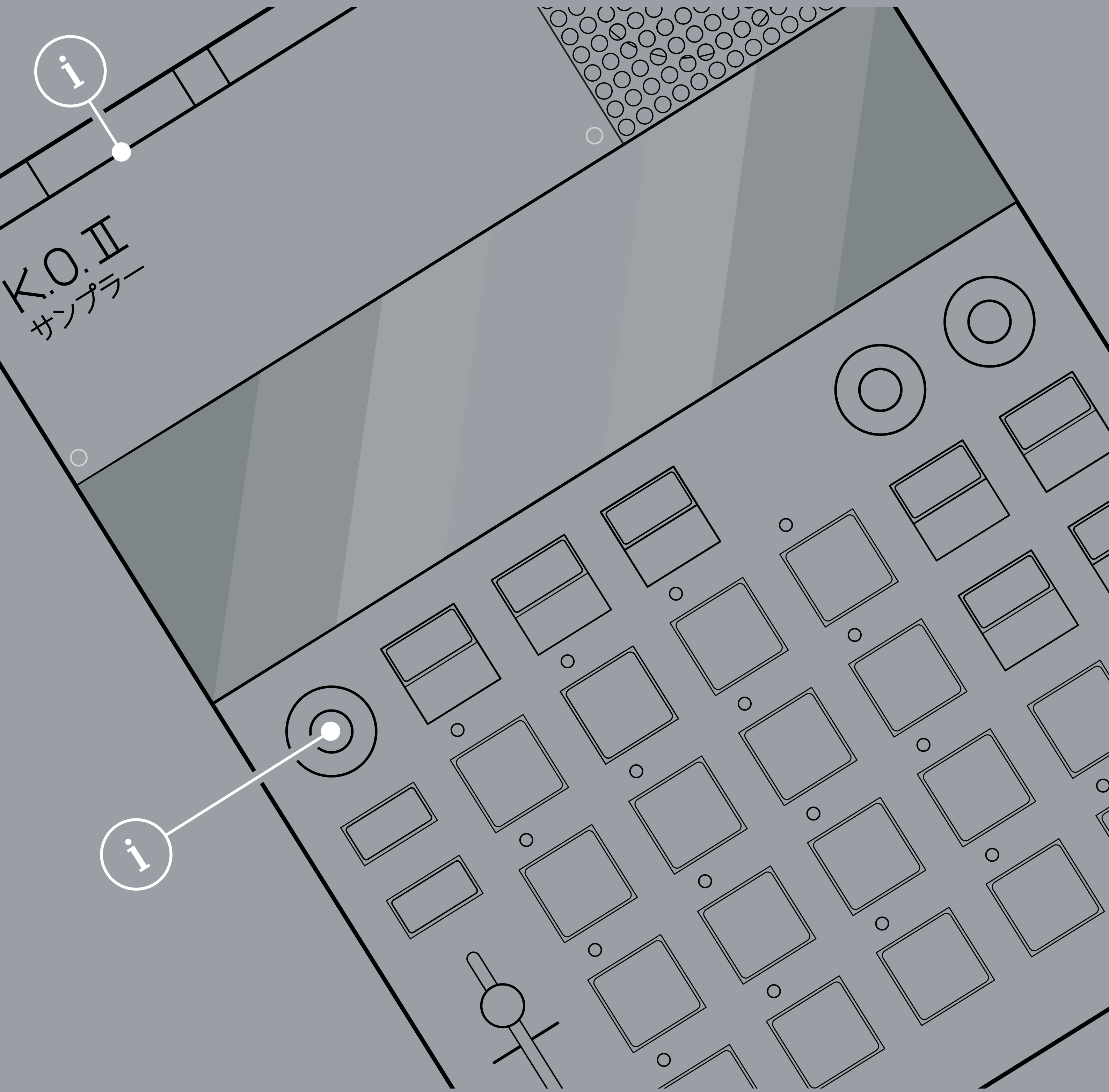
K.O.II EP-133

A Guidebook

A Notebook

A Reference Book



SYNTHDAWG
PRODUCER GUIDES

An unofficial reference for
Teenage Engineering's KO.II
EP133 sampler composer.

OS
2.0.0

Contents

1

Overview

Located in Sweden, Teenage Engineering are a design company with a heritage of developing a range of synthesizers, products and audio devices that follow a less traditional path. Aimed at portability, style and immediacy they released the OP-1 Synthesizer in 2011 and followed up 11 years later with a redesigned and updated field version. In 2018 the OP-Z synthesizer and sequencer was introduced and a series of miniature Pocket Operators released along the way. The EP-133 KOII released in 2023 could be described as a big brother to the PO range and more of an intermediate device to the higher end OP's. The KOII, as it will simply be referred in this guide, is a 64MB Sampler Composer and is position as an affordable yet feature rich audio production device. While at first glance the KOII seems to fit into the groove box category of audio tools, under the hood is a much more unique instrument accompanied by a quirky workflow. The hybrid display is a combination of fixed symbols which illuminate based on its state as well as a 3 digit dynamic display. But hey, Teenage Engineering have never been predictable. The KOII can play and record samples, sequence loops and patterns and apply a range of effects. It has a built in microphone and speaker and can be battery or USB-C powered to provide support to on the fly and portable music production. The device has 6 Stereo or 12 mono voices and can hold up to 999 samples depending on the sample size and memory availability. Audio, MIDI and Sync In and Out connectivity enables integration with a range of compatible audio devices. Remember that this notebook is a guide and reference and is designed to also give space to capturing your own learnings and notes through the process. The KOII is a fun, innovative and creative sampler that will give any music producer or artist a fast and efficient production tool and this book aims to keep you interested and inspired along the journey.

1.1 How to Use This Notebook

NOTES

Write Your Notes Here

This book combines a formal reference and guide along with your own notes and comments, bringing together detailed walkthroughs, illustrations, tips alongside your own comments, captured in the wide margins. The format is aimed to cover multiple learning styles and appeal to a wide community.

KOII vs EP-133.

These are interchangeable names used for the KOII …errr… the EP-133. In this manual we will use the name KOII purely for simplicity and consistency.

(X) Knobs

Rounded parentheses represent the context sensitive, White Volume, Orange X and Black Y potentiometer knobs. Some tables also show the colours representing the specific rotary control. Knobs can be rotated clockwise across a fixed and defined value range 0-100%.

[Erase] Buttons

Square parentheses contain functions that are selectable by a specifically defined and dedicated hardware button or pad either held or pressed.

Hold [Shift] + Press [Erase]

Functions which require two button selections simultaneously are shown with the + symbol between. These may be prefaced with the instructions to 'Hold' i.e. Long press, 'Press' i.e. Tap or quick press or Turn i.e. Rotate the knob.

If a secondary parameter is in focus, its name is used in the instruction. For example [Shift] + [System] uses the Erase / System button. Sometimes a generic 'Pad' name i.e. [Pad] is used when referring to a group of options.

{Fader} Control

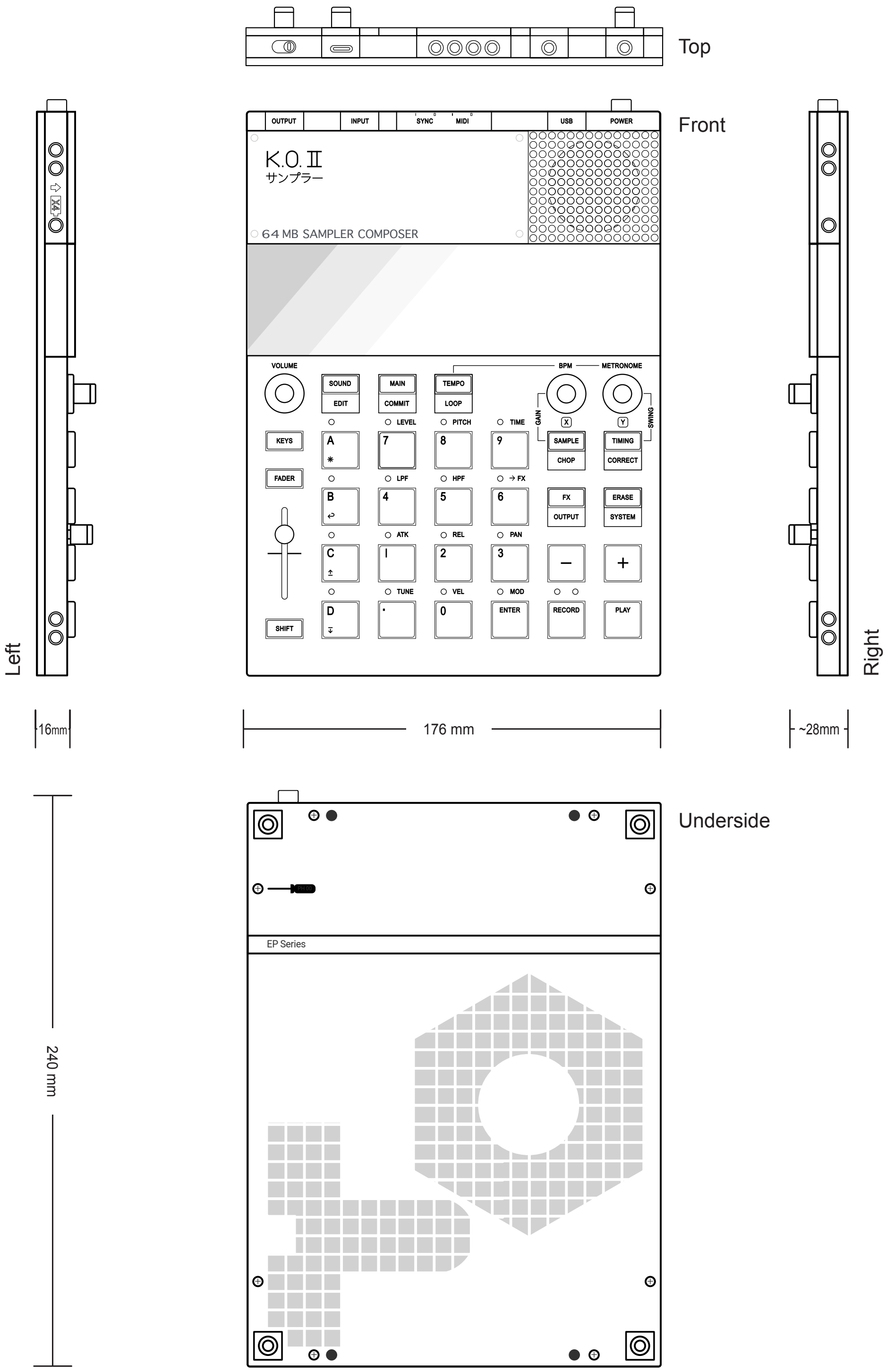
The fader function is described within the curly parenthesis along with the generic {Fader} and its function used i.e. 'Level'.

Example Commands

Press / Turn	*Button*	*Pad*	*White Knob*	*Orange Knob*	*Black Knob*	*Fader*
	SOUND [Sound]	ENTER [Enter]	(Vol)	(X)	(Y)	Level
Shift +	*Button*	*Group A*	*Group B*	*Group C*	*Group D*	LED
	COMMIT [Commit]	[…]	[Undo]	[Copy]	[Paste]	On / Flash
Tips	Tips, Hints and more generic, good to know info contained in the grey bands					

1.2 Overview

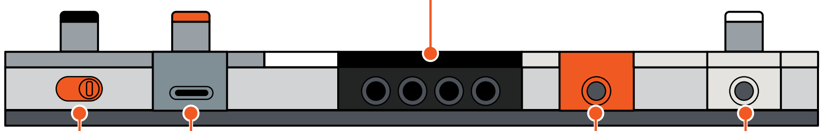
The physical dimension in ‘mm’ are shown. Knobs and switches are included in the illustration dimensions. The rear panel also contains 8 small crosspoint screws to access the inner circuitry of the KOII. There are 4 rubber feet located on the underside panel as an aid for surface grip.



1 Overview

General Overview

Sync & MIDI
External Sync and MIDI (Type A) Input and Output Interface.



Power Switch
Turns the unit On or Off

USB-C
Communication interface and alternate power supply input

Input
External Audio input.

Output
External Audio output.

Audio, Sync and MIDI Connections are 3.5mm - 1/8th Mini Jacks. Audio inputs and outputs are TRS stereo.

Battery Cover
4 x AAA Battery slot located under the cover.

Display
Icon and super segment hybrid display.

Volume
Main output audio level.

Keys
Chromatic keys mode.

Fader
Assignable fader control.

Shift
Hold to access secondary button functions.

Internal Speaker & Mic
Master Audio Output. Mic is located under speaker cover.

X Control Knob
Orange - X, Multi Functional, including tempo.

Y Control Knob
Black - Y, Multi Functional, including metronome.

Function Options
Multi-functional selection buttons.

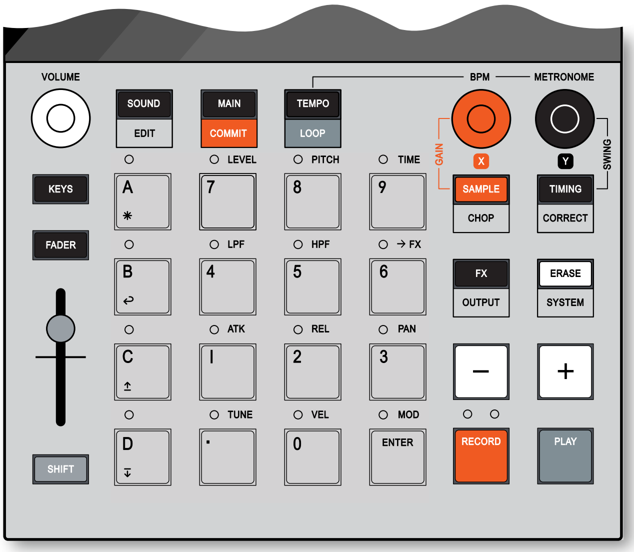
Navigation Buttons
Navigate and increment, decrement parameters.

Play / Record
Transport Buttons.

Group Buttons
4x Group selection and function buttons.

Pad Buttons
12x Pad buttons. Controls sounds and fader function.

Functional Overview



Sound Mode
Manage, assign and edit samples and sounds.

Main Mode
'Home' for projects and patterns. Performing, recording, arranging.

Tempo Mode
Global BPM and metronome options. Toggle loop mode.

Sampling Mode
Record audio samples from the line in, mic or input device.

Timing Mode
Sequencer note intervals, swing and quantization settings.

FX Section
Master effects for groups and for live inputs.

Erase
Erases selection i.e. sample etc. Also access to system options.

Fader Assignment
Assign a parameter to the fader as labeled on the selected pad.

Fader
Adjusts the assigned parameter. Default is group volume level.

Keys Mode
Pads or ext keyboard play a sound over a 12 note pitch range.

Shift
Access to secondary options labelled under the button.

Play
Start/stop transport to play a pattern or activate a recording.

Record
Arm or activate pattern recording mode.

Step / Increment
Step through pattern. Also increment various parameters.

Step / Decrement
Step back in a pattern. Also decrement various parameters.

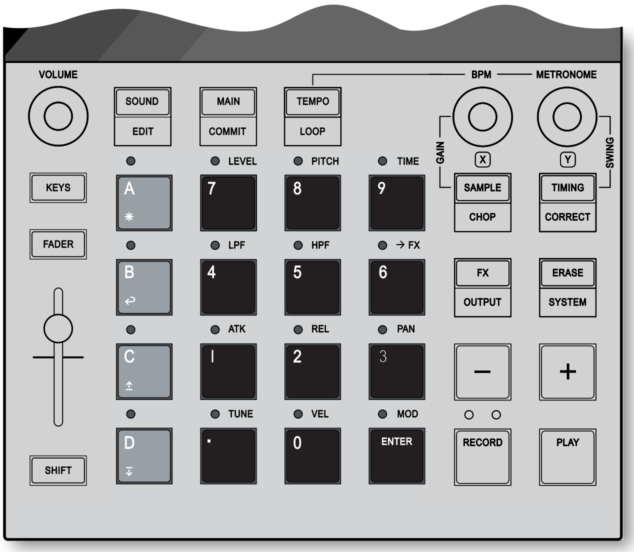
Main Volume Knob
Global volume level control.

X Control Knob
Adjusts 'X' defined parameter. Hold shift for precise increments.

Y Control Knob
Adjusts 'Y' defined parameter. Hold shift for precise increments.

1 Overview

Pad & Group Overview



Group A
Group containing 99 Patterns and 12 pad sounds.

Group B
Group containing 99 Patterns and 12 pad sounds. Undo as secondary option.

Group C
Group containing 99 Patterns and 12 pad sounds. Copy as secondary option.

Group D
Group containing 99 Patterns and 12 pad sounds. Paste as secondary option.

TUNE
Pad .
Contains a sound for patterns and playback. Also selection pad.

VEL
Pad 0
Contains a sound for patterns and playback. Also selection pad.

ATK
Pad 1
Contains a sound for patterns and playback. Also selection pad.

REL
Pad 2
Contains a sound for patterns and playback. Also selection pad.

PAN
Pad 3
Contains a sound for patterns and playback. Also selection pad.

LPF
Pad 4
Contains a sound for patterns and playback. Also selection pad.

HPF
Pad 5
Contains a sound for patterns and playback. Also selection pad.

→ FX
Pad 6
Contains a sound for patterns and playback. Also selection pad.

LEVEL
Pad 7
Contains a sound for patterns and playback. Also selection pad.

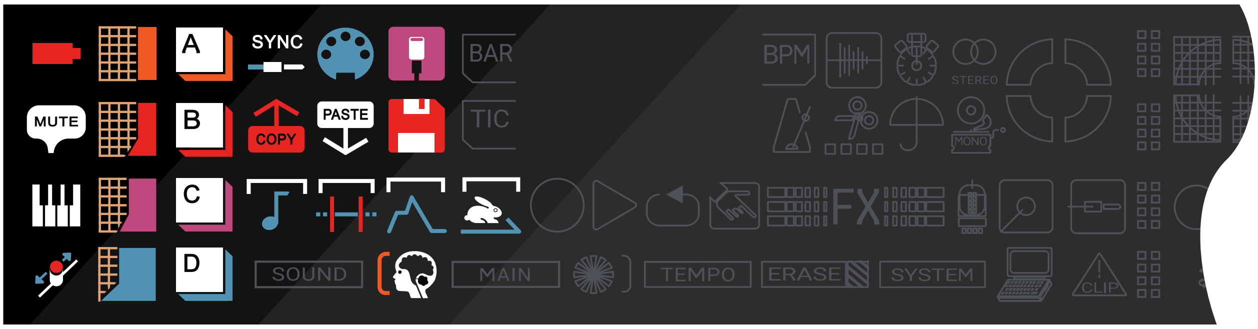
PITCH
Pad 8
Contains a sound for patterns and playback. Also selection pad.

TIME
Pad 9
Contains a sound for patterns and playback. Also selection pad.

MOD
Pad Enter
Contains a sound for patterns and playback. Also selection pad.

Pad Sensitivity
Pads can trigger sounds and apply numeric data entry. For sounds they are velocity sensitive when pressing the lower half of the physical button.

Display Overview



Battery Power State
Lit: Battery power on. Unlit - USB Power.
Flash: Battery power low.

Mute
Lit: Mute for a group or when 2 or more pads are in mute groups.

Keys Mode
Lit: Pads play a sample in chromatic mode.
Unlit: Pads play a sample per pad.

Fader
Lit: Fader automation or lev muted state.

Fader Value - Highest
Lit: Current fader value at highest level

Fader Value - High
Lit: Current fader value at high level

Fader Value - Low
Lit: Current fader value at low level

Fader Value - Lowest
Lit: Current fader value at lowest Level

Group A
Lit: Group A active - Button LED On

Group D
Lit: Group B active - Button LED On

Group C
Lit: Group C active - Button LED On

Group D
Lit: Group D active - Button LED On

Sync Activity
Lit: Sync input activity is being received.

MIDI Activity
Lit: MIDI activity is being sent or received.

USB MIDI
Lit: USB communication is connected.
Flash: MIDI activity is being sent or received.

Copy
Lit: Bar, Pattern or Sound has been copied.

Paste
Lit: Bar, Pattern or Sound has been pasted.

Umbrella
Lit: Undo option is available.

Time
Lit: Sample time stretch options active.

Envelope
Lit: Sample envelope options active.

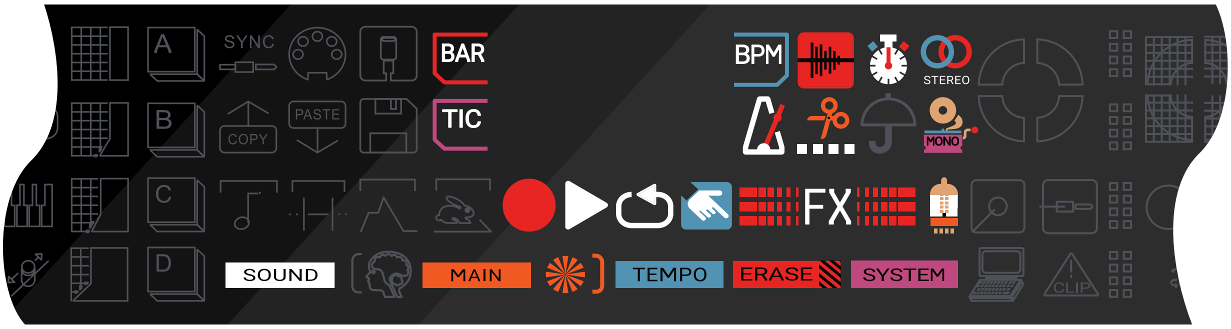
Trim
Lit: Sample trim options active.

Sound
Lit: Sample sound edit mode active.

Floppy!
Lit: Data is being saved.
Flash: Sound has been erased.

Auto Save
Lit: System is automatically saving the state.

Display Overview



Erase
Lit: Something is being erased.

System
Lit: System setting options active.

Tempo
Lit: Tempo mode active to change BPM.

Sound
Lit: Sound mode ON.

Main
Lit: Main mode ON.

Bar Mode
Lit: Navigating in the quantized grid.

TIC Mode
Lit: Navigating in the grid free time - tics.

BPM
Lit: Display shows beats per minute.

New Pattern
Lit: Selecting an empty pattern.

Record
Lit: Recording in progress.
Flash: Armed for recording.

Play
Lit: Play in progress.

Loop
Lit: Loop playback is active.

FX
Lit: Current FX is being activated.

FX
Lit: Current FX is being activated.

FX
Lit: FX Mode options are active.

Master Output Compressor
Lit: Master compressor engaged.
Flash: Master compressor activated.

Numpad Active
Lit: Pads operate as numerical entry pads.

Stereo
Lit: Current sample is in stereo.

Mono
Lit: Current sample is in mono.

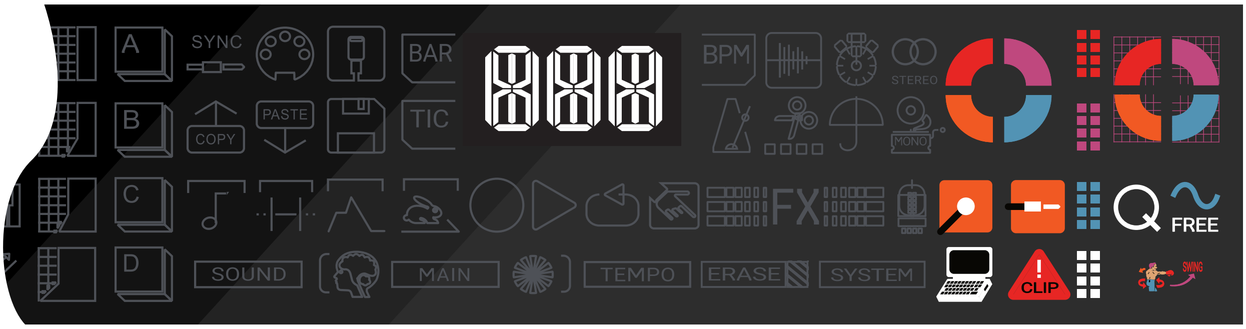
Time
Lit: Current sound has time stretch enabled.

Sample Mode
Lit: Sampling is in progress.

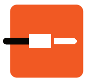
Chop
Lit: Sample chopping in progress.

Metronome
Flash: Blinks in time with metronome BPM.

Display Overview



Mic
Lit: Microphone is in use, threshold exceeded.



Line In
Lit: Line Input is in use, threshold exceeded.

Clip
Lit: Audio input level is clipping

Computer
Lit: KOII Device is connected and communicating to / from a computer.

High Level
Lit: Volume of current track at high.
Flash: Vol of current track approaching high.

High Mid Level
Lit: Volume of current track at high/mid level.
Flash: Vol of current track approaching high/mid.

Low Mid Level
Lit: Volume of current track at low/mid level.
Flash: Vol of current track above low.

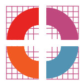
Low Level
Lit: Volume of current track is low.
Flash: Vol of current track close to low.

Quantize
Lit: Quantize mode active.

Free
Lit: Free time mode activate.

Swing
Lit: Swing is enabled.

X Level
Lit: Illuminates in a rotary direction to indicate the current value of the 'X' Control parameter.



Y Level
Lit: Illuminates in a rotary direction to indicate the current value of the 'Y' Control parameter.

0-25% Parameter Value = bottom left quadrant.
Value increases clockwise.
75-100% Parameter Value = bottom right quadrant.

3-Digital Display
A 3 Digit display is also contained within the centre of the hybrid display. This typically displays alpha-numeric data.

The main display when playing will show: Bar : Beat : Step

Screen Indicators

1.3 Architecture

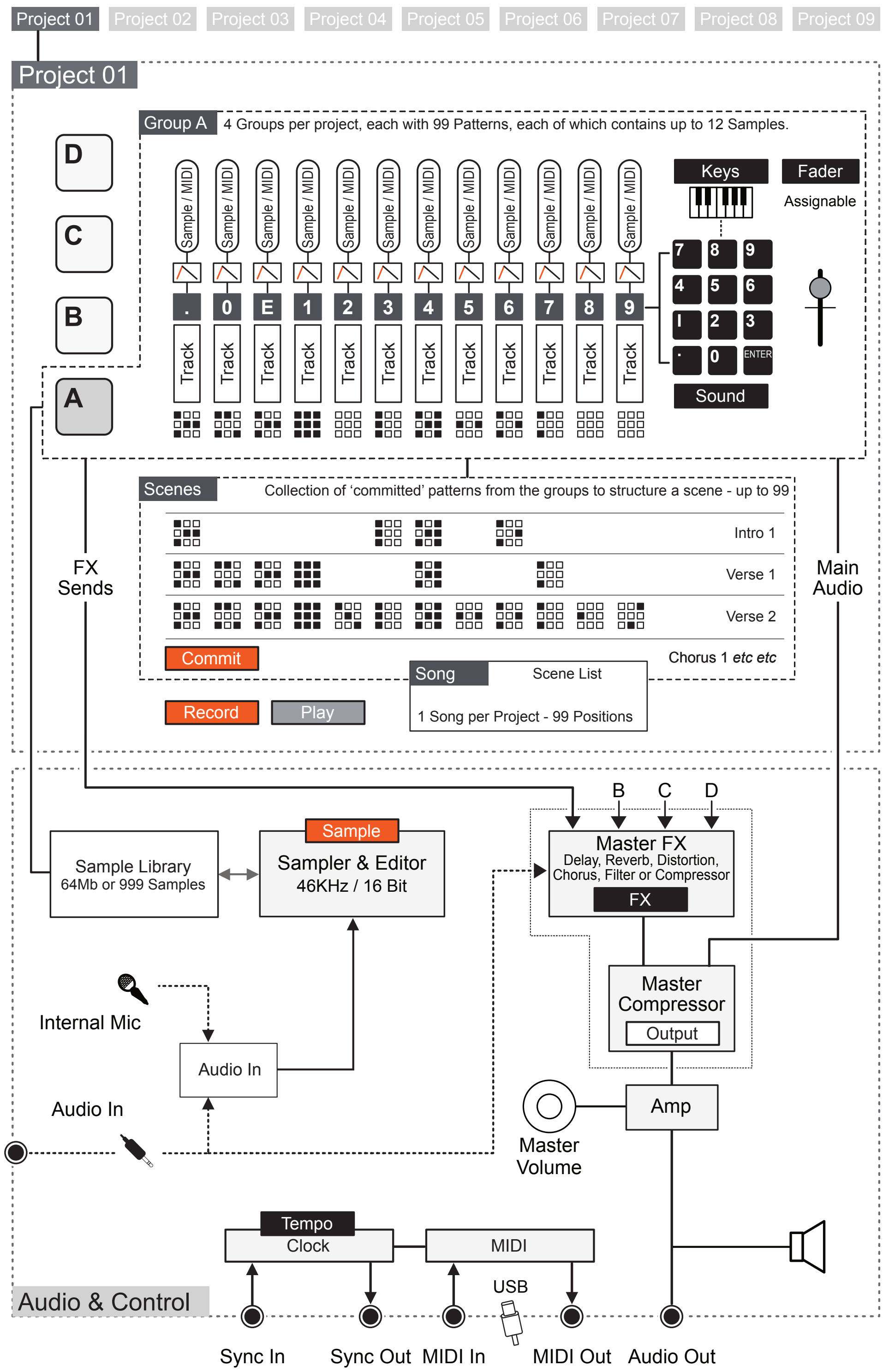
NOTES

The KOII has some features that are an aid to administer the structural components rather than drive the creative process. A summary of the specification of features is as follows:-

- Sample Library Size - 64Mb or 999 Samples.
- 9 Projects.
- 80,000 Notes per project.
- 4 Groups per project.
- 12 Samples per Group.
- 99 Patterns per group, each pattern with 12 tracks (Samples & MIDI)
- 1 to 99 Bar pattern length per group.
- 99 Scenes per project.
- 6 Send FX / Effects + 12 Punch In FX / Effects

NOTES

The flow diagram below is for illustration purposes only and does not serve as an accurate schematic or circuit diagram of the KOII circuitry.

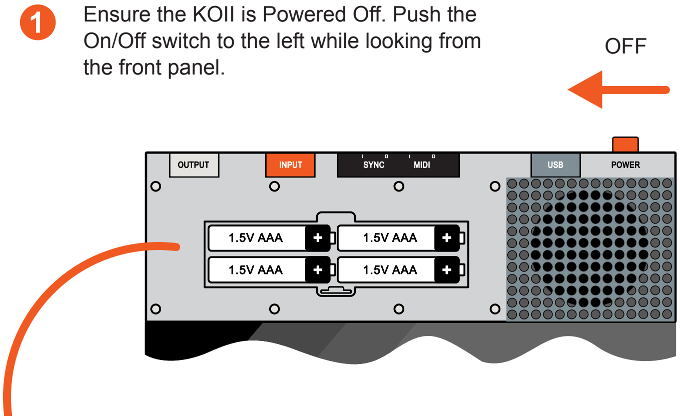


1.4 Power Supply

The KOII power can be supplied by 4 x AAA Batteries, ideal for travel and portability or by using a connected USB-C power supply. The battery compartment is located under the front plate.

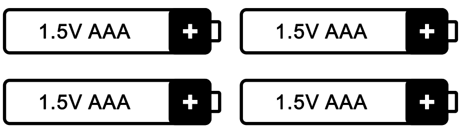
Battery Option

1. Ensure the KOII is Powered Off. Push the On/Off switch to the left while looking from the front panel.

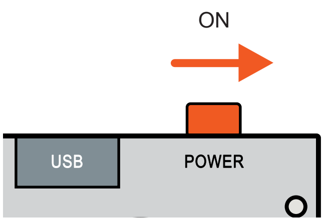


2. Remove the battery cover. Be careful not to break or bend the cover as it is fixed in multiple points.

3. Insert 4 x AAA Batteries into the battery compartment. The + positive battery terminal should be located to the right as stated in the device.



4. Reposition and fit the battery cover back over the battery compartment. Ensure all of the brick style mounts are aligned to locate and attach the cover.

5. 

Turn the power to the device On. Push the On/Off switch to the right while looking from the front panel.

6. The display will illuminate and show the battery icon when battery powered. If the battery symbol flashes the battery power is low and they should be replaced.

NOTES

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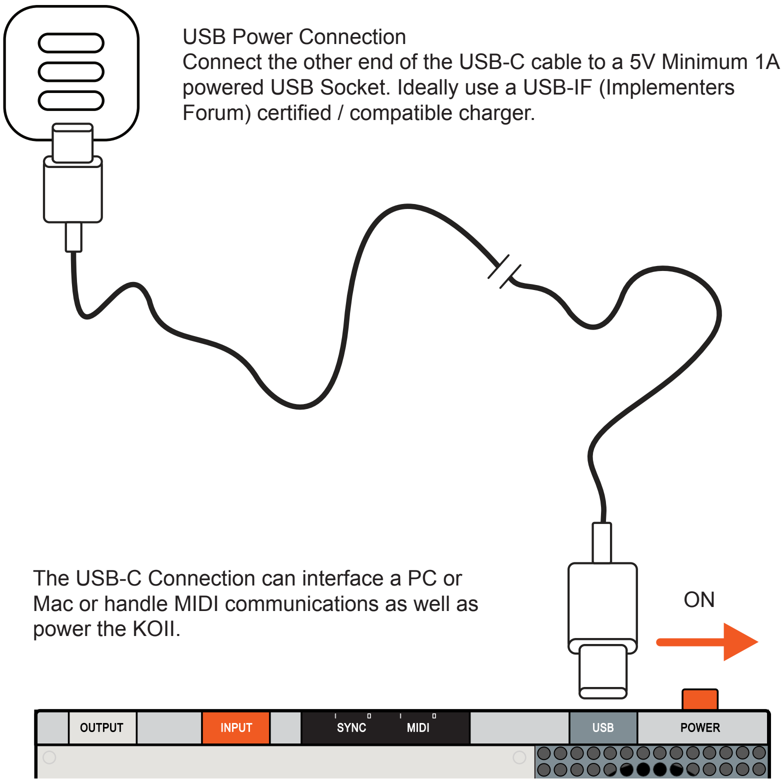
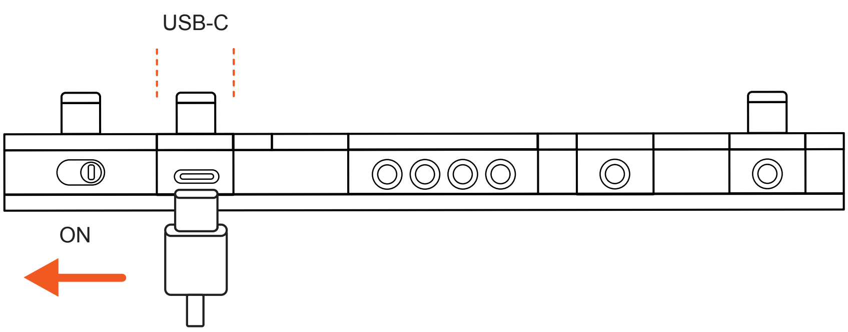
USB-C Powered Option

An alternative source of power is to use a USB-C Cable along with a USB power supply. This could be a mains powered USB adapter, another device such as a PC or Mac or a portable USB power source. Note that the KOII is a low power user so a portable battery pack should ideally have no auto off functionality or have the ability for auto off to be disabled.





The display will illuminate immediately on power up. The battery icon will be off when powered by the USB connection and will be lit orange when battery power is active.

2

Quick Start Reference

The KOII from the off can be a quirky and unusual device which takes a little time to get familiar with. Adopting the terminology, music production processes and developing a workflow will come as you spend more time with the device. This section is designed to speed up this process and not spend too much time on the deep dive or intimate details but to concentrate only on steps that start to make music. The KOII out of the box includes a quick start guide and can be a useful reference. This section starts with a quick reference guide, structured based on workflows rather than the device functions. This part is maybe a useful few pages to print or carry side by side with the KOII. This section will concentrate on some elements of a full song. It is a quick start to using the KOII to make music but does not cover everything it can do in detail. This section therefore lays the foundation for creating musical tracks that could normally go on to form a full arrangement. Three tutorials are included. These are written without any deep dives or detailed explanations, but designed to purely get you started for making some music with the KOII. They may be a little wordy as they walkthrough the basics and aim to help people new to the KOII get started. Equally this section can be skipped if you are a more experienced user. This covers the basics and only the fundamental steps needed to achieve a goal. This will be expanded further in upcoming sections of this book to add more advanced elements and give more detail into the specific element of the song. Remember that the KOII does have a song mode which allows the construction of multiple song positions of a full arrangement. As such this section will only give examples on how to create three parts and the assignment to scenes to add variations when played manually as a performance.

2 Quick Start

2.1 Quick Reference Guide

NOTES

A reference to keep side by side with the KOII when getting started.

Projects

Function	Command	Mode	Description
Open Project	Hold MAIN/COMMIT + Hold Pad 1-9	Main	Hold Main with Pad 1-9 until the selected project number i.e P 1 is displayed. A total of 9 projects are available.
Erase Project	Hold ERASE/SYSTEM + Hold MAIN/COMMIT + Hold Pad	Main	Hold Erase and Main with Pad 1-9. The 'PRJ' indicator will flash then 'DEL' will indicate the project has been deleted.
Lock Mode	Hold MAIN/COMMIT + Power Up	Main	Lock mode will allow normal editing but will not auto save the project. Patterns, Scenes, Pads etc are protected.

Tempo & Time Signature

Function	Command	Mode	Description
Tempo BPM	Press TEMPO/LOOP, Turn Orange X Knob	Tempo	X Knob will adjust the Tempo BPM when in Tempo mode between 60 - 180 BPM.
Tempo BPM	Hold TEMPO/LOOP + Press Pad	Tempo	Type in a value using pads to set the Tempo BPM when in Tempo mode between 40 - 399 BPM.
Tempo BPM	− / +	Tempo	Adjusts the Tempo BPM in Tempo mode. In Main mode it applies a tempo / pitch shift while playing.
Tap Tempo	Tap TEMPO/LOOP	Tempo	Tap the Tempo button 4 taps to set the tempo BPM.
Tempo Match	Hold SAMPLE/CHOP + Hold TEMPO/LOOP	Tempo	Analyse the Tempo of incoming audio and sets the global tempo to this BPM.
Metronome Vol	Press TEMPO/LOOP, Turn Black Y Knob	Tempo	Y Knob will adjust the Metronome volume level when in Tempo mode. Set to minimum level to mute the metronome.
Time Signature	Hold MAIN/COMMIT + Press TEMPO/LOOP		Selects the time signature option. The upper and lower time signature values changed with the X & Y Knobs
Time Signature	Turn Orange X Knob, Turn Black Y Knob		Orange X Knob adjusts the time signature upper beats per bar. Black Y Knob will adjust the lower note interval.

Sounds

Function	Command	Mode	Description
Select Sample	Hold SOUND/EDIT + Press Pad 0-9	Sound	Select a pad to load a sample to. Hold Sound and use Pads 0-9 to select a sample by its number 1-999.
Change Sample	− + Use [Shift] + Buttons to change x10	Sound	Select a pad to change. In Sound mode this will increment up or down in the library and change the assigned sample.
Clear Pad	Hold SOUND/EDIT + Empty Sound Slot	Sound	Set a pad to an empty slot to clear the pads sound. Empty sound slots will flash the number when selected e.g. 999.
Erase Sound	Hold ERASE/SYSTEM + Hold SOUND/EDIT	Sound	Erase the current sample from the sound library permanently. Flashes ‘SND’ then will display ‘DEL’ to confirm erase.
Set Amplitude	Turn	Sound	The X Knob adjusts the sounds amplitude volume level for the selected pad.
Set Pitch	Turn	Sound	The Y Knob adjusts the sounds pitch setting for the selected pad.
Play Sound	Press	Sound, Main, Tempo	Tap a pad with a sound to audition and play the assigned sound.
Copy Pad	Hold SHIFT + Press C	Sound	In sound mode and a pad to copy must be selected. Copies pad sound to the clipboard.
Paste Pad	Hold SHIFT + Press D	Sound	In sound mode and a destination pad must be selected. Pastes from the clipboard to the pad.
Sound Edit	Hold SHIFT + Press SOUND/EDIT − +		Select Sound Edit Mode. Navigate menu options with -/+ buttons. Display will show which menu is selected.
Playback Mode	Turn	Sound Edit ‘SND’	In Sound edit mode select the ‘SND’ Menu using -/+. Sub menu options are One Shot, Key poly playback and Legato.
Pan Sound	Turn	Sound Edit ‘SND’	In Sound edit mode select the ‘SND’ Menu using -/+. Turn Y Knob to adjust the panning of the sound audio.

NOTES

Sounds (Continued)

Function	Command	Mode	Description
Trim Sample	Turn In, Turn Out	Sound Edit ‘TRI’	In Sound edit mode select the ‘TRI’ Menu using -/+. In/Start and Out/End positions are adjustable with X & Y Knobs.
Envelope	Turn Attack, Turn Release	Sound Edit ‘ENV’	In Sound edit mode select the ‘ENV’ Menu using -/+. Attack time and Release time are adjustable with X & Y Knobs.
Time Stretch	Turn Type, Turn Stretch	Sound Edit ‘TIM’	In Sound edit mode select the ‘TIM’ Menu using -/+. X Knob selects the type, Off, BPM or Bar. Y Knob to stretch.
Save Edits	SHIFT + SOUND EDIT *> 2 Sec*		Save sound after editing to the existing slot number.

NOTES

Groups

Function	Command	Mode	Description
Select a Group	Press A	Main, Sound, Tempo	Press the group button A, B, C or D to select. The button LED will be lit.
Mute Level	Hold FADER + LEVEL 7	Main, Sound, Tempo	Fader for the Level parameter i.e. Pad 7, should be set to minimum level. Level is the default parameter for the fader.
Solo Group	Hold FX OUTPUT + Hold A	Main	Hold one or more group buttons plus the FX button to solo the selected groups. Release buttons to revert to normal.
Group Volume	Press A +	Main, Sound, Tempo	Holding the group plus adjusting the fader changes the group volume level.
Copy Sounds	Hold SHIFT + Press C Press C	Sound	Must be in sound mode. Copies all of the group sound to the clipboard.
Paste Sounds	Hold SHIFT + Press D	Sound	Must be in sound mode. Pastes from the clipboard to the group.
Choke Group	LEVEL 7 Press to toggle pad in or out of mute group	Sound Edit 'GRP'	In Sound edit mode select the 'GRP' Menu using -/+. Select a pad to add to or remove from mute / choke group.
Assign to Fader	Hold FADER + HPF 5	Main, Sound, Tempo	Select a pad to assign to the fader. Label above the pad defines which pad selects which parameter e.g. Pad 5 is HPF.
Clear Fader	Hold ERASE SYSTEM + FADER	Main, Sound, Tempo	Clear all current automation, events and parameter values for the fader. Display will flash 'FDR' then 'DEL to confirm.
Reset Fader	Hold SHIFT + Press FADER	Main	Reset to default fader assignments i.e. only LEV assigned. Display will show 'RES' for Reset.
External Audio	Turn IN Turn IFX	Main	In Main mode, Orange (X) adjusts level and Black (Y) adjusts the FX send amount. Both are for the incoming audio.

NOTES

Sampling

Function	Command	Mode	Description
Sampling Mode	SAMPLE / CHOP	Sample	Selects sampling mode. Default audio source is the Mic unless an audio 'input' is connected in which case this is used. Resampling internal sources is an option.
Sample Record	TIME 9	Sample	Pads trigger recording to the selected pad. Flashing LED's are empty slots, solid LED's indicate an active sample.
Hands Free Record	Hold SHIFT + TIME 9	Sample	Arm a pads ready for recording to the selected pad. Flashing LED's are empty slots, solid LED's indicate active sample.
Input Level	Turn	Sample	Adjusts the mic or input audio level.
Threshold	Turn	Sample	Adjusts the mic or input audio threshold level. Recording starts when the pad is held and the threshold is exceeded.
Sample Slot	Hold SOUND / EDIT + Press I Pad 0-9	Sound	Select a specific slot, typically empty from the library in which to record. Empty library slots have a flashing slot number.
Chop Mode	Hold SHIFT + Press SAMPLE / CHOP	Chop	Sample chop mode. Used for slicing the recorded sample across pads.
Auto Chop	Press C Target Group Press − +	Chop	In chop mode. Automatically slices the sample across the selected number of pads in the chosen target group.
Live Chop	LEVEL 7 Each pad pressed will record a slice.	Chop	In chop mode. Manually slices the sample to each pad pressed. Sample plays while pressing pads.

NOTES

Pattern Setup

Function	Command	Mode	Description
Select Pattern	Hold A + Press 7 Pad 0-9	Main	Select the pattern number using the numeric pads between 01 - 99. Patterns labelled as 'group & pattern #' i.e. A.07
Change Pattern	Hold A + Press – +	Main	Steps through and selects the prior or next pattern incrementally by pattern number.
Erase Pattern	Hold ERASE/SYSTEM + A	Main	Erases the currently selected pattern. Will display a flashing 'PTN' followed by 'DEL' to confirm erase.
Copy Pattern	Hold SHIFT + Press C Press C	Main	Copies the currently selected <u>pattern</u> to the clipboard
Copy Bar	Hold SHIFT + Press C	Main	Copies the currently active <u>bar</u> to the clipboard
Paste	Hold SHIFT + Press D	Main	Pastes the bar or pattern currently in the clipboard into the current bar location or pattern
Pattern Length	Press RECORD Press – +	Main	The pattern length, adjusted in 1 bar increments. Hold buttons together if live recording is in progress.
Duplicate Length	Press RECORD Hold SHIFT + Press – +	Main	Doubles the pattern and also duplicates the notes into the extended length.
Note Interval	Press TIMING/CORRECT Turn	Timing	Resolution of note interval for recording into the pattern. Options: 1/32nd, 1/16th, 1/16th Triplets, 1/8th, 1/8th Triplets.
Swing	Press TIMING/CORRECT Turn	Timing	Adjust the note swing 50-75 for recording. 50 is no swing. Only operates on 1/8th and 1/16th note intervals.
Quantize	Press TIMING/CORRECT Press – +	Timing	Switch between quantize on, which automatically fixes notes to the grid or free mode to record timing as played.
Keys Mode	Press KEYS	Main, Sound, Tempo	Switches the pads to play a sound chromatically, like a keyboard. Scale is selectable. Default is pad per sound.

NOTES

Pattern Setup (Continued)

Function	Command	Mode	Description
Keys Root Note	Hold KEYS, Press 7	Main	Select the root note in keys mode based on the pad / note selected.
Keys Octave	Hold KEYS + Press − +	Main	Transpose up or down the octave range on each press of the -/+ buttons.

Live Recording

Function	Command	Mode	Description
Live Recording	Hold RECORD + Press PLAY	Main	Triggers live recording immediately from the bar start. This allows recording using pads while the sequence plays live.
Live Recording	Press RECORD, Press PLAY	Main	Starts live recording after a count in of 4 from the bar start. This allows recording using pads while the sequence plays live.
Record from Start	Press RECORD, Hold SHIFT + Press PLAY	Main	Starts live recording from pattern start after a count in of 4. This allows recording using pads while the sequence plays live.
Arm Recording	Press RECORD, Press 7	Main	Recording is armed after pressing Record. Then pressing any pad will start the live recording.
Stop Recording	Press PLAY	Main	Stops live recording if in progress. Playback is also stopped.
Stop Recording	Press RECORD	Main	Stops live recording if in progress. Playback continues to play uninterrupted.
Overdub	Press PLAY, Hold RECORD	Main	While playing, hold record to overdub record. Any new recording is added to the existing pattern.
Live Erase	Hold ERASE SYSTEM + Hold 7	Main	While playing or recording, hold erase and the pad to erase. The steps will be erased for the parts and duration held.
Note Repeat	Hold TIMING CORRECT + Hold 7	Main	Hold timing and play any pad sound. The note will repeat while held at the note interval timing set.

Step Recording

Function	Command	Mode	Description
Navigating Pattern	Press − Press +	Main	In main mode. Pattern stopped, the -/+ button will step through the pattern based on the note interval resolution.
Record a Step	Hold RECORD + Press 7 Any Pad	Main	Records a pad note into the current step. Pad LED will be lit when a note is active. Position will stay on current step.
Erase a Step	Hold ERASE SYSTEM + Press 7 Any Pad	Main	Erases a pad note from the current step. Pad LED will be unlit when a note is not present. Position will stay on current step.

Recording Parameter Automation

Function	Command	Mode	Description
Step Record	Hold RECORD + fader	Main	With the sequencer stopped, records the values of the fader assigned parameters per step. Latches value to the step.
Live Record	Hold RECORD + fader	Main	With live recording in progress, records a change in value of the fader assigned parameters across all recorded steps.

Post Recording Edits

Function	Command	Mode	Description
Nudge	Hold SHIFT + Hold 7 + Press − +	Main	Nudges the pads note forward or back in ticks (24 per step) in free mode and the note step interval when in quantize mode.
Note Velocity	Hold SHIFT + Turn (orange knob)	Main	Adjusts the velocity setting of all notes for the selected pad and step.
Note Duration	Hold SHIFT + Turn (black knob)	Main	Adjusts the note duration between 1 tick min and 1 bar max for the selected pad and step. Also dependant on envelope.
Live Quantize	Hold SHIFT + Hold TIMING CORRECT Hold 7	Main	While playing. Select 'Correct' then hold the pad to quantize each pad note as played to the defined note interval.
Pad Quantize	Hold SHIFT + Hold TIMING CORRECT Hold 7	Main	While stopped. Select 'Correct', then press the pad to quantize all pad notes to the defined note interval.

NOTES

Scenes

Function	Command	Mode	Description
View Scene	Hold MAIN / COMMIT	Main	View the currently selected scene. Naming convention S.XX e.g. S.01, S.02.
Change Scene	Hold MAIN / COMMIT + Press − +	Main	Incrementally change to the next available scene up or down.
Select Scene	Hold MAIN / COMMIT + Press I Pads 0-9	Main	Select an available scene based on its defined number. Use the pads to select the scene number 01-99.
Erase Scene	Hold MAIN / COMMIT + Press ERASE / SYSTEM	Main	Erase the current scene from the project.
Commit Scene	Hold SHIFT + Press MAIN / COMMIT	Main	‘Commit’ creates a new scene by duplicating the current state to a newly active scene, retaining the prior scene.

Songs

Function	Command	Mode	Description
Song Mode	Hold MAIN / COMMIT + ENTER	Main	Main + Enter will open the song mode editor, indicated by the L.xx displayed.
Play Song	Hold MAIN / COMMIT + Press PLAY	Any	Main + Play from any mode will play the entire song from the start. Pressing play in song mode plays the song slot and shift + play plays the song from the start.
Select Scene	Hold SHIFT + Press − +	Song	Assigns a selected scene to the current song slot. Scenes must be available and pre-prepared.
Add Song Slot	Hold SHIFT + Press A *	Song	In song mode, add a slot. To delete hold shift + C and to insert Shift + D.
Navigate Song	Press − +	Song	Navigate through song slots

NOTES

Effects & Loop

Function	Command	Mode	Description
Capture Loop	Hold SHIFT + Hold TEMPO / LOOP	Main	Capture and play the loop. The loop duration and range is based on the time of holding the buttons.
Loop Length	Turn	Main	While loop is active, turn Y to adjust the playback duration. Used as a performance aid.
Loop Slide	Turn or Tap − +	Main	While loop is active, turn X to adjust the playback position. Used as a performance aid. Also use [+] [-].
Exit Loop	Press TEMPO / LOOP	Main	Exit the loop immediately and continue to play the pattern while in synchronisation.
Exit Loop	Press MAIN / COMMIT	Main	Exit the loop on the next bar and continue to play the pattern while in sync.
Varispeed	Hold − Hold +	Main	Shift the Pitch and Speed on-the-fly while playing patterns while the buttons are held. Clears on release.
Punch In Effect	Hold FX / OUTPUT + Hold 7	Main	Trigger a punch in effect when while holding and applying velocity with Pads. Pad selected represents a specific effect.
Master FX	Press FX / OUTPUT Press − +	Main	Select the effect to assign to the master section. Options: Delay, Reverb, Distortion, Chorus, Filter, Compressor
FX Send	Press FX / OUTPUT	Main	Adjust the send amount of audio routed to the global master effect.
FX Parameters	Turn Turn	Main	The X and Y Knobs adjust two of the defined parameters dedicated to the effect assigned to the master output.
Compressor	Hold SHIFT + Press FX / OUTPUT	Main	Select the master output, labelled ‘OUT’ compressor. This is the output effect and not the send effect compressor
Comp Params	Turn Drive Turn Speed	Main	Master Output selected. The X and Y Knobs adjust the drive and speed of the master output compressor.

Effects & Loop (Continued)

Function	Command	Mode	Description
Sidechain	Hold SHIFT + Press FX/OUTPUT	Main	Select the master output, navigate with -/+ to the sidechain option labelled 'S.Ch'.
Sidechain Params	Turn (Length), Turn (Shape)	Main	The X and Y Knobs adjust the length and shape of the sidechain from the MIDI sidechain trigger..

MIDI, Sync, System

Function	Command	Mode	Description
MIDI Channel	Hold SHIFT + Press SOUND/EDIT, Turn	Sound Edit 'MID'	In the MIDI Sub-Menu. Set the MIDI Out channel with Knob X for the selected pad. Channels 1-16.
MIDI Root	Hold SHIFT + Press SOUND/EDIT, Turn	Sound Edit 'MID'	In the MIDI Sub-menu. Set the MIDI Root Note.
MIDI Sync	Set in the System > MID	Main	MIDI Sync is setup in the MIDI Configuration settings for the system.
Sync	Set in the System > SYN	Main	Sync is setup in the SYNC Configuration settings for the system.
System Menu	Hold SHIFT + Press ERASE/SYSTEM	Main	Access to the system settings menu. Use the -/+ and the 'Enter' pad to navigate. Shift + Enter to backup in the menu.

NOTES

Summary of Punch-In Effects

Headline	Pad
Random Pitch	.
Sample Swap	0
Ratchet	Enter
Beat Repeat	1
Tape Stop	2
Modulated Filter	3
LPF	4
HPF	5
FX Sender	6
Tremolo / Gate	7
Pitch Bender	8
LoFi	9

Assignable Master Effect Available Parameters

Effect	Type	Orange (X)	Black (Y)
DLY Delay	Send	Length.	Feedback.
RVB Reverb	Send	Length.	Colour.
DST Distortion	Send	Drive.	Colour.
CHO Chorus	Send	Modulation.	Feedback.
FLT Filter	Send	Cutoff.	Resonance.
CMP Compressor	Send	Drive.	Speed.
OFF	Send	-	-
OUT Compressor	Master Inline	Drive.	Speed.

Automatically Save	No manual command needed. Projects auto save when not playing.

Some controls can be changed at a higher resolution for more exact adjustments. Use [Shift] when turning a Knob to set the value in a more precise way.

2.2 Tutorial 1 - Make Some Beats

NOTES

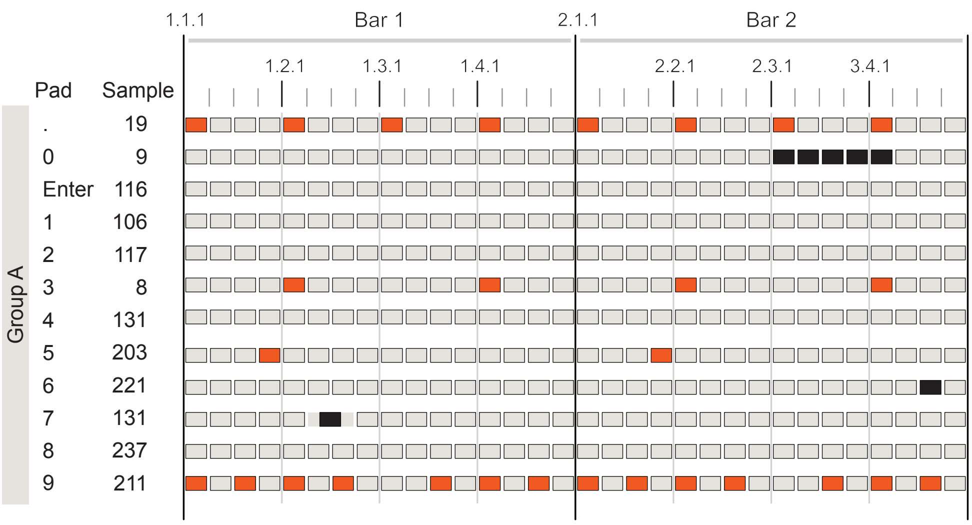
What a better place to start than making a beat. Orientated on the percussion and drums of a track plus an alternative or two using scenes to demonstrate how to create progressive variations. Remember that there are multiple workflows that can be adopted and you will choose your own style. These tutorials only demonstrate one workflow option.

How to Build Beats **Tutorial 1**
3 Key Learning Objectives.

1. Assemble sounds and create a drum kit
2. Sequence and record patterns.
3. Use scenes to create variations.

Beat Overview

The following beat will be created using Group A and the incumbent pads. It is possible to use an existing sample populated project, although this tutorial will create a kit from scratch and use a blank Project #8 to start. Think of a pad as the host to an individual track.



Pattern Length: Ln 2, 2 Bars.
Note Interval: 1/16th Notes.
Time Signature: 4/4
Tempo: 120 BPM

Note Active: Scene 1 & 2
Note Active: Scene 2 Only

NOTES

Lets start by setting up the requirements for the patterns. This will all be set-up for Group A. This typically is the group used for drums and percussion. Some settings may already be set as defaults but we can check to ensure you are familiar with these elements. We will start in a blank new Project 8. At various stages in the workflow or when a command has been completed, Press [Main] to return to the default 'home' operating mode to stay orientated.

SETTING THE PROJECT & PATTERN STRUCTURE

1. Open Project 8. Hold [Main] + Hold [8] Pad. This will flash then P 8 will be selected. Projects 1-5 are already populated with sounds and one of these can be used if a ready made kit is preferred.

2. Select a Group, Press [A] to select group A. All pads used will be edited within this group. Group A LED is lit to indicate selection.

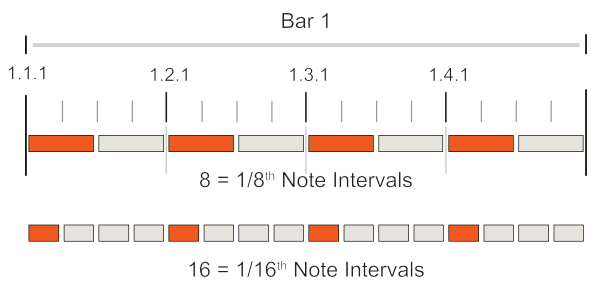
3. This example operates in 4/4 time signature. Lets check the setting. Hold [Main] + Press [Tempo]. This should display 4/4. Use the Orange (X) or Black (Y) knobs to change if needed.

4. The note interval is the resolution of the pattern sequencer and will determine the timing between notes. Lets set this to standard 1/8th Notes for now. Press [Timing], then Turn Orange (X) knob until '8' is displayed.

5. Now lets set the tempo beats per minute. Lets try 120 BPM. Press [Tempo] to enter this mode then Turn Orange (X) knob. The Tempo can be changed in finer settings using [Shift] with the knob.

6. We want to build this beat over 2 Bars. The KOII pattern length is set in intervals of 1 bar with the default length set to 1 Bar. Press [Record], then Press [-] or [+] to adjust the length in bars. Ln.1 is default so set to Ln.2.

7. Press [Record] again to exit record mode.



Continuing our tutorial, now lets build the kit for the percussion. We don't need to populate every pad with a sound but it is good practice to build a kit across all pads based on the general KOII convention. This will allow samples to be changed for defined instruments i.e. you could try various kicks located in Pad '.' for example. The diagram shown previously for this tutorial identifies all the kit samples from the factory library by number, loaded to each pad for this example. Sound Mode is used through the majority of steps in this workflow.

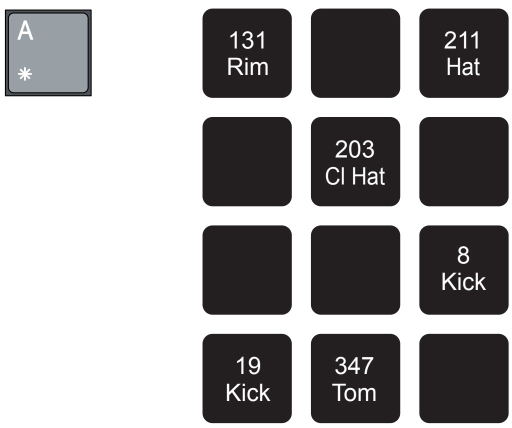
CREATING A DRUM KIT USING THE GROUP PADS

1. While still in Group A, Select each pad in order. This process will show how to load the sample to play and to contribute to the kit.

2. Select 'Sound Mode', Press [Sound]. The display will show 'SOUND' icon and the sample library number loaded into the selected pad is displayed too. Loading an empty sound library slot to a pad will effectively create an empty sound. As a tip, keeping slot 999 empty is useful for this purpose.

3. Adding samples to the pad sounds. Only the Sounds used in the beat pattern will be described in order to keep this tutorial simple. The process followed is the same for each sample, the process for the first sample below should therefore be repeated for others.

   I. Select Pad [.], LED is lit to show it's selection. The currently loaded sample, if any will be shown by its library slot number on the display. We want to load a kick sample. Sample # 19 is the 'Afterparty Kick' in the library. To select 19, Hold [Sound] + Press Pads [1] & [9]. Pressing [-] or [+] will also increment sample selection.



In [Sound] Mode, holding a pad will scroll the sample name across the KOII alpha-numeric display. Normally only the sample number is displayed for the selected pad.

NOTES

II. Select Pad [0], LED is lit. Hold [Sample] + Pad [3], [4], [7] to assign the Sample # 347, 'Tom Skin' to the pad.

III. Select Pad [3], LED is lit. Hold [Sample] + Pad [8] to assign the Sample # 8, 'Kick Verb' to the pad.

IV. Select Pad [5], LED is lit. Hold [Sample] + Pad [2], [0], [3] to assign the Sample # 203, 'Closed Hat Lo' to the pad.

V. Select Pad [7], LED is lit. Hold [Sample] + Pad [1], [3], [1] to assign the Sample # 131, 'NT Rimshot B' to the pad.

VI. Select Pad [9], LED is lit. Hold [Sample] + Pad [2], [1], [1] to assign the Sample # 211, 'Hat Closed 8x8' to the pad.

4. Pressing the pads will now trigger the sample assigned. Sample pads can be triggered in Main, Sound or Tempo mode.

In [Sound] Mode, using [-] [+] will increment the sound selection in single step increments. To jump sample selection in steps of 10, hold [Shift] + [-] or [+].

Now to sequence some steps. Live or Step Recording can be chosen, but examples for both will be used in this tutorial. We will work first with the default, Scene 1 for Group A. Each track is represented by a Pad. This section will guide individually for the Live and Step Recorded parts. The user preference will determine which mode works best for you.

LIVE RECORDING SCENE 1.

1. Lets begin with checking what Scene we are working in. Hold [Main] and the Scene will be displayed. This should begin with ‘S’ and we would expect to be in Scene 01 labelled S.01 by default. If another scene is displayed and active, Hold [Main] + [-] / [+] to change it to S.01.

2. Starting with the Pad [.], lets create a kick pattern using Live Recording. A typical 4 on the floor beat will be good enough.

3. Ensure Quantize is on. This will be shown with the ‘Q’ Icon displayed. To change between Free and Quantize, Press [Timing], then use [-] or [+] to toggle Quantize or Free. Quantize records notes adjusted to the absolute grid intervals while Free will record exactly as played, even if this is off grid.

4. It is often useful in live recording to use the metronome. Press [Tempo] then adjust the Black (Y) Knob for the metronome volume. This can be turned down later to mute if needed.

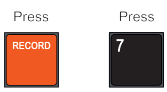
5. Hold [Shift] + [-] until the display shows 1.1.1, i.e. at the pattern start. Normally navigation and recording is performed in the current bar.

6. In [Main] Mode, Press [Record]. This arms live recording.

7. The sequencer will start recording when the first pad is played. As soon as Pad [.] is triggered the first note is recorded into the downbeat on Step 1.1.1 and the sequencer will continue to loop the pattern playback.

8. Subsequent beats can be recorded when played. Press Pad [.] on each on beat, 1.2.1, 1.3.1, 1.4.1, 2.1.1, 2.2.1, 2.3.1, 2.4.1. Slight misaligned playing style will be quantized to the 1/8th grid set previously.

9. Pad [9] contains a hat sample. Don’t select the pad yet. This can be recorded slightly differently to add more of a human feel and to speed the process. Select ‘Free’ instead of Quantize, Press [Timing] then [+]. Also while in the timing menu set the note interval to 16, i.e. 1/16th resolution.



Live Recording can be armed with [Record], start with a [Pad]. Other live recording modes can start with [Record] + [Play] for immediate start or [Record] then [Play] for a count in start.

9. Press [Main] to return 'home' and press [Record] to arm again.

10. Hold [Timing] + Hold [9], adjusting lower button pressure will record the notes and velocity while the pattern plays. Punch in repeated hat patterns to record through the 2 Bars, lifting the pad where you want to keep gaps.

11. Press [Play] to stop. Repeat these steps for Pad 3 and Pad 5 or use Step Recording. Press [Play] to hear the results so far.

Pressing [Play] will play the results so far. Lets create a new scene and add some further beats to create a variation. We will now focus on Step recording as the method used for this example to give a more precise control.

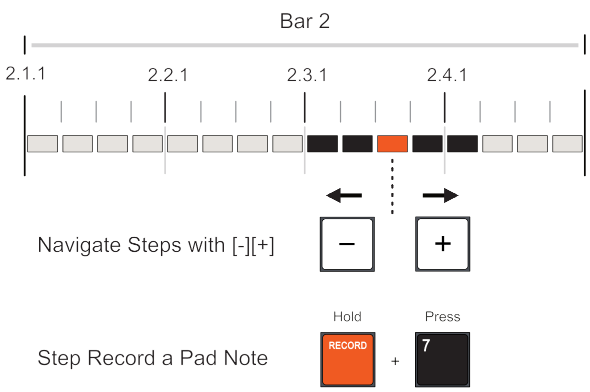
STEP RECORDING SCENE 2.

1. Commit the current scene to S.01 and duplicate its current state to a new scene S.02. Hold [Shift] + Hold [Commit]. Scene number will flash and S.02 will be created as the new, active scene. This allows any progression and development from scene 1 to continue into scene 2.

2. These tracks will add a short trill of notes at the end of bar 2 using Pad 0, an accent hit, Pad 6 and an off grid hit, Pad 7.

3. Ensure Quantize is off, i.e. set to 'Free'. This will be shown with the 'Free' Icon. To change between Free and Quantize, Press [Timing], then use [-] or [+] to toggle Quantize or Free. Note Interval Timing should retained as 16.

4. Press [Main] to ensure orientation back to the 'Home' mode.



NOTES

5. Use Step Recording to record the notes. Navigate to select a step in the sequence using [-] or [+]. Ensure Record is off. If it is on, Press [Record].

    I. Select Pad [0], LED is lit to show selection. Navigate to Step 2.3.1. Hold [Record] + Tap [0]. This will step record a note into the current step. Record notes also to 2.3.2, 2.3.3, 2.3.4, 2.4.1.

    II. Select Pad [6], LED is lit to show selection. Navigate to Step 2.4.3. Hold [Record] + Tap [6]. This will step record a note into the current step.

    III. Select Pad [7], LED is lit to show selection. Navigate to Step 1.2.3. Hold [Record] + Tap [7]. This will step record a note into the current step. To nudge the note off grid, Hold [Shift] + [7] + [+] or [-]. Nudge to a value of .39, ticks. This will play this note slightly earlier in the $1/16^{th}$ grid.

    IV. If a note is to be erased, select the step, then Press [Erase] + [Pad].

Playing the scenes now to listen to the two versions.

PLAYING THE SCENES.

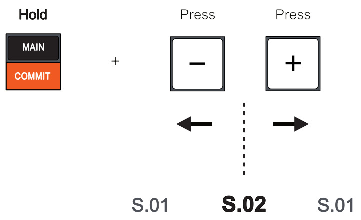
1. Hold [Main] + Press [-] to navigate to S.01.

2. Press [Play]. Scene 1 will play and loop playback.

3. While playing, Hold [Main] + Press [+] to navigate to S.02. Scene 2 containing the 3 additional tracks will now play.

4. Manually iterate between scenes as they play. Hold [Main] + Press [0] - [9] pads to select the scene number 01-99 directly. This is the easier option when working with higher number of scenes.

Navigate Scenes with [-] [+]



A scene must be created and exist first to allow it to be selected

2.3 Tutorial 2 - Add a Bass line

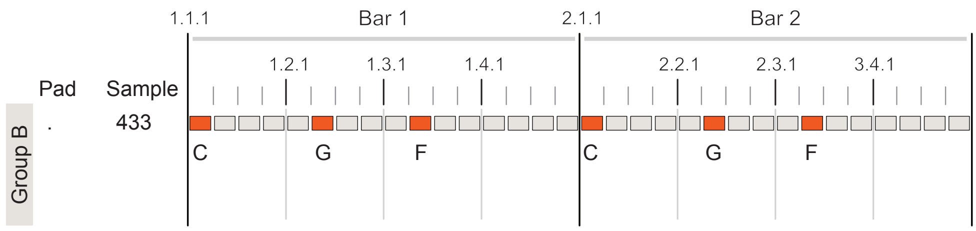
We have a beat and now what about a bass line? A simple option may work best with this beat but feel free to make it as complex or details as you wish. Experimentation is encouraged. But lets aim for achieving the key learning objectives as a start point.

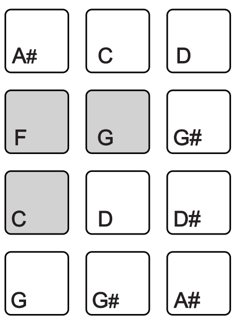
Add a Bass Line **Tutorial 2**
3 Key Learning Objectives.

1. Keys Mode for Chromatic Patterns
2. Changing System Menu options
3. Adding effects.

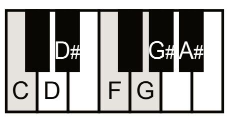
Bass Overview

The following bass line will be created using Group B and the incumbent pads. We will use only one sample but this will be recorded using 'Keys Mode' which allows us to play and record notes chromatically or from a selected scale. While the melody chosen has only three notes, feel free to use your own notes progressions. The process applied is the same.





Minor Scale



NOTES

To add a bass lets add a sample to a pad in the Group B and set the Scale to a Minor Scale instead of the default 12T. This will be an ultra simple bass line, after all we aim to teach and not to create a worldwide smash hit.

SETTING THE SOUND AND PATTERN STRUCTURE

1. Select a Group, Press [B] to select group B. All pads to be edited will be hosted in this group. Group B LED is lit.
2. We want to build this bass line over 2 Bars. The KOII pattern length is set in bars, default 1 Bar. Press [Record], then Press [-] or [+] to adjust the length in bars from Ln.1 to Ln.2.
3. Press [Record] again to exit record mode.
4. Load a sound to Pad, Press [.] and then Hold [Sound] + Press [4], [3], [3] to select the sample, 433 named ‘Mud B’.

SELECTING KEYS MODE

1. The scale is set in the system settings.
   I. Press [Shift] + [System] to open the System Menu.
   II. Press [-] or [+] to navigate at the menu level. In this example, select ‘PAD’ for the Pad menu options.
   III. Once selected, Press [Enter] to confirm. The ‘Enter’ option is located at the bottom right pad.
   IV. Press [-] or [+] to navigate at the menu level. In this example, select ‘SCA’ to select scale. Press [Enter].
   V. Press [-] or [+] to navigate the available scales. When ‘MIN’ is displayed Press [Enter]. This will select the Minor Scale.
   VI. Pressing [Shift] + [Enter] in the system menu will navigate back up the menu levels if required.
2. Press [Main] to select the ‘home’ mode.
3. Select Keys Mode. Press [Keys]. The Keyboard icon will be displayed.
4. The Pads can now be played as 12 notes on a keyboard. The available notes will be based on the selected scale i.e. Minor Scale. An external MIDI keyboard could also be used.

NOTES

Lets use Live Recording to capture a few notes for our bass line. Use any notes you wish and perhaps even try different scales. The example uses C-F-G to add a little unpredictability to the tune.

RECORDING A BASS LINE

1. As an option, commit the current scene and create a new one. Alternatively use the current scene.
2. Ensure Quantize is on. This will be shown with the ‘Q’ Icon. To change between Free and Quantize, Press [Timing], then use [-] or [+] to toggle Quantize or Free. $1/8^{th}$ Notes set as note interval i.e. 8.
3. In [Main] Mode, Press [Record]. This arms live recording. Hold [Shift] + [-] until the display shows 1.1.1, i.e. locate to the pattern start. Normally navigation and recording is performed in the current bar.
4. The sequencer will start recording when the first pad is played. This time we are in Keys mode and as such we will record notes. Pad 1 will represent C, Pad 4 is F and Pad 5 is G.
5. Record a C-F-G Pattern through each of the 2 bars using Pad 1, 4 and 5. The notes should be played at steps 1.2.2, 1.3.2, 2.1.1, 2.2.2, 2.3.2. Try other note sequences when playing and not recording to rehearse and find a bass line to your liking.
6. Press [Main] to orient back ‘home’. Press [Play] to hear both the bass line and the drum beat.

ADDING CHARACTER THROUGH EFFECTS

1. More character can be added to the bass sound with a send effect. In this case adding some distortion.
2. Press [FX] and navigate through the effects using [-] or [+]. Select the ‘DST’ Distortion option. In this mode the Orange (X) knob changes the drive and Black (Y) Knob adjust the colour parameter. These can be tweaked now or later to hone in on a desired sound.
3. While in [FX] Mode, the Fader will change the FX send level value. This is the audio amount sent from the main track to the effect loaded into the master FX slot. Adjust the send using the {Fader} to approx 70%. Try this while playing the track.

2.4 Tutorial 3 - Drone Pad

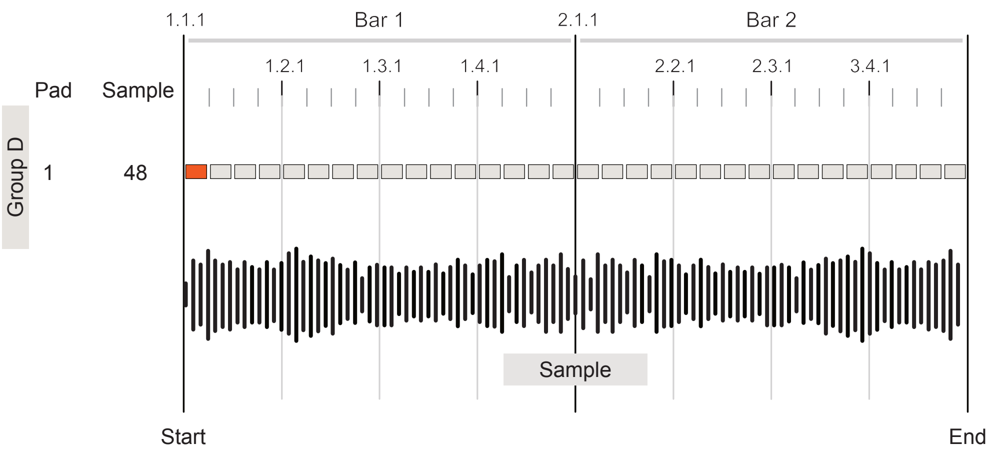
Creating a melody would follow a similar process to the bass line. This tutorial therefore shifts direction slightly to add a background drone style sound. The intent is to learn how to sample audio then using that to create a sound. It may not make people rush to Spotify to listen to our song, but will help develop your workflow knowledge even further.

Drone Pad **Tutorial 3**
3 Key Learning Objectives.

1. Recording external audio
2. Recording pattern automation.
3. Mixing tracks.

Sound Overview

The following sound will be created using Group D and the incumbent pads. The sound we use will be recorded with the mic for this example. Some minor edits will also be introduced. Feel free to record in another synth or external audio using the line input.



Sample via the Internal Mic a Vocal ‘Hummmmmmmmmm…’

NOTES

Lets start here by creating a new sound in Group D. Here we want to develop a long, drone type sound. Maybe not something that fits too well with the beat and track so far but will give insights into the elements to fulfil this objective. Maybe view this as an independent tutorial from 1 & 2…. Or…Take up the challenge to make it fit!!

SAMPLING TO CREATE A NEW SOUND.

1. Make sure you are oriented to the main home location, Press [Main].

2. Select Group D, Press [D].

3. Lets record a sound directly into a Pad and hence into a library slot. To make it easy for everyone we will record using the on board mic and perform a long 'hummmmm' vocal. Equally this could be a synth chord, iPhone sound or instrument through the audio input. Process to record and edit a recorded sample is the same.

4. Press [Sample] to enter audio sampling mode. If no audio input is physically connected 'MIC' is displayed. Adjust Orange (X) knob to set the audio input level while making a sound into the mic. Try 3 quadrants, 75% on the level.

5. Adjust Black (Y) Knob and set to about 25%, i.e. 1 quadrant on the level. This is the threshold and audio must be above this level to trigger recording.

6. All pads will be flashing to suggest they are free to record into. To record the audio, Hold a pad. Hold Pad [1] and perform the audio 'hummmmmmm' into the mic. When the threshold audio level is exceeded recording will start. The timer will run. Release the pad after about 5 Seconds of sampling. The pad button will be lit solid to indicate that an audio sample is present.

7. Press [Main] Mode to return home.

8. Pressing Pad [1] will play the sampled sound while ever the pad is held.

9. Lets perform some minor edits to the sound.

    I. Press [Shift] + [Edit] for Sound Edit mode. Navigate to 'TRI' using [-] and [+]. This is the trim option where the Orange (X) Knob and Black (Y) Knob adjust the sample start and end positions respectively. With a Tempo of 120, 1 Bar would be 2 Seconds i.e. BPM / 60 Seconds = 2 Beats per Sec. Set sample end to 4 Seconds for 2 bars duration.

    II. Press [Sound] and adjust the Pitch using the Black (Y) Knob to suit. Also the Orange (X) Knob can be used to change the samples amplitude.

NOTES

We can quickly add the sound as the first step in our 2 bar pattern and then change its character using effects and automation.

EFFECTS AND AUTOMATION.

1. Select Group D, Press [D]. Make sure you are oriented to the main home location, Press [Main]. Navigate to step 1.1.1 of the pattern.

2. Press [Record] to arm live recording. Hold [1] for the 2 bar duration.

3. A long drone is added to the step for the duration of the 2 Bar length.

4. Lets add some effects. Press [FX] and navigate to the Reverb ‘RVB” option using [-] and [+]. Adjust the fader to send the drone sound to the reverb and also adjust Orange (X) and Black (Y) knobs for the reverb effect. This can be revisited later and maybe experiment with other effects.

5. Automation adjusts the parameters while playing through the pattern. These can be recorded to be applied automatically as the pattern plays.

    I. Set the fader to control ‘Pitch’. Hold [Fader] + Press [8]. Adjustment to the fader position will now affect the tuning. While playing hold [Record] + Adjust {Fader} to record the automation

    II. Set the fader to control ‘Mod’. Hold [Fader] + Press [Enter]. Adjustment to the fader position will now affect the tuning. While playing hold [Record] + Adjust {Fader} to record the automation

    III. Set the fader to control ‘Pad’. Hold [Fader] + Press [3]. Adjustment to the fader position will now affect the tuning. While stopped select step 1.3.2. Hold [Record] + Adjust {Fader} to bottom. Repeat for step 2.3.2 but move fader to top. This will record fader position only for the steps selected.

    IV. Experiment with more automation options.

NOTES

You may have three tutorials now that can (or maybe cannot !!) play together as a single track snippet. If you want to mix the levels of each group to get a nice balance this can be done at group level.

MIXING 3 GROUP LEVELS.

1. This is an iterative process and can be performed during the workflow or at the end of the production process.
2. To adjust Group A Level, Hold [A] + Adjust {Fader}.
3. To adjust Group B Level, Hold [B] + Adjust {Fader}.
4. To adjust Group D Level, Hold [D] + Adjust {Fader}.
5. This can also be used to mute groups while working on others. To temporarily solo a group hold [FX] + Group [A], [B], [C], or [D].
6. The progress so far can be stored and move to a duplicate version for further development using [Commit].

3

Projects & Global

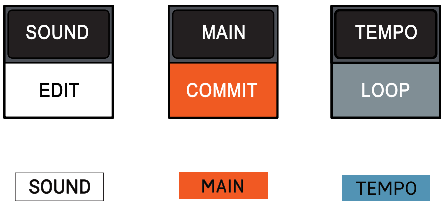
Every soup, pho or ramen needs a bowl to serve it up and make it easy to eat. In the KOII a project is a container which holds all the ingredients of a song arrangement including groups, sounds and scenes. A project is the highest level within the KOII structural hierarchy. KOII has 9 project slots which helps to build full music productions inside the box. Projects will be automatically saved at points during the normal workflow and the previous project will be recalled automatically on start up. Project management is therefore a fairly straight forward task and the currently working project is always available. If a project operates at the highest level, then it is logical that there will also be a number of functions that operate globally across the project. There are 3 primary operating modes in the KOII with the 'Main' Mode considered the default home state and the mode which is used for project activities. The Main Mode is the environment which controls scenes and patterns as well as projects. The 'Tempo' mode allows setting of the overall speed of the pattern playback in beats per minute and also controls the metronome option. The KOII also has a number of system settings but these will be covered elsewhere although they will be referenced if relevant in the context of the description. While the urge is to jump past the first couple of notebook chapter sections and get straight to the creative elements is strong, it is advised to take a little time to get an overview and understand the core structure of projects as a starting point. This will help speed up the workflow and understand the workings of the KOII more holistically and of course help get the best from the device.

NOTES

3.1 Introduction to the 3 Main Modes

The KOII operates in one of 3 main modes, each allowing control of specific elements within the selected mode. Each mode is selected using one of the 3 buttons located directly under the screen.

Mode Selection Buttons



Displayed Mode Icons

Mode Summary - Primary Button Options

Sound	Main	Tempo
Operates at the sound level which enables assignment and management of samples. Setup and editing of basic sound parameters.	The main ‘home’ mode is the primary operating mode. Live performing, arranging and recording. Also manage projects, scenes and patterns.	Tempo mode is where the time based elements such as beats per minute are set along with associated functions such as the metronome.

These modes are working environments for operating the KOII. The primary mode is ‘Main’ where most of the general playback and operating functions are performed. Main mode is where most activities are performed and is considered the ‘home’ location. Step in and out of the Sound and Tempo modes for configuration and for setting up of a project.

When operating with the KOII, one of these modes will typically be selected, although other modes also exist.

NOTES

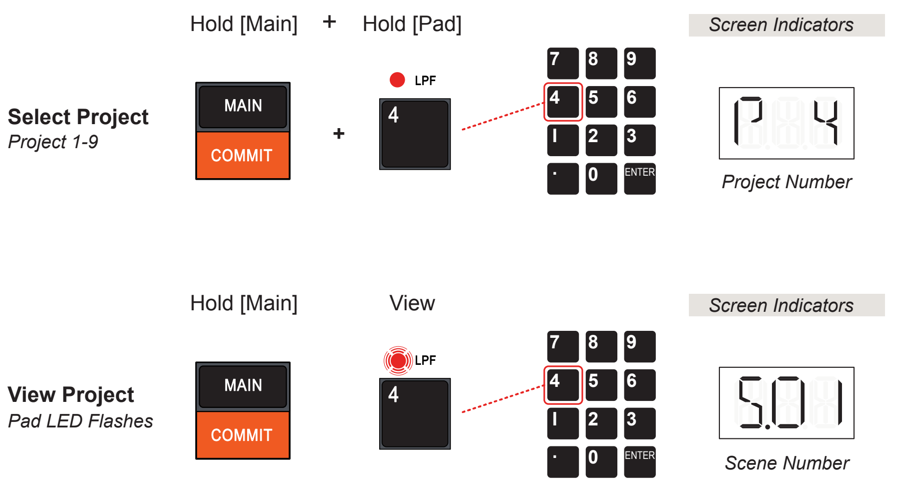
3.2 Choosing a Project

KOII has 9 projects each with a maximum of 80,000 notes. Think of a project as a container for the components of an entire song production.

The KOII when new out of the box and started for the first time will be configured with the first 1-5 projects populated with sounds. Project slots 6-9 will be empty. There are no 'factory' pre-configured patterns or scenes supplied.

SELECTING A PROJECT

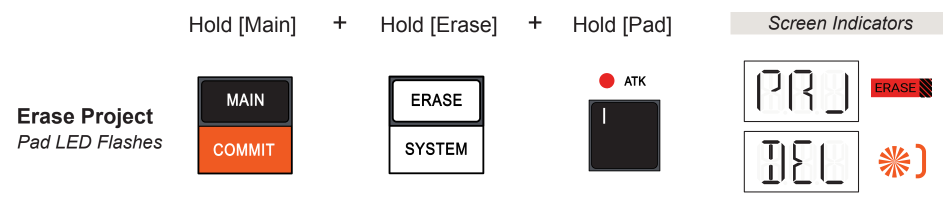
1. Ensure patterns are not playing, Press [Play] to stop playback
2. Hold [Main] to view which Scene and Project is active. The digital display will show a Scene number S.01, S.02 etc. The flashing pad LED represents the currently active project, for example Pad 8 represents Project 8.
3. Hold [Main] + Hold [Pad]. The chosen pad will represent the project to select 1-9. It is important to keep both buttons held.
4. The display will indicate the selected project P 1, P 2 etc while the representative pad button LED will illuminate for the selected project.
5. Release the [Main] and [Pad] buttons.
6. The new project is now active.



NOTES

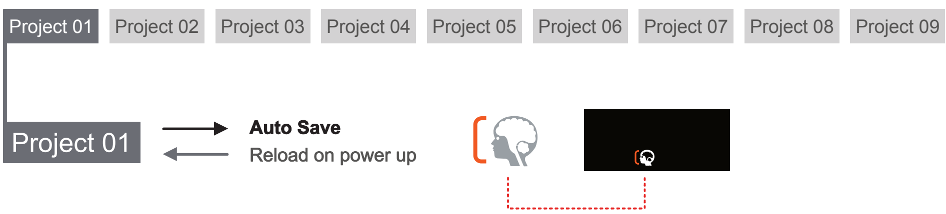
ERASING A PROJECT

1. Ensure the KOII is not playing. Press [Play] to stop if it is currently playing.
2. Hold [Main] + Hold [Erase] + Hold [Pad] 1-9 representing the project to erase. Select buttons in quick succession and keep them all held.
3. The display will indicate:
   - The ‘Erase’ symbol will be lit while holding buttons.
   - The digital display will flash with ‘PRJ’ followed by displaying ‘DEL’. Note that until the pad for the project is held the screen will display.
   - The ‘New Pattern’ display symbol will be lit. This will remain until a pattern is created / populated.
4. The project is now blank and ready for populating with sounds and creating patterns. Erasing a project cannot be undone.



Saving a Project

The active working project is automatically saved at intervals when the KOII is not playing. The ‘Auto-Save’ symbol will illuminate on the display while saving. The previous working project is reloaded on power up. There is no manual option to save a project.



3.3 Global Tempo

KOII has a global tempo range between 40-399 Beats per Minute or BPM. The tempo knob can control between 60-180 BPM directly while the wider range is only accessible by manually entering the tempo via the pads.

ADJUSTING THE TEMPO

1. Press [Tempo] to select tempo mode.

2. The ‘Tempo’ symbol will illuminate on the screen and the digital display will show the current tempo BPM with the ‘BPM’ symbol illuminated.

3. Turn (X) Orange Knob to adjust the beats per minute to a value between 60-180. The X Knob controls tempo and Y Knob adjusts the metronome volume when in ‘Tempo’ Mode. Hold [Shift] + Turn (X) for point value adjustment. Also adjustments are possible with the [-] and [+] buttons in this mode.

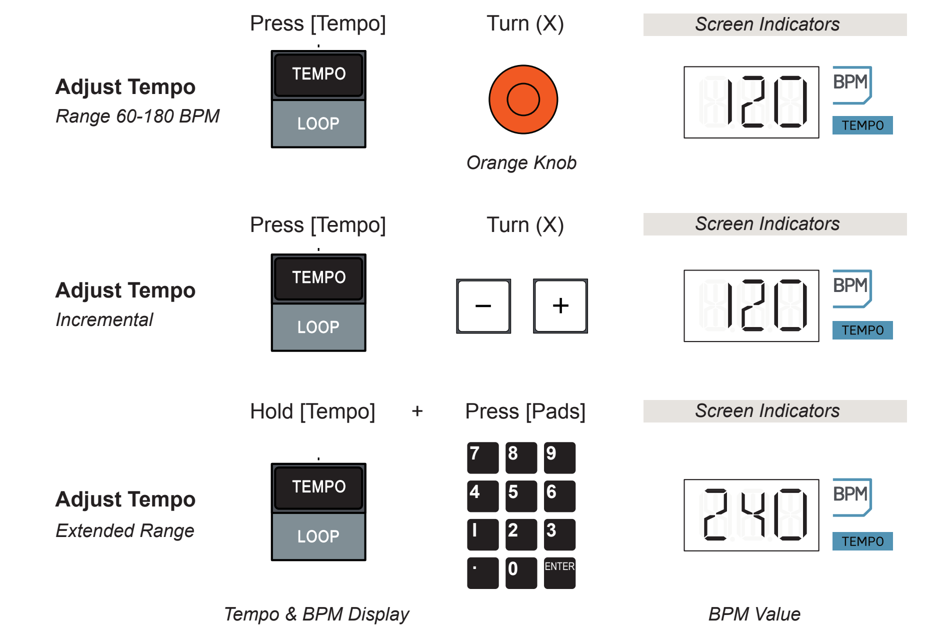
4. Alternatively set tempo by manual entry:-

   - Hold [Tempo] + Press [Pad] 1-9 to enter a tempo value manually e.g. for 120 BPM Hold [Tempo] + Press [1], Press [2], Press [0].

   - To enter point values, use the [.] pad when entering numbers, e.g. for 120.5 BPM, Hold [Tempo] + Press [1], Press [2], Press [0], Press [.], Press [5].

   - The pads revert to numeric entry pads while holding the [Tempo] button, indicated by the ‘Numpad’ symbol on the display.

   - Using the number pad allows a wider range between 40 - 399 BPM.



NOTES

TAPPING A TEMPO ENTRY

1. Press [Tempo] to view the current value.
2. The ‘Tempo’ symbol will illuminate on the screen and the digital display will show the current tempo BPM with the ‘BPM’ symbol illuminated.
3. Tap [Tempo] a minimum of 4 times in time with the beat to record.
4. The KOII will capture the timing of each tap and establish a tempo from the tapped rhythm.
5. A tempo BPM from the range will be set as the global tempo BPM.

NOTES

Tempo Match

A useful feature of the KOII is to monitor the audio input and to automatically determine a tempo from the audio scanned. This is based on the internal mic audio or audio from a device connected to the input of the KOII.

AUTOMATICALLY SETTING THE TEMPO BASED ON AUDIO IN

1. Play the external audio track. This can be physically connected to the input connection or played through the internal microphone. This process can be applied for normal playback or during sampling.
2. Hold [Sample] + Hold [Tempo] to analyse the audio. The display will show two dots while analysing and will display the sampling icon.
3. When the tempo has been established it will appear on the display and will be set as the global project tempo. The buttons can now be lifted.

Hold [Sample] + Hold [Tempo]

Screen Indicators

Tempo Match
Analyse & Set BPM

While playing external audio

While tempo matching generally works well on the KOII there are some occasions where the tempo is analysed correctly but then is applied to the global tempo at 1BPM lower that the analysed tempo of the external audio. This may be a system bug and is therefore something to be aware of.

NOTES

3.4 Metronome

The metronome is an aid to beat creation, generating an audible tick and visual beat that represents the tempo. This allows playback and recording accompaniment in time with the tempo. By default the metronome will automatically be audible when recording only. This behaviour can be changed in the KOII system settings.

Metronome Configuration Options - System Settings

Menu	Option	Description
Seq > Met	ON	Metronome will be audible in time with the tempo only when recording or playing, [Play] or [Record] + [Play].
Seq > Met	REC	Metronome will be audible in time with the tempo only when recording using [Record] + [Play]. *Default mode*.
Seq > Met	CNT	Metronome will be audible in time with the tempo at record during count-in only using [Record], then [Play].

CHANGING THE METRONOME BEHAVIOUR

1. The behaviour of the metronome is based on the system configuration. By default this mode is ‘REC’ where the metronome is audible when recording is active.

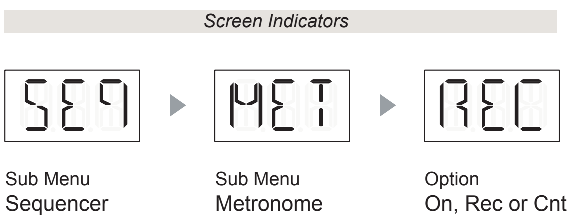
2. Hold [Shift] + Press [System] to open the system configuration settings.

3. The metronome is located in the sequencer sub menu. To select this menu Press [+] or [-] to navigate until ‘SEQ’ is in view, then Press [Enter] Pad.

4. The metronome ‘MET’ option is available in this sub-menu. If it is not visible, Press [+] or [-] to navigate options until ‘MET’ is in view, then Press [Enter] Pad.

5. The options available are ‘ON’, ‘REC’ or ‘CNT’. When navigating the currently selected option name will be static and the other available but de-selected options will flash. Press [+] or [-] to navigate options until the desired option is in view, then Press [Enter] Pad.

6. Once set, Press [Main] to exit the menu and return to the main view.



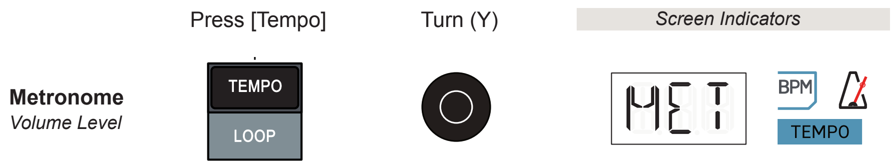
SETTING THE METRONOME AUDIO LEVEL

1. The metronome will play based on the behaviour in the system settings. By default this is ‘REC’ where the metronome is audible when recording using [Record] + [Play]. The metronome will only be audible at the appropriate point in recording based on the settings for metronome.

2. The metronome volume can also be adjusted when in tempo mode. Select tempo mode, Press [Tempo].

3. Turn Black (Y) Knob to adjust the metronome volume. The display will indicate ‘MET’ for Metronome while adjusting the knob.

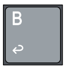
4. The metronome logo will illuminate and flash in time with the tempo. The Metronome level will be indicated by the ‘Y’ Level gauge on the display.



NOTES

3.5 Undo Command

Various editing options and functions in the KOII can be undone using the dedicated ‘Undo’ command. The umbrella icon will be displayed when an undo action can be performed. Use the [Shift] + [B] button to perform an undo command while ever the umbrella is lit. When an undo command is initiated, the command most recently performed is reversed and the KOII state will revert to its most recently prior state before the last action.

	Hold [Shift]	+	Press [B]	*Screen Indicator*
Undo *Undo last command*		+		 *Undo is available when lit*

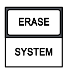
NOTES

3.6 Erase Command

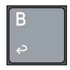
Some functions in the KOII can be erased using the dedicated ‘Erase’ button. This differs from the undo command in that undo steps back to the prior state in a chain of commands while erase clears / resets the state of the element in focus. For example a sample or pattern can be removed using the erase option.

Hold [Erase] + Hold *Target* | *Screen Indicator*

Erase
Remove item in focus



+



Example: Project

Lit while [Erase] is held

When target is deleted

Summary of Erase Command Options *(Sequencer stopped unless stated)*

Erase Target	Command
Clear A Project	Hold [Main] + [Erase] + Hold Pad [1] … [9] PRJ will blink followed by DEL confirming erase
Clear A Pattern	Hold [Erase] + Hold [A], [B], [C] or [D] PTN will blink followed by DEL confirming erase
Clear Current Scene	Hold [Erase] + Hold [Main] SCN will blink followed by DEL confirming erase
Clear Current Sound from Device Permanent erase	Hold [Erase] + Hold [Sound] Number and SND will blink followed by DEL
Erase Track Notes While stopped	Hold [Erase] + Hold Pad [.] … [9] TRK will blink followed by DEL confirming erase
Erase Track Notes While playing	Hold [Erase] + Hold Pads [.] … [9] Erase notes / pads at the points played
Clear / Reset All Fader Automations Reset to current positions while stopped *	Hold [Erase] + Hold [Fader] FDR will blink followed by DEL confirming erase
Erase Fader Movements Reset to actual position while playing live *	Hold [Erase] + Hold [Fader] FDR will blink followed by DEL confirming erase

** Undo command can be applied*

3.7 Copy & Paste Commands

NOTES

Some functions offer the option to copy an element to the internal clipboard and paste it elsewhere. These will depend on the function and context in which you are working. These are located as secondary options of the [C] and [D] Group Buttons which initiate the Copy and Paste commands respectively.

Hold [Shift] + Hold [C]

Screen Indicator

Copy
Copy to clipboard

Copy Command Initiated

Hold [Shift] + Hold [D]

Screen Indicator

Paste
Paste from clipboard

Paste into Target Destination

Summary of Copy / Paste Command Options - *Selected Mode indicated*

Copy / Paste - Mode	Command
Current Pattern - [Main]	Copy: Hold [Shift] + Press Group [C], [C] Paste: Hold [Shift] + Press Group [D] *
Current Bar - [Main]	Copy: Hold [Shift] + Press Group [C] Paste: Hold [Shift] + Press Group [D] *
Individual Sound - [Sound]	Copy: Hold [Shift] + Press Group [C] Paste: Hold [Shift] + Press Group [D] *
All Group Sounds - [Sound]	Copy: Hold [Shift] + Press Group [C], [C] Paste: Hold [Shift] + Press Group [D] *

Select the source, ie Pad 4. Then the element can be copied to the clipboard. Select the target destination i.e. Pad 5. The Paste command will then copy from the clipboard and paste into this destination. Copy / Paste also works across groups.

** Undo command can be applied after paste*

NOTES

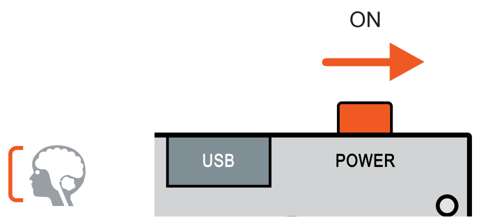
3.8 AutoSave & Lock Mode

There is no manual save option for the KOII. Saving is performed automatically and periodically as edits are made. The auto save icon will be lit when an auto save is being performed.

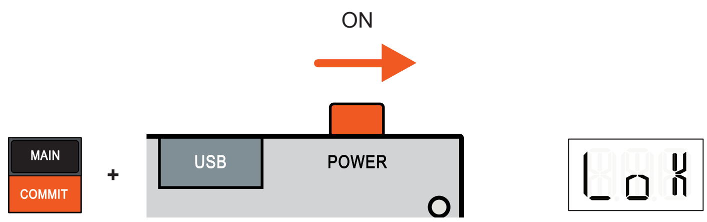
Lock mode allows tho KOII to operate in a protected mode to prevent edits overwriting the current state. Changes to projects, patterns, scenes, pads and settings can be made in lock mode but are not auto saved. This is useful for live performances and sets where only playback and temporary changes are required and the integrity of the original projects are protected from overwriting with these changes.

USING THE KOII IN LOCK MODE

1. Ensure the KOII is powered OFF.
2. Hold [Main] + Power Up the KOII by sliding the On/Off switch to ON.
3. The display will show ‘LOK’ on power up for a short period.
4. The projects can now operate as normal and edits can be made in the current loaded project. Saving is disabled.
5. When powered OFF and back ON, the state of the original project is recalled. This will avoid auto saving any temporary changes.



Auto Save icon is displayed when the KOII is saving. This is the normal operating mode when powered up for the device. Lock Mode is an alternate option.

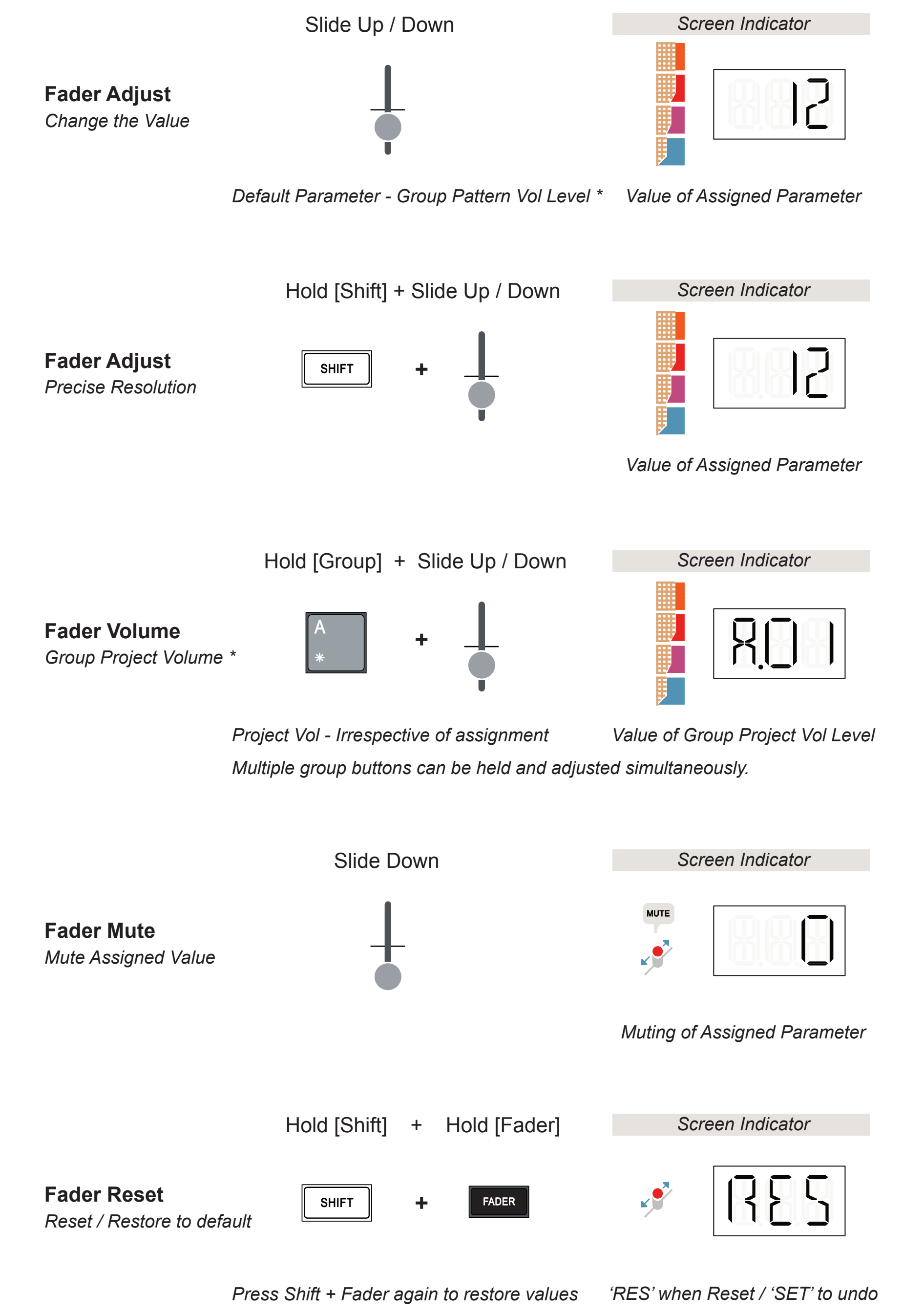


Turn the power to the device On while holding [Main] to start up in Lock Mode. Power Off then On to return to the original state and start as normal.

3.9 Fader Control

The KOII fader is a multi functional feature and operates at the selected group level. The fader controls parameter values for many functions. Fader control can be configured, the default is the selected group volume level.

Fader Operation

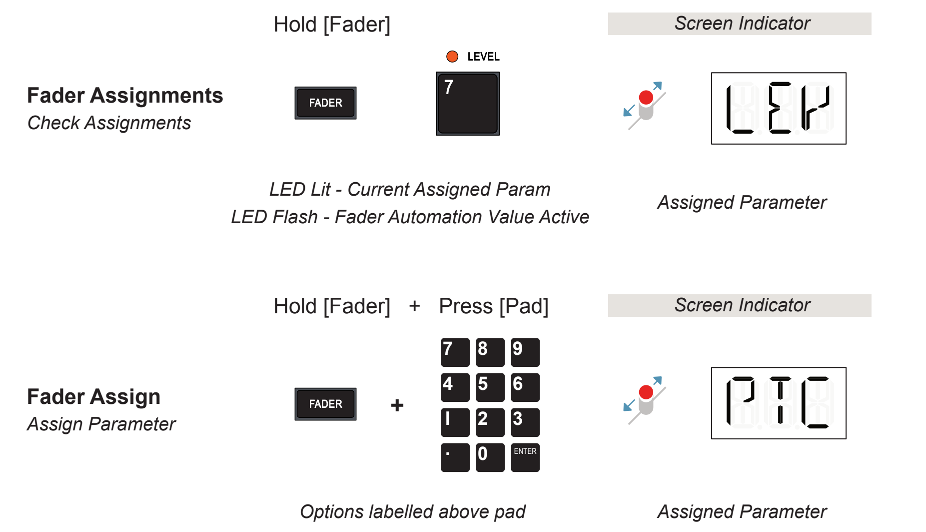


** Group project volume level applies to all patterns and scenes in the project group and cannot be automated. This is different to the Level 'LEV' default assignment which controls the group patterns volume and can be automated.*

NOTES

Fader control can be assigned to a parameter. This can be configured based on the labels indicated above the pads. Value changes can also be recorded in a pattern. The assignments are based on the selected group and not individual sounds.

Fader Configuration



The Fader Reset / Restore function can even be used creatively to mangle and go crazy with fader settings before resetting the values using [Shift] + [Fader]. Pressing this button combo again will restore the previous settings.

Automation refers to the automated control of parameter changes within a production. This is handled in the KOII by recording the manual parameter changes, typically with the fader, into a pattern.

Automation can be recorded in live recording or step recording mode. Live recording will record any movements of the fader at momentary values across the pattern. In step recording the fader position will be recorded and latched (until the next value change) to each selected step.

Fader Automation

Fader Record
Automation

Hold [Record] + Slide Up / Down

Records to the pattern live / active step

Screen Indicator

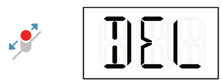
Value of Assigned Parameter

Fader Erase
Clear Automation / Values

Hold [Erase] + Hold [Fader]

Resets fader group and pattern

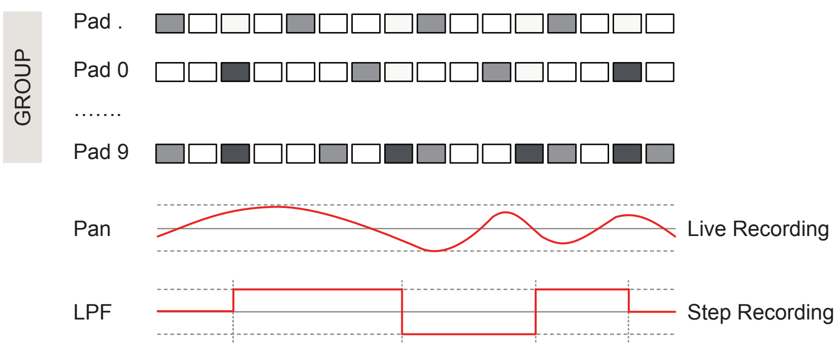
Screen Indicator



'FDR' flashes then 'DEL'

Hold the [Fader] button to check the state. Pads with flashing orange LED's contains automated values. Pad LED's with a static lit orange LED indicate the currently selected fader assignment. Pads with no LED lit have no fader assignments or automations. This will be respective to the group selected.

Recording Automation Example



When step recording, the value is set for a step, but will latch to that value through the pattern until the next value change.

NOTES

NOTES

Notes on erasing the fader automation.

Fader automation can be applied to Multiple parameters. Also the settings can be easily erased, however the behaviour is different depending on how the erase is performed. This applies to all fader assignments.

- With the sequencer stopped. Hold [Erase] + Hold [Fader] until ‘FDR’ flashes then ‘DEL’ is displayed to erase the automation. The current fader position values will be set once the erase command has been completed.
- With the sequencer playing. Hold [Erase] + Hold [Fader] until ‘FDR’ flashes then ‘DEL’ is displayed to erase the automation. A snapshot of the fader values at the point the erase and fader buttons are pressed will be set once the erase command has been completed.
- Both the scenarios can be undone as can resetting the fader.

Fader Assignment Summary

Fader Assignment Options per Group - See Pad Labels

Parameter	Pad	Label	Description	Range	Default
Tune	.	TUNE	Transpose tuning in semitones	-12 to +12	0
Velocity	0	VEL	Velocity level	0 to 100	100
Mod	Enter	MOD	Modulation amount applied. Introduces a vibrato effect	0 to 100	0
Attack	1	ATK	Attack time (X) of the group sound envelope.	-99 to 99	0
Release	2	REL	Release time (Y) of the group sound envelope.	-99 to 99	0
Pan	3	PAN	Panning of group.	L99 to R99	Mid
Low Pass Filter	4	LPF	Filter cutoff	animated	
High Pass Filter	5	HPF	Filter cutoff	animated	
Effect Send	6	→FX	Amount of group audio sent to the FX section.	0 to 100	0
Level (Default)	7	LEVEL	Pattern volume. Different to the group project volume.	0 to 100	100
Pitch	8	PITCH	Tuning of pitch	-5.0 to 5.0	0.0
Time	9	TIME	Time stretching of group	0.10 to 2.00	1.00

Try setting multiple parameter combinations to automate or use as a live performance feature.
For example combining the LP and HP Filters for a Band Pass function.

4

Samples & Sounds

The KOII can hold up to 999 samples (or 64MB of samples) and is supplied with 300 samples leaving 699 for user samples. A sample is the central audio source for every sound. The sound can then be shaped and changed by editing the sounds parameters. This includes trimming a sample range, manipulating the playback mode or time and shaping its envelope. A sample can also be chopped and sliced to play across the pads but this will be covered in the sample recording section of this guide. Samples are located in the on-board sound library and organised by genre of the sound type. User samples can be recorded directly into the KOII or transferred in from an external PC or Mac. A TE online, web MIDI compatible sample transfer tool is a useful aid to managing samples between the KOII and the PC or Mac. This will transfer .WAV and .AIF samples into a compatible format for the KOII. Samples can also be recorded directly at a rate for stereo / mono at 46.875 kHz / 16-bit giving around 12 mins of mono and 6 mins of stereo audio storage. Always bear in mind that higher sample rates will account for more of the 64MB memory. The lowest spec samples will be more memory efficient but will be reduced in audio quality. Finally, sound edits can be saved into the numbered slot once editing is complete.

4.1 Sound v Sample

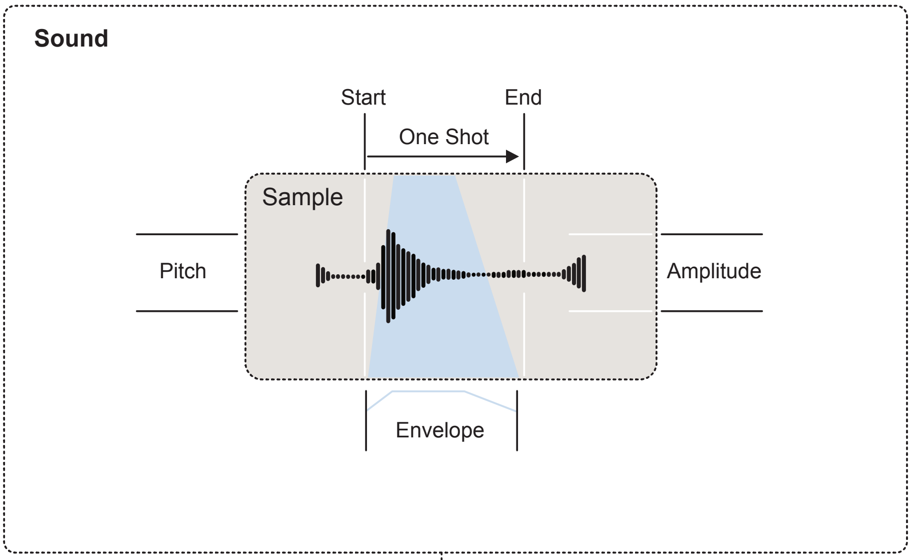
The terms sound and sample are often used interchangeably. A sample is a snapshot of recorded audio in its rawest form and is the fundamental audio element used in a sound. A sound adds a number of additional settings to change, enhance or edit the sample. This could be simply trimming the start and end or more elaborate parameters such as time stretching to adjust the tempo of the sample match the project BPM.

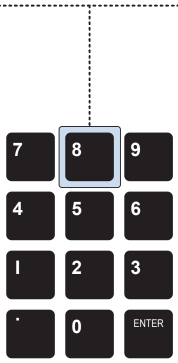
Think of a sample as the core audio element, unaltered and stored in the library and when loaded and parameters assigned becomes a sound.

Example: Sample v Sound

Example of a sample with some associated parameters to create a unique sound.

Edits in Sound Mode are stored as sounds in the current project and do not affect or change the original sample itself.





A sound can be assigned to a pad to allow pattern recording and playback.

NOTES

4.2 Auditioning the Pad Sounds

Pads are used to manually play and record patterns. Pads are populated with sounds. A project should therefore be loaded with a kit of sounds assigned to the pads. Out of the box, KOII Projects 1-5 are pre-populated but you can change these or add sounds to a blank project.

PLAYING THE PADS

1. Ensure the project is open. The quick option is to load a preconfigured project. Example for a factory project, Hold [Main] + Hold Pad [1] - [5]. These factory projects have pads that are loaded with the sounds.

2. Auditioning pads will operate in any of the 3 main modes.

3. To change the pad group, press [A], [B], [C] or [D].

4. Press [Pad] to play the loaded sound for the specific pad. This should be audible when playing and visible on the display audio meter. Sounds can be loaded to each of the 12 pads.

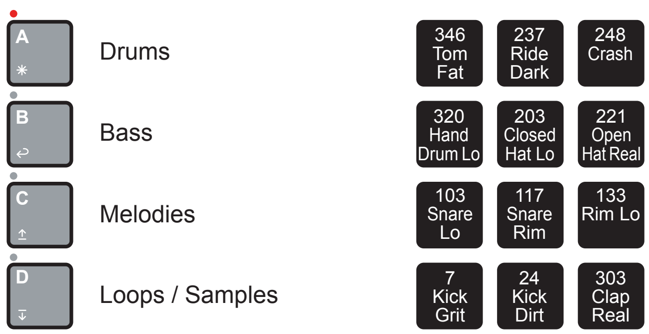
5. Audio level is set for the pad group by default using the fader control. Turn this up if the sounds cannot be heard. Note that the fader can be manually assigned to other parameters but by default is assigned to ‘LEV’, the pattern group volume level.

6. To view the sample number and name loaded into a pad, Press [Sound] to select ‘Sound Mode’. Hold [Pad] to view the sample number and while held will also display the sample name.

Example: Group Convention & 12 Pad Assignments

Tap Group [A] - [D]

Tap Pad [.] - [9] to play each sound



Group Conventions

Sample Number and Name

To view name, Hold [Pad] in ‘Sound Mode’

NOTES

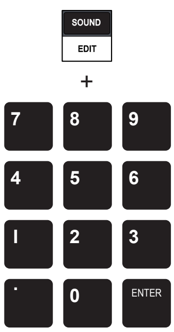
4.3 Loading & Erasing Samples

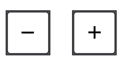
The KOII is supplied with 300 on-board samples and more can be added. Up to 64MB of memory or 999 samples is the capacity available. Samples can therefore be loaded into a pad or replace existing pad samples.

Once a sample is loaded, it can be edited along with a variety of parameters in order to shape the sound. Activities to load or edit sounds is performed in 'Sound Mode'. Samples can also be loaded to pads using the web browser based EP Sample tool.

LOADING A SAMPLE TO A PAD

1. Select Sound Mode, Press [Sound]. The display will show the sound mode symbol.
2. Select the pad group, press [A], [B], [C] or [D].
3. Select the pad to edit. Press [Pad].
4. The display will show the number of the sample currently loaded to the pad and will also indicate if the sample is mono or stereo by the displayed symbol. The number is based on the sample library location. Keep the pad held to also see its name.
5. To assign or change a sample for the selected pad, either:-
    - Press [+] or [-] to navigate and select from the available samples. Each is presented by its sample slot number. Hold [Shift] + Press [+] or [-] to navigate in larger increments of 10.
    - Hold [Sound] + Press [Pad]s i.e. Type the sample number using the pads. Example, Hold [Sound] + [1], [9], [9] for sample 199. If no sample is found in the selected number location, its number will flash.





Select by sample number or iterate up and down through the sample list

Sample number is displayed. If number is flashing, the sample slot is empty.

or

Sample format is also shown

NOTES

Samples are stored based on a standard KOII convention to help organising samples by genre. This is used as a guide and samples can in reality be located in any slot. It does however make it easier to recall samples and speeds up the workflow by sticking to the defined convention as well as ensuring compatibility with the TE KOII tool set. The KOII has 300, 33.7MB of samples in the Teenage Engineering supplied factory library.

Sample Library and EP Sample Tool Convention

Sample Slot Numbers	Genre	Description
1 - 99	Kicks	A variety of kick samples. The factory supplied library consists of samples 1-31.
100 - 199	Snares	A variety of snare samples. The factory supplied library consists of samples 100-145.
200 - 299	Cymbals	A variety of cymbal samples. The factory supplied library consists of samples 200-253.
300 - 399	Percussion	A variety of perc samples. The factory supplied library consists of samples 300-357.
400 - 499	Bass	A variety of bass samples. The factory supplied library consists of samples 400-459.
500 - 599	Melody	A variety of melodic samples. The factory supplied library consists of samples 500-559.
600 - 699	Loop	No factory samples provided in this category.
700 - 799	User 1	No factory samples provided in this category.
800 -899	User 2	No factory samples provided in this category.
900 - 999	SFX	No factory samples provided in this category.

ERASING A SOUND PERMANENTLY FROM THE LIBRARY

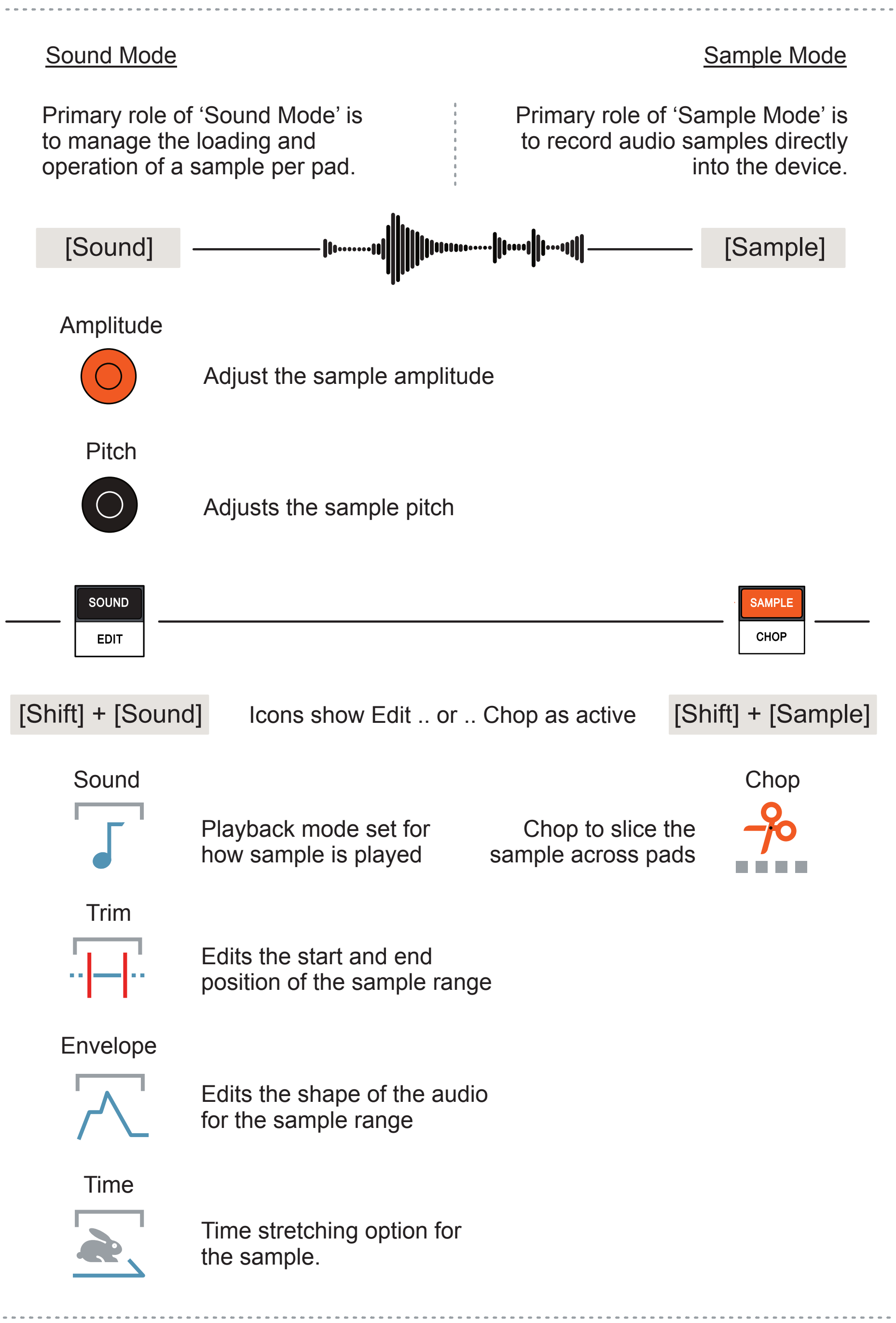
1. Select Sound Mode, Press [Sound]. The display will show the sound mode symbol.
2. Select a sound. For example select a [Pad] with a sound or navigate using [+] [-] through the sound library. The number will be displayed.
3. Hold [Erase] + Hold [Sound].
4. While holding the buttons, the display will show the ‘Erase’ icon and flash ‘SND” followed by ‘DEL’ to confirm that the sound is deleted from the KOII on board library.
5. The undo command cannot be used to restore the sample.

4.4 Sample Editing Options

The KOII is supplied with 300 on-board samples and more can be added. A sound can be edited and further elements applied when operating in ‘Sound’ / ‘Edit’ Modes. These affect how a sample is played within its own pad. In addition a sample can be sliced across multiple pads to create a kit from a single sample. This chop feature is available in the Sampler as this is a function typically used directly after recording a new sample.

NOTES

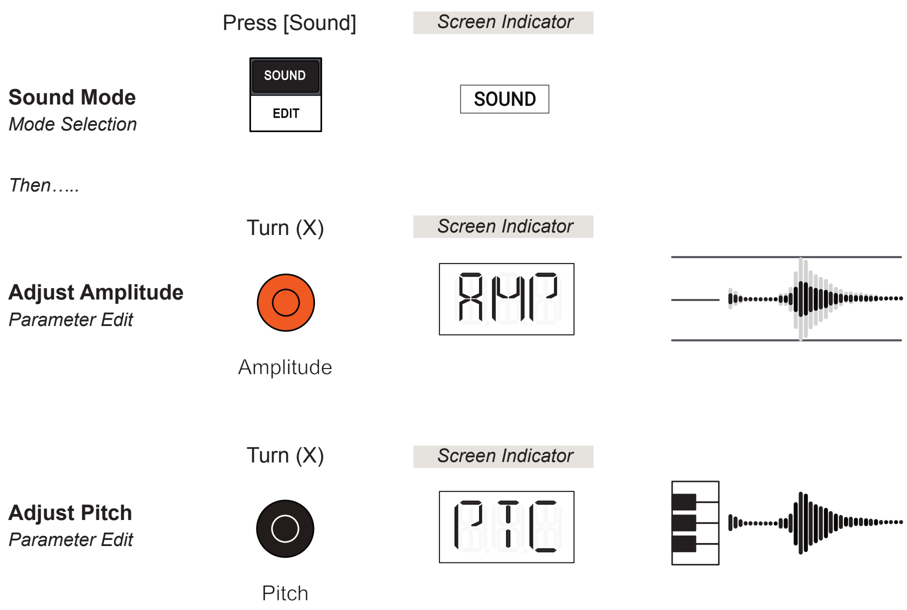
Sample Editing Options



NOTES

4.5 Sound Mode Amplitude & Pitch

As well as being the mode in which sounds are loaded to pads, the 'Sound Mode' offers immediate access to amplitude and pitch adjustment. These are controlled using the dynamic X and Y Knobs.



ADJUSTING THE AMPLITUDE OR PITCH

1. Select a pad with a sound loaded to edit.
2. Select Sound Mode, Press [Sound]. The display will show the sound mode symbol and the sample number on the alpha-numeric display.
3. To change the amplitude, Turn Orange (X) Knob. The display will shown 'AMP' while the parameter is being edited. Amplitude affects the overall volume level of the sample.
4. To change the pitch, Turn Black (Y) Knob. The display will shown 'PTC' while the parameter is being edited. Pitch affects the frequency and hence the musical tone of the sample.
5. It is advised that as there are no parameter values displayed other than the general X and Y indicators, to make the adjustments by ear.

NOTES

4.6 Sound Edit Mode Options

The ability to edit a sound further is also possible in ‘Sound Edit Mode’. This offers more detailed editing on the start and end position, envelope attack and release, time stretching and selecting the playback mode. MIDI settings are accessed here too. This mode is accessible as the secondary option on the [Sound] button i.e. Edit. Any edits can be saved to the same slot number.

	Hold [Shift]	+	Press [Edit]	*Screen Indicator*
Sound Edit Mode *Mode Selection*	SHIFT	+	SOUND / EDIT	 *Icon for current edit mode option*

Then Press

	Press [-] or [+]	
Sound Edit Mode *Option Selection*	− +	Cycle through the available edit options on each press of the +/- buttons.

	Hold [Shift]	+	Hold [Edit]	*Screen Indicator*
Save Sound Edit *Save to slot number*	SHIFT	+	SOUND / EDIT *> 2 Seconds*	SAV *Save only when play is stopped*

Available Sound Edit Mode Options

Option	Display	Description
SND - Sound	SND	Sound playback options. Selects how the sample is played; One Shot, Key - Polyphonic playback or Legato - monophonic.
TRI - Trim	TRI	Adjusts the start and the end length positions of the selected sample.
ENV - Envelope	ENV	Attack phase which ramps up to the final sound level from a note on trigger and also sets the release ramp down when note is off.
TIM - Time	TIM	Affects the time stretching of the sample in order to match the tempo BPM without altering the sample pitch.
MID - MIDI	MID	MIDI setting to set communication channel and root note for connecting an external device such as a keyboard *.
GRP - Mute Group	GRP	A mute group, also called a choke group, clusters pads together to allow only one sound from the group to play at once.

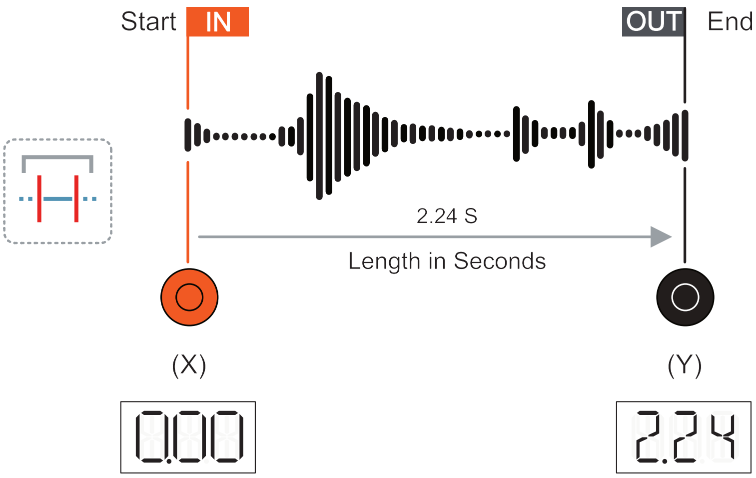
* MIDI Settings will be covered elsewhere in this guide book and this section will concentrate on the sound / sample specific editing options.

NOTES

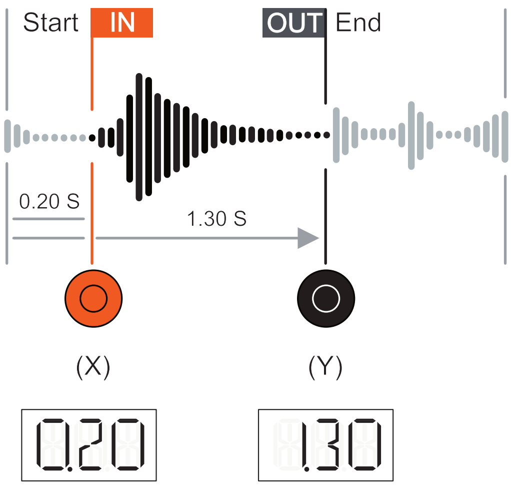
4.7 Trimming a Sound Start & End

The ability to edit a sounds start and end position based on the sample loaded into a pad is possible in 'Sound Edit' mode. This is accessible as the secondary option on the [Sound] button.

Original Sample



Edited Sample



The KOII does not show a graphical illustration of the sample on the display. The start and end positions are displayed as numerical values when edited. These represent the positions in seconds from the original sample start.

NOTES

TRIMMING A SOUND START & END

1. Select a pad with a sound to edit.

2. Select Sound Edit Mode, Hold [Shift] + Press [Edit]. The display will show the sound edit mode symbol and the alpha-numeric display will show the selected option. Edit is secondary option on ‘Sound’ button.

3. To change the selected option, Press [+] or [-]. Each press will cycle through each option incrementally. The display will indicate the current option selected. <u>Ensure ‘TRI’ Trim is selected</u>.

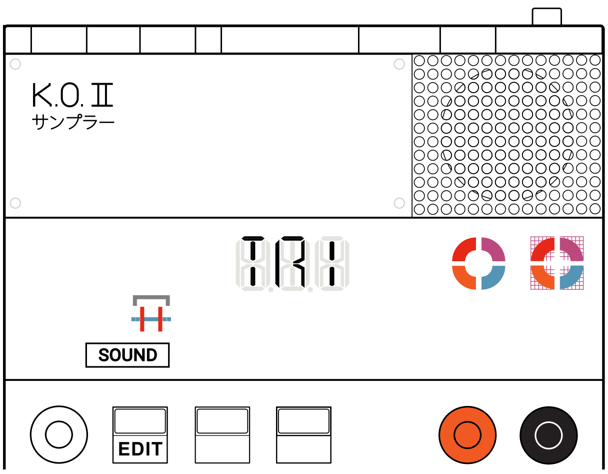
4. Adjust the ‘In’ setting. Turn the Orange (X) knob. This will determine the playback start position and is set in seconds from the original ‘0.00’ position. The lower ‘In’ setting cannot be set above the ‘Out’ value.

5. Adjust the ‘Out’ setting. Turn the Black (Y) knob. This will determine the playback length by setting the end position, set in seconds at a location below its original end position. The upper ‘Out’ setting cannot be set below the current ‘In’ value.

6. Using [Shift] when setting the values will offer a more precise adjustment.

7. Tapping the pad will audition the sound. This is useful when setting the range by ear through the editing process. The two X & Y quadrant icons will also visually represent the sample trim positions.

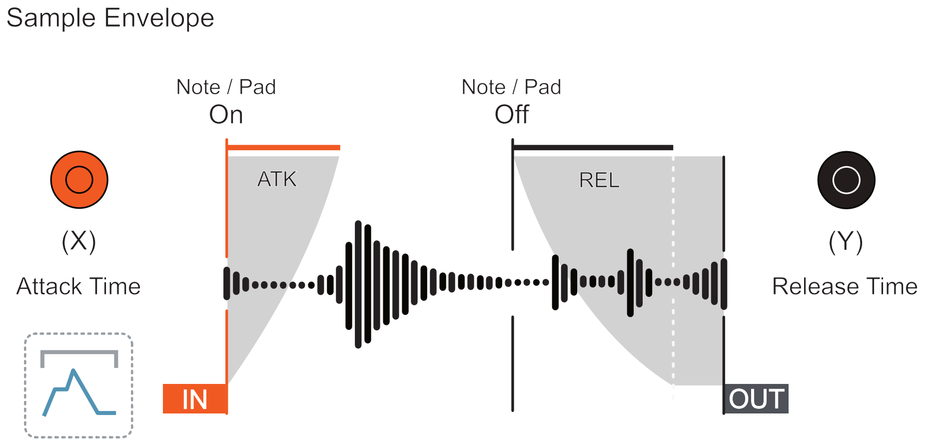
8. Note that the trim in and out positions can also be set from within the EP Sample Tool. This will set the In and Out positions in real time reflected in the KOII device.



NOTES

4.8 Shaping the Sound Envelope

The sound can be shaped with the Attack and Release Envelope - ENV. This controls the fade in of the initial part of the sound and the fade out of the sound after the note trigger / pad is released. For example, Percussive sounds may need to be sharp and punchy while pad sounds will be smooth and long, evolving over time.



The KOII does not show a graphical illustration of the sample on the display. The attack time and release time ramps are indicated by the X and Y quadrant icons on the display and are not displayed as numerical values.

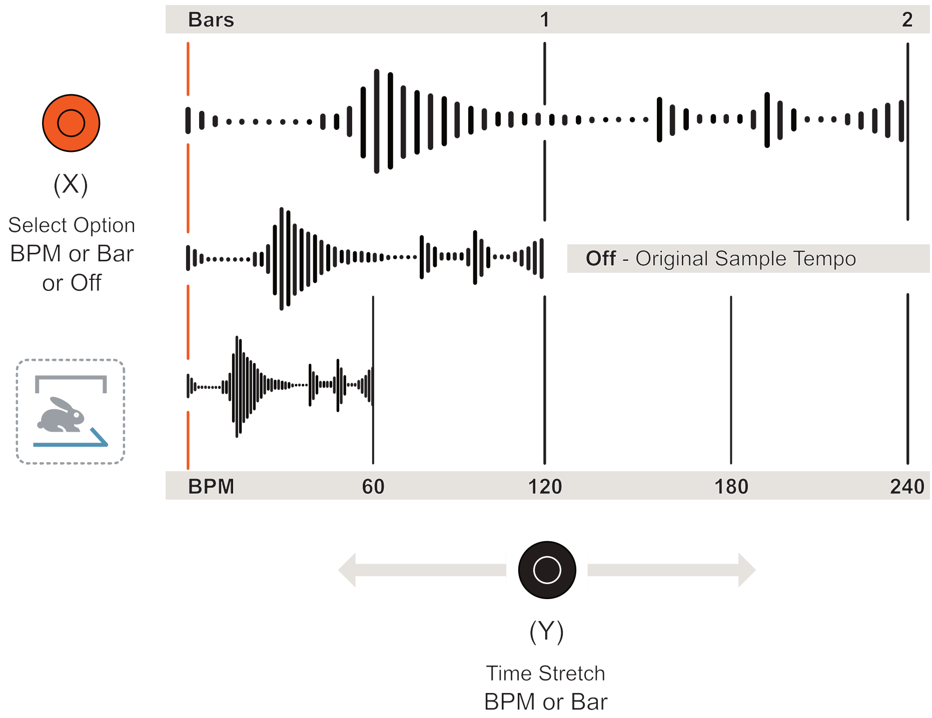
SHAPING A SOUND ENVELOPE

1. Select a pad with a sound to edit.
2. Select Sound Edit Mode, Hold [Shift] + Press [Edit]. The display will show the sound edit mode symbol and the alpha-numeric display will show the selected option. Edit is secondary option on 'Sound' button.
3. To change the selected option, Press [+] or [-]. Each press will cycle through each option incrementally. The display will indicate the current option selected. Ensure 'ENV' Envelope is selected.
4. Adjust the Attack time. Turn the Orange (X) knob. This will determine the time for sound to ramp up from zero to maximum when a pad or note is triggered. The X Quadrant icon displays the status.
5. Adjust the Release time. Turn the Black (Y) knob. This will determine the time for sound to ramp down to minimum when a pad or note trigger is lifted or released. The Y Quadrant icon displays the status.
6. Tweak and iterate by ear to get the desired sound.

4.9 Time Stretching a Sound

Time stretching is a common technique used in sampling and modern audio production. It essentially will stretch or shrink the length of a sample without affecting its pitch. This is an aid to matching a sample to the tempo of a project while retaining its tonal and musical characteristics.

KOII Time stretch function can adjust the sample by Length - in Bars to match a division scale or by Tempo - in BPM to match the global tempo. Samples loaded to a pad will initially play to the project tempo. Time stretching can also be set to OFF.



BPM Tempo Changes. Increasing the BPM will effectively extend the sample length and therefore play it slower. Lowering the BPM will reduce the sample length, playing it faster. Tempo range is 60-180 BPM.

Bar Changes. Increasing the number of bars for the sample will extend its length and therefore play slower. Reducing the number of bars will reduce its length and play faster. The increments are fixed to a set of options. As a tip, adjusting the time signature can also help select an odd bar length eg 3 bars stretch by using a setting of 4 plus a time signature of 3/4.

1/4	1/2	1	2	4

NOTES

NOTES

TIME STRETCHING A SAMPLE

1. Select a pad with a sound to edit.

2. Select Sound Edit Mode, Hold [Shift] + Press [Edit]. The display will show the sound edit mode symbol and the alpha-numeric display will show the selected option.

3. To change the selected option, Press [+] or [-]. Each press will cycle through each option incrementally. The display will indicate the current option selected. Ensure 'TIM' Time is selected.

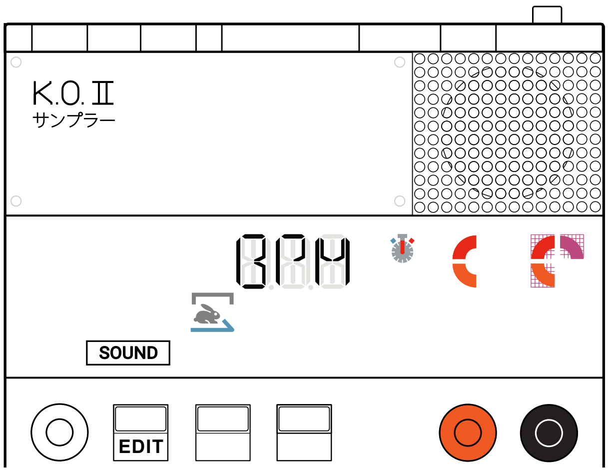
4. Select the Time Mode required. Turn the Orange (X) knob. This will determine If this function is Off - X Quadrant Icon Off, BPM - X Quadrant Icon 50% or Bar - X Quadrant Icon Full. The Display also states, OFF, BPM or BAR when selecting an option.

5. Once the mode is selected the sample can be stretched or reduced. Turn the Black (Y) knob. The Y Quadrant icon displays the status.

    - Changing in BPM will require the Tempo BPM to be adjusted to that of the original sample, if known. This will align to the project tempo. Also it is possible to tweak the sound by ear which is also useful for creative experimentation to different BPM values.

    - Changing in Bars or Increments will adjust the length by fixed amounts. This is more useful when working to defined grid and a timeline for the track. Useful for percussive loops to fit a time slot.

6. The 'time stretch active' clock icon is displayed if the selected sample has time stretching applied.

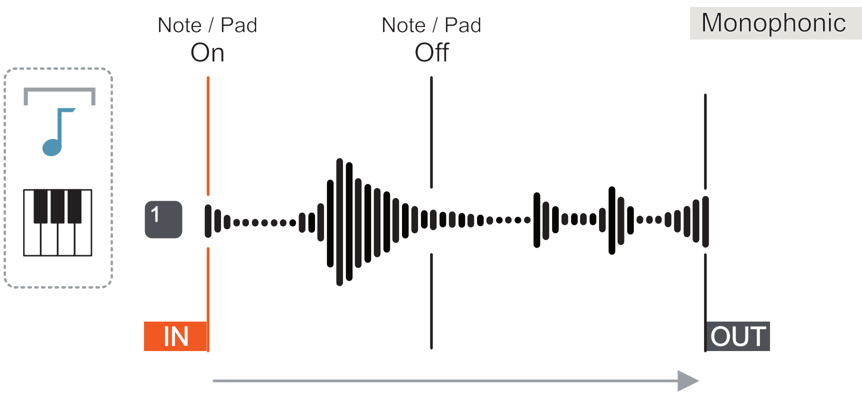


4.10 Play Mode

Once a sample is edited or tweaked the possibility to set the play mode is available. The playback mode determines how the sample will be played when triggered by a note or pad. There are three options available, configurable in ‘Sound Edit Mode’. The diagrams and examples illustrated are better explained when the KOII operation is set to [Keys] Mode. Keys mode plays a selected single pad sample chromatically across the other pads.

	Press [Keys]	*Screen Indicator*
Keys Mode *Chromatic Playback*		

ONE = One-Shot Play Mode



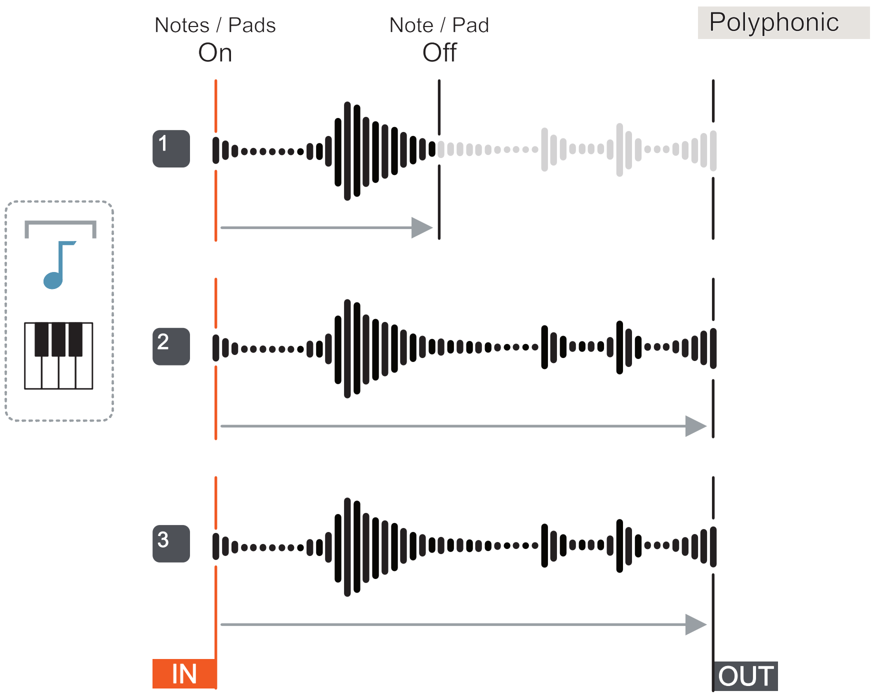
Once triggered, the sound plays once, fully to the end, even if the note trigger is lifted off. Playing the same note will always restart from the defined sample start position. The most recent note played will take precedence over prior notes. The play mode is set for selected pad sample.

This same behaviour will also apply when out of keyboard mode and playing multiple pads. Each individual pad’s play mode setting will then apply.

NOTES

NOTES

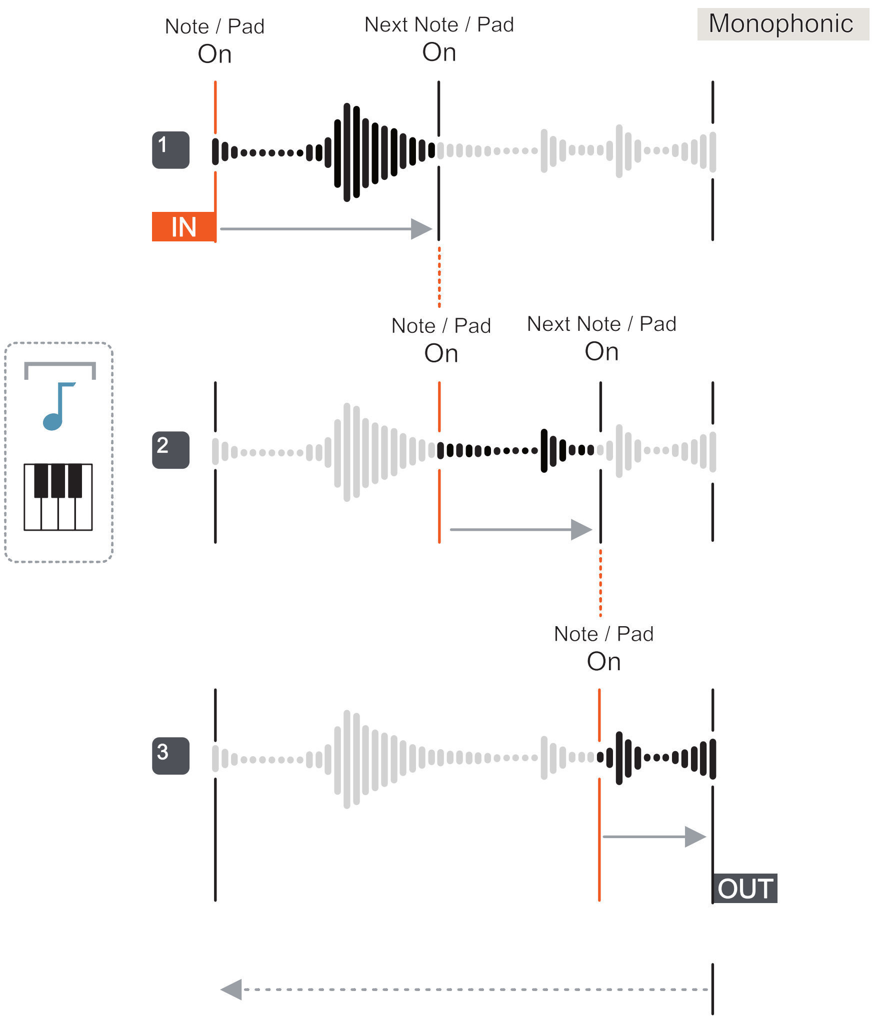
KEY = Key Multi Play Mode



Multiple pads can triggered with the same sample for example to play a chord in keys mode. When triggered at the same time, they will all play together. Sounds will play to the end once while the notes are held and If any note trigger is lifted off, then that sound will stop.

This same behaviour will also apply when out of keys mode and playing multiple pads. Each individual pad's play mode setting will then apply.

LEG = Legato Play Mode



In keys mode, pads can triggered with the same sample one at a time. When a pad is triggered it will play to the end, once while the note is held. If any other note is triggered it will stop and the new note will continue. Subsequent notes triggered while the sample is playing do not restart, but the new note will take over playing the sample from the current position of the trigger. When a sample reaches its end position then it can restart from the beginning.

When out of keys mode and playing multiple pads the individual play mode per sound applies. This will then operate similar to keys mode where entire samples can be played individually.

NOTES

Comments on Monophonic v Polyphonic Application

Here is a good place to offer a few quick notes to clarify these definitions.

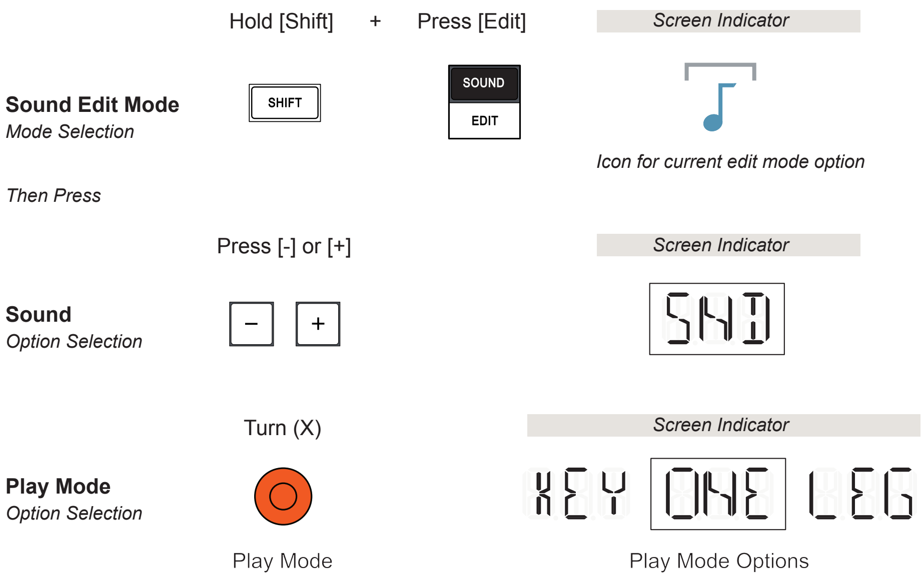
Monophonic sounds can only play individually. It is not possible to play multiples of the same note, at the same time. This is normally set to a single sound engine called a ‘voice’ in a synth or audio device. These are useful for percussion, effects and bass sounds.

A number of voices can also be applied and allow multiples of the same note to be played together, at the same time. This is called Polyphony. This is where chords can be played, ideal for melodies and pad sounds. For example, to play a triad chord, three voices would be needed and so forth.

Some devices have limited polyphony and manage the voice allocation based on a set of rules, for example stealing the first voice to play the last through a cycle of up to, say 4 voices.

The EP133 KOII is described at an overall device level of having 16 Mono / 12 Stereo Voices from OS version 2.0. Time stretching, high pitched samples and stereo samples add an extra load on the polyphony. In OS version prior the polyphony was limited to 12 Mono / 6 stereo.

Setting up the play modes is set up in the ‘Sound Edit Mode’ for each individual pad sound. When in keys mode the selected sound play mode is reflected across that sounds note range.



SELECTING THE PLAY MODE PER SOUND

1. Select a pad with a sound to edit.

2. Select Sound Edit Mode, Hold [Shift] + Press [Edit]. The display will show the sound edit mode symbol and the alpha-numeric display will show the selected option.

3. To change the selected option, Press [+] or [-]. Each press will cycle through each option incrementally. The display will indicate the current option selected. <u>Ensure ‘SND’ Sound is selected</u>.

4. Select the Play Mode required. Turn the Orange (X) knob. This will determine If this function is;

    - ONE - One Shot, X Quadrant Icon Off
    - KEY - Key mode, X Quadrant Icon 50%
    - LEG - Legato, X Quadrant Icon Full.

NOTES

NOTES

4.11 Mute Group / Choke Group

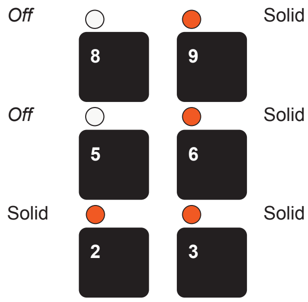
A mute group will allow control over the choking / muting of all sounds in the group other than the last one triggered. In a real life example, playing an open hi hat will be choked / muted once the hi hat is closed. A single hi hat cannot be physically played open and closed at the same time, its one or the other.

The KOII mute group emulates this behaviour by only allowing the most recently triggered sound in the group to be played while muting all others in the same group. Only one mute group per A,B,C or D group exists in the KOII. This can be configured in the 'Sample Edit Mode'.

Example: Mute Group Behaviour (Mute Group Mode)

Flashing pads indicate that the mute group is empty.

Unlit pads indicate the sound is not in the active mute group. Press to select in the group.



Solid lit pads indicate the sound is in the active mute group. Press to deselect.

The last pad sound triggered in a mute group will play and all others in the group are muted. In this example the mute group contains Pads 2, 3, 6 and 9. Pads 5 & 8 are not assigned in the group. Play 6 and 2, 3 and 9 and will mute.

Hold [Shift] + Press [Edit]

Screen Indicator

Sound Edit Mode
Mode Selection

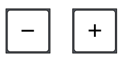
Icon for current edit mode option

Then Press

Press [-] or [+]

Screen Indicator

Sound
Option Selection



Mute Group
Sound Selection

Mute Group Active, No pads assigned - *Flash*

Pad is Assigned to the Mute Group - Solid

Pad is Not Assigned to the Mute Group - Off

Press [Pad] to toggle selection in or out of the group.

SETTING UP A MUTE GROUP

1. Select the group A, B, C or D. Only one mute group is available per pad group in the KOII.

2. Select Sound Edit Mode, Hold [Shift] + Press [Edit]. The display will show the sound edit mode symbol and the alpha-numeric display will show the selected option.

3. To change the selected option, Press [+] or [-]. Each press will cycle through each option incrementally. The display will indicate the current option selected. <u>Ensure 'GRP' Mute Group is selected</u>.

4. All Pads will flash if the mute group mode is active, but no pads are assigned yet into the group. The Mute Group is empty.

5. Once the mode is selected the pads are used to select which sound is in the group and which are not assigned to the group. Press [Pad] to toggle a sound into the group or to remove from the group.

    - Pad LED Off / Unlit. The Mute Group is active, with assigned pads.This specific Pad sound is NOT assigned to the mute group. These sounds are ignored by the mute / choke group behaviour. Tapping this [PAD] will assign it into the group.

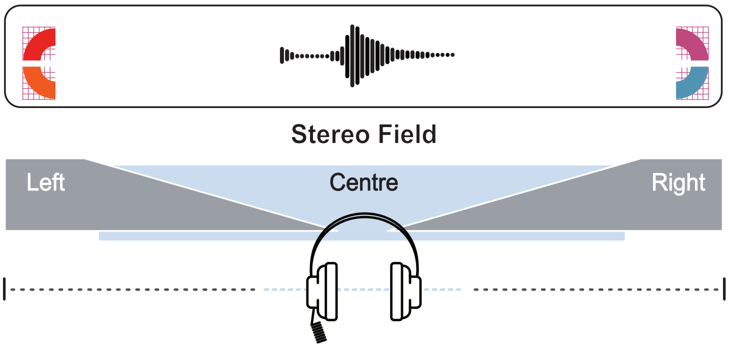
    - Pad LED Solid Lit. The Mute Group is active, with assigned pads.This specific Pad sound IS assigned to the mute group. These sounds operate to the mute / choke group behaviour. This means only the last played sound in the mute group will be audible and when triggered will mute / choke any other sounds playback. Tapping this [PAD] will remove it from the group.

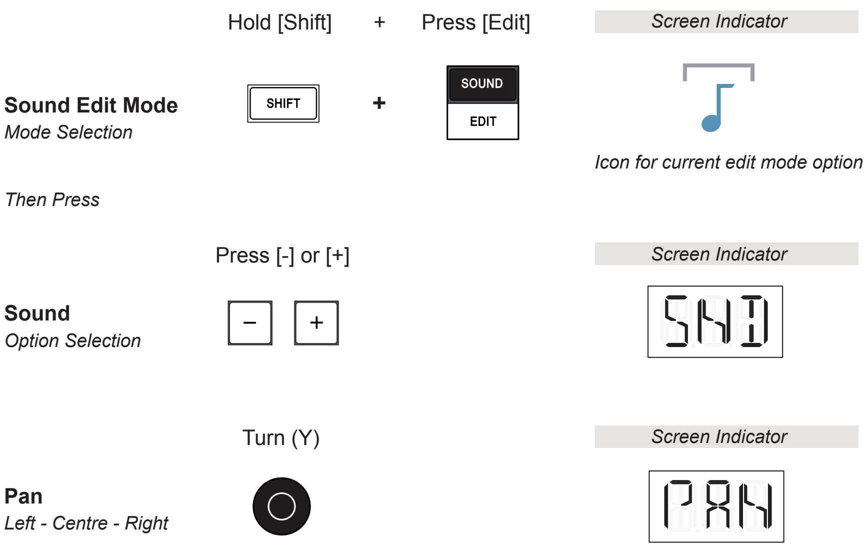
    - Pad LED's Flash. No pads are assigned to this Mute Group or all assigned pads have been removed.

NOTES

4.12 Panning a Sound

The final sound edit option which can be set for each individual pad assigned sound is the pan function. Pan, shortened from the word 'panorama' describes how the audio is positioned within the left and right sides of the stereo field. This is useful to widen the sound range, for example keeping bass central and maybe panning percussion wider left and right to offer a more interesting sound experience.





The Pan adjustment is found in the SND section of the sound edit menu settings alongside the play mode settings. Where Play Mode is selected with the Orange (X) knob, the Pan setting uses the Black (Y) Knob.

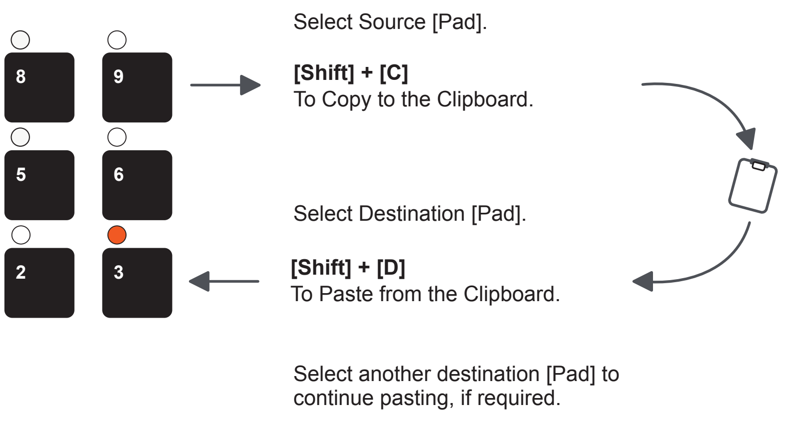
4.13 Copying / Pasting a Sound

A sound which includes a sample and any applied edits can be copied from one pad to another. This will copy the sound from one pad to another pad but will not copy between sound slots in the library.

NOTES

COPYING A SOUND FROM PAD TO PAD

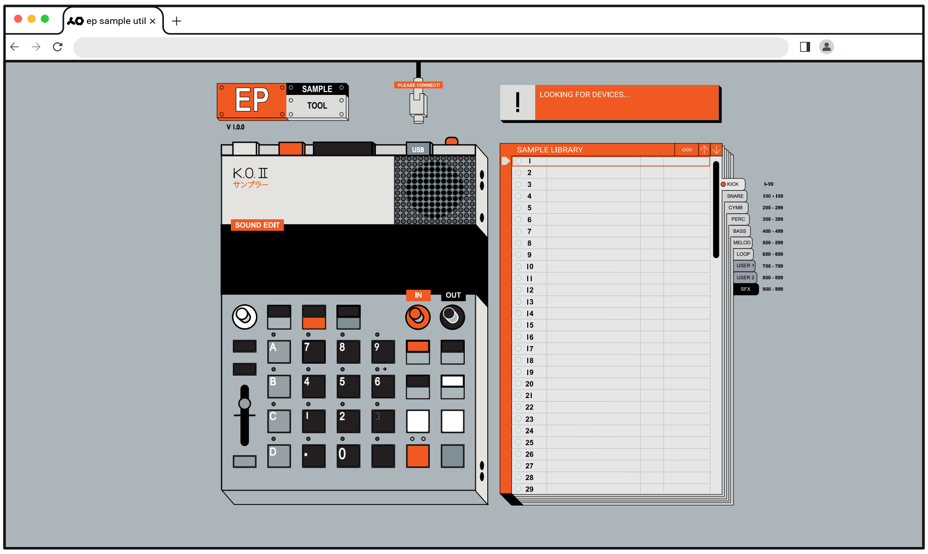
1. Select Sound Mode, Press [Sound].
2. Select the pad which contains the source sound to copy from, Press to select a Pad [.] to [9].
3. To copy the sound to the clipboard, Hold [Shift] + Hold [C] group button. The display will show the ‘Copy’ icon and indicate ‘PAD’ has been copied.
4. Select the destination to paste the copied pad into, Press Pad [.] to [9]. An empty pad will be shown as _ _ _ on the display. If a sound already exists its sample number will be displayed.
5. To paste the sound from the clipboard, Hold [Shift] + Hold [D] group button. The display will show the ‘Paste’ icon and indicate that a ‘PAD’ has been pasted into the selected pad.
6. Any existing sound on the destination pad will be overwritten. However the undo command operates in this mode and can be used to restore the prior state. Press [Shift] + Press [B] group button to initiate Undo.



NOTES

4.14 EP Sample Tool

Other than the factory supplied library, samples can either be recorded directly into the KOII or user samples transferred in using the online EP Sample Tool. Some basic editing can also be performed for the samples within the EP Sample Tool.



The EP Sample tool operates in a Web MIDI compatible the browser. These include Google Chrome v43 and above and Microsoft Edge v79 and above and Firefox. Others that may work include Baidu and Opera. Refer to the browser specification and documentation for further information.

CONNECTING THE EP SAMPLE TOOL

1. Connect the KOII to a web connected PC or Mac using the USB connection.
2. Open the preferred Web MIDI Browser and open the EP Sample Tool.
3. The page should show a connected device and the sample library will be populated to match the connected devices sample library.
4. Click on a sample or hold the spacebar in the web page sample library to trigger the sound, played on the KOII device. Navigate up and down with the keyboard arrows on the PC/Mac.

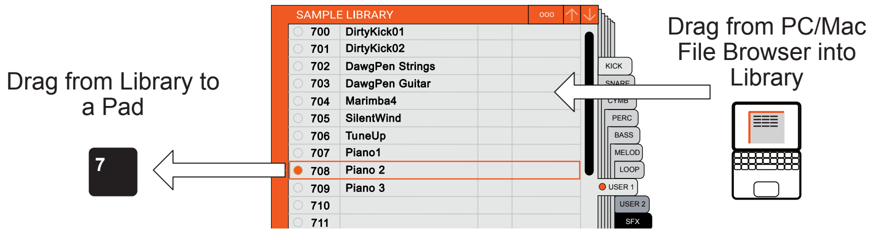
NOTES

LOADING A SAMPLE TO A PAD USING THE EP SAMPLE TOOL

1. Connect the KOII to a PC / Mac using the USB and open the EP Sample Tool in the compatible web MIDI browser.

2. Select the sample genre tab on the sample browser section. For example, click ‘Kick’. The list of available samples is presented. Use the scroll bar to navigate the list.

3. On the browser, click and drag the desired sample from the Sample Library list and drop onto the target pad to load into. The target pad will highlight when active as the destination. A group can also be selected in the EP Sample Tool.



UPLOADING USER SAMPLES TO THE KOII

1. Connect the KOII to a PC / Mac using the USB and open the EP Sample Tool in the compatible web MIDI browser.

2. Select the sample genre tab on the sample browser section. For example, click ‘User 1’. This is the destination for the samples to be transferred to and represents the KOII library.

3. The list of available samples are presented. Use the scroll bar to navigate the list. Upload will require an available empty slot.

4. On the PC/Mac file browser open a window with the source samples

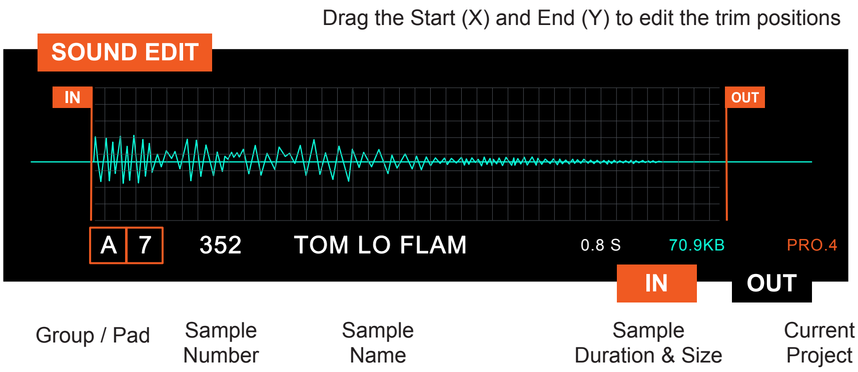
5. Click and drag the desired sample from the PC/Mac file browser and into Sample Library list and drop onto and empty sample slot.

6. The transfer bar will show progress and the file will be uploaded to the KOII sample library. The total memory used is also indicated.

7. If required, samples can be deleted by pressing ‘X’ on the sample in a the library slot list. Samples can also be downloaded to the PC/Mac by pressing the ‘↓’ on the sample.

The EP Sample Tool also provides some additional sample information and offers some limited editing options. Essentially the trim start and end position of the sample can be edited directly in the EP Sample Tool which will affect the sample in the KOII device. These are the same as the X-Y positions set onboard the device itself.

EP Sample Tool



EDITING START / END POSITION IN THE EP SAMPLE TOOL

1. Connect the KOII to a web connected PC or Mac using the USB connection.
2. Open the preferred Web MIDI Browser and open the EP Sample Tool.
3. The page should show a connected device.
4. Press a Pad or Select in the Browser EP Sample Tool a Pad. The Sample will be triggered and the pad highlighted as selected.
5. The sample details will be presented on the browser screen. This includes the sample number, name, duration in seconds and file size. The total device memory usage is also indicated in the pie chart at the top of the library section.
6. To edit the start position, drag the ‘IN’ marker on the PC/Mac Browser EP Sample Tool to the position desired. This will affect the start position in the KOII directly. This is the same as the trim ‘X’ parameter.
7. To edit the end position, drag the ‘OUT’ marker on the PC/Mac Browser EP Sample Tool to the position desired. This will affect the end position in the KOII directly. This is same as the trim ‘Y’ parameter.
8. The In X & Out Y positions are retained in the KOII for the sample even if the KOII is disconnected from the EP Sample Tool.

NOTES

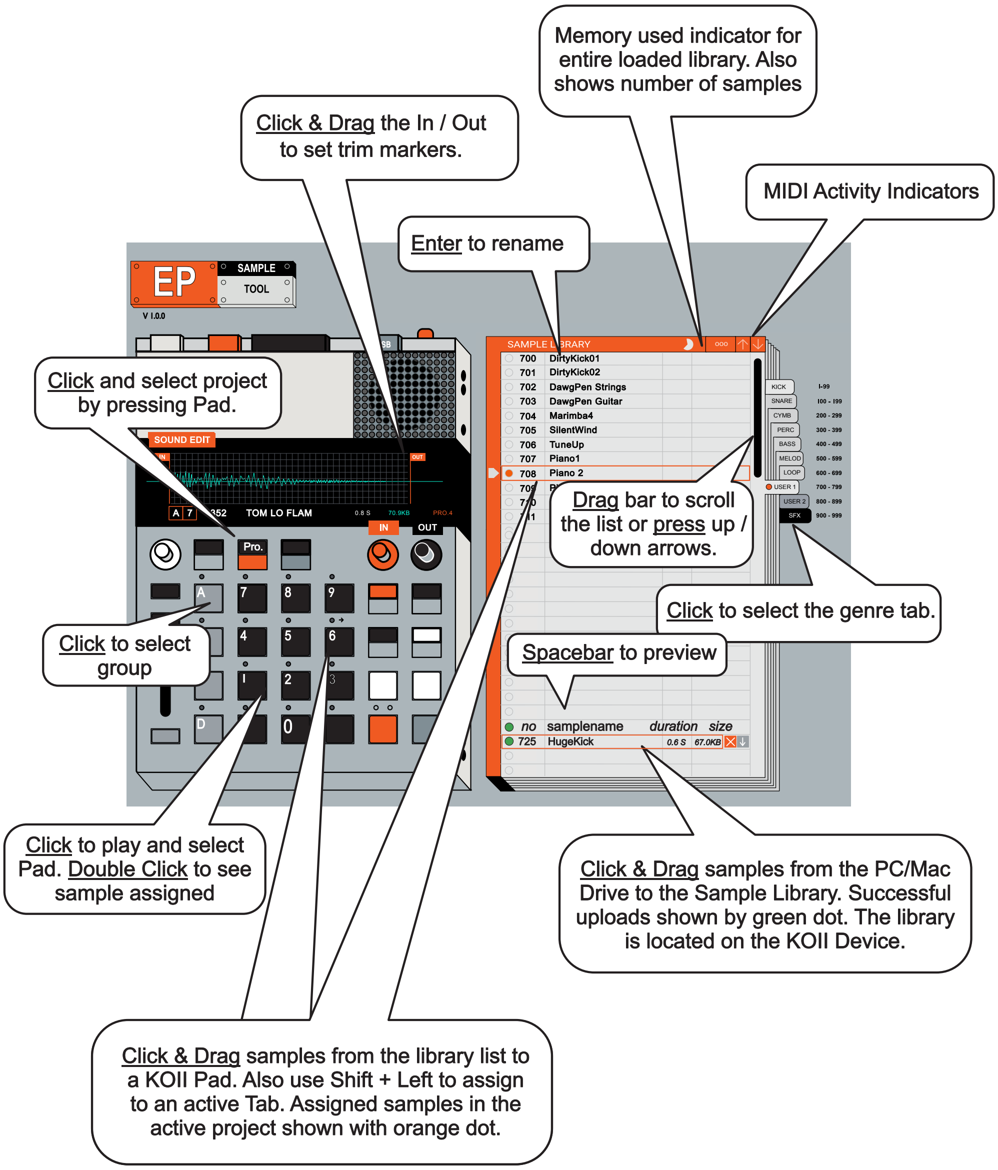
Summary of EP Sample Tool Commands

The following commands are used in the EP sample tool from a PC or Mac. Some specific commands may be different between a PC and Mac and hence why only generic mouse / trackpad command are provided.

- Multiple samples can be dragged into the library from a PC/Mac and will be located in sequence to the first slot dropped into. Use shift to select multiple samples.
- Multiple (up to 12) samples can be selected collectively and dragged to a pad group. Assigned pads have a dotted red border.
- Use CMD or CTRL to select first or last row in the selected tab and OPTION or ALT to select the genre tab.
- Dropping samples from the PC/Mac directly to a pad is possible. This will also copy to the library and then only the first sample is assigned to the target pad.
- The sample selected can be downloaded by clicking its down arrow and can be deleted from the library by clicking the ‘X’.

NOTES

Illustrated summary of EP Sample Tool.



5

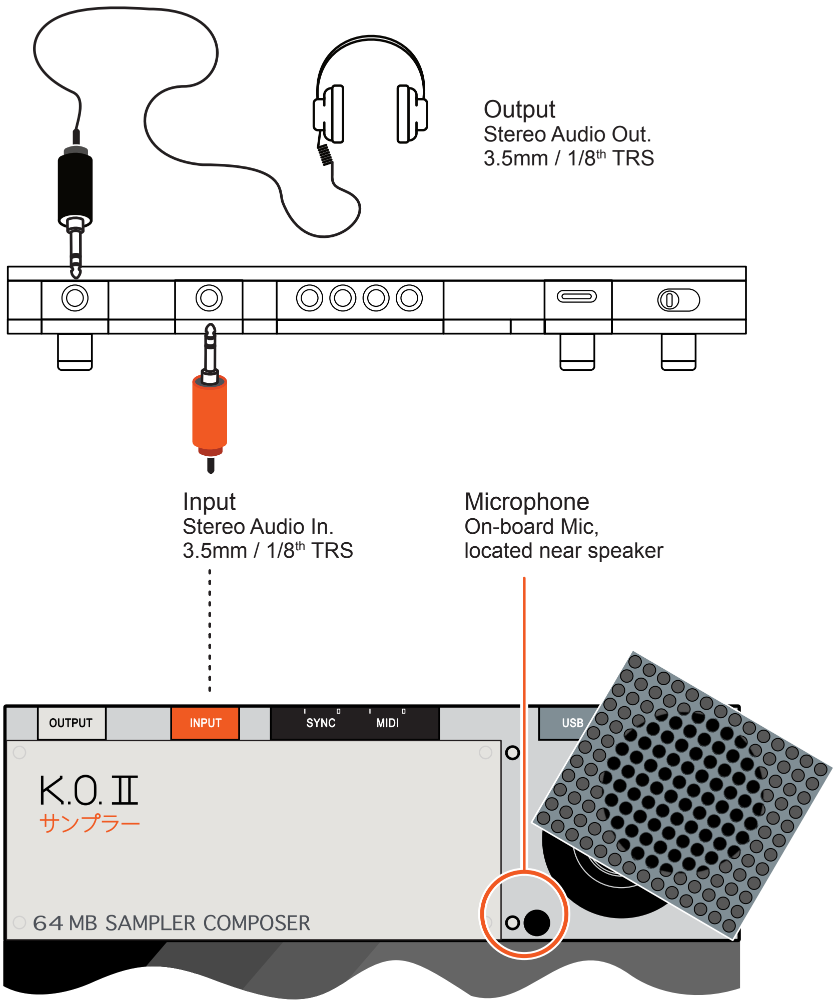
Sampling & Audio

Sampling is a technique thats been a mainstay of music production for many years. Dating back as far as the 1940's with experimental projects and later gained common and wide scale recognition through the 80's. Today the ability to sample and capture audio can be performed easily in many table top and hand held devices. Sampling is the creative technique of capturing audio snapshots which can then be used in the process of creating new compositions. This can be pieces of music (although remember to respect and honour copyright rules and laws!), found sounds from the everyday environment or other musical instruments or human voices. The KOII aims to offer a sampling option that is quick and easy allowing a seamless workflow from sampling audio to making a composition. These principles are at the heart of the KOII and are the reason why it is called a 'Sampler Composer'. So far the processes covered are focussed on the use of the on board sample library and editing existing samples. The library is still a relevant topic here but the focus shifts to capturing and recording samples to add to the library and create unique and custom sounds. Samples can be recorded into the KOII from the on board microphone or an audio device connected to the stereo line input. In addition the KOII can interface easily with other Teenage Engineering field devices such as the CM-15 Microphone. The option to layer effects by resampling patterns and sounds is possible. A number of editing features that are specifically orientated to the audio sampling process are also covered in this section including chop where an audio sample can be sliced across the pad range.

5.1 Audio Inputs & Output

The first thing to understand on the KOII is what physical inputs allow for capturing incoming audio. Also what are the physical audio outs.

Audio Inputs & Outputs



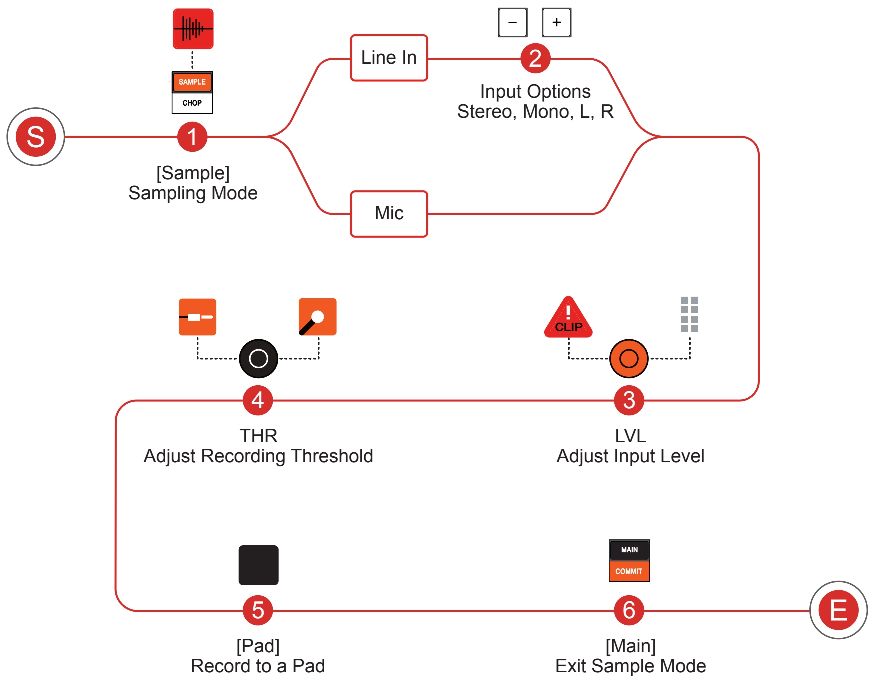
Battery and Speaker Cover are Removable

Note: In some situations when the USB cable is connected between a PC/Mac and the KOII and the browser PE Sample Tool is open, the KOII may switch to 'Main Mode' automatically after sampling is completed. Normal operation when not connected to the EP sample tool is for the KOII to remain in sample mode after sampling and main mode would need to be selected manually in order to exit sampling mode.

NOTES

5.2 Basic Sampling Workflow

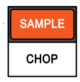
Sampling can be performed from an external device, connected to the audio Line In of the KOII. This is a TRS Stereo input but sampling can be selected as mono. In addition an internal microphone located near the speaker can also be used as the audio input. This is the default recording source when there is nothing plugged into the Line In socket.



Press [Sample]

Screen Indicators

Sampling Mode
Mode Selection



Icons for sampling option

Then Press

Press [Pad]

Record a Sample
Start / End Recording

Pads are in Sampling Mode - *Flash*

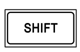
Pad contains a recording - Solid

Sample Time

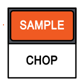
Alternatively

Hold [Shift] + Tap [Pad] Tap [Sample]

Record a Sample
Hands free sampling



+

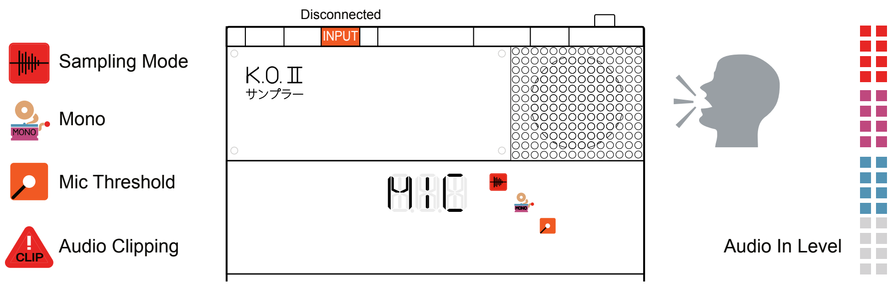


To stop sampling

NOTES

5.3 Sampling from the Internal Microphone

The default recording source is the microphone, recording in mono. This is the on board mic located under the speaker cover. This will be automatically presented if the Audio 'Input' socket is open and unplugged.



SAMPLING WITH THE MIC

1. Ensure that the Line In connection is open / disconnected. When line in is open, the mic is the selected audio option.

2. Press [Sample] to enter sampling mode. The Pad LED's will flash and the sampling icon is displayed along with the sample mono icon.

3. Turn the Orange (X) Knob to adjust the recording input level. Speak into the mic, located near the speaker. If the audio is too loud the clipping icon may be triggered. Set the audio high enough to get a loud, but unclipped recording. Level indicator will show incoming audio.

4. Turn the Black (Y) Knob to adjust the recording threshold level. This is the audio level that, when exceeded, will start recording when the pad is held. Any audio below this level will not trigger the recording, even if the pad is held. The display mic icon is lit when the mic input threshold is exceeded.

5. Hold a [Pad] to start recording. The sample will be recorded from the microphone and into the selected pad. Recording is stopped when releasing the pad. Alternatively, hands free recording can be performed. Hold [Shift] + [Pad]. Press [Sample] to stop. Sample time is displayed.

6. When the pad is released or sampling stopped, its Pad LED will be lit solid orange or if hands free recording. The sample will be saved into the first available slot in the sample library and named by the slot number:-

   Example: Slot 32 - Named **Sample #32.PCM**
   Download: Name: **032 sample.wav** Mono, 46.9kHz, 16bit

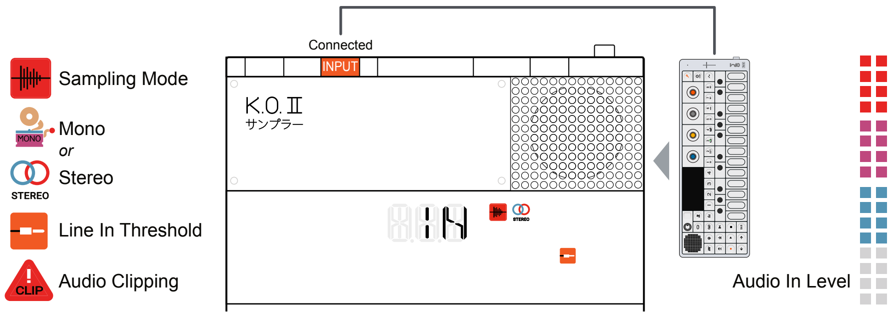
7. The pad can now be played by pressing it. The pad can be overwritten and re-recorded by repeating steps 2 & 5.

8. Press [Main] to exit sampling mode and to revert to the main page.

NOTES

5.4 Sampling from the Line In

If a TRS Cable is connected to the Audio ‘Input’ socket then the KOII will default to recording the Line In signal. An external device should be connected from which to generate the audio input.



SAMPLING WITH THE LINE IN

1. Ensure that an external device is connected to the the Line In input.

2. Press [Sample] to enter sampling mode. The Pad LED’s will flash and the sampling icon is displayed along with the sample mono or stereo icon.

3. Press [+] or [-] to select the audio option required.

    - IN - Mono summed input is the audio source recorded. Mono icon is illuminated on the display.

    - IN - Stereo Input is the audio source recorded. Stereo icon is illuminated on the display.

    - RSP - Mono resampling. Samples internal audio. Mono icon is illuminated on the display.

    - RSP - Stereo resampling. Samples internal audio. Stereo icon is illuminated on the display.

    - L.IN - Mono left channel will be the audio source recorded. Mono icon is illuminated on the display.

    - R.IN - Mono right channel will be the audio source recorded. Mono icon is illuminated on the display.

4. Turn the Orange (X) Knob to adjust the recording input level if required. Play the external device. If the audio is too loud the clipping icon may be triggered. Set the audio high enough to get a loud, but unclipped recording. The level indicator will show the incoming audio.

5. Turn the Black (Y) Knob to adjust the recording threshold level if required. This is the audio level that, when exceeded, will start recording after the pad is held. Any audio below this level will not trigger the recording start, even if the pad is held. The display line icon is lit when the threshold for the input is exceeded.

6. Start playing or make ready for playing the external audio device.

7. Hold a [Pad] to start recording. The sample will be recorded from the line input and into the specific pad location held and will cease recording when the pad is released. Alternatively, hands free recording can be performed. Hold [Shift] + [Pad] to start recording then to stop press [Sample]. Sample time is displayed.

8. When the pad is released, the Pad LED will be lit solid orange. The sample will be saved into the first available slot in the sample library and named by the slot number:-

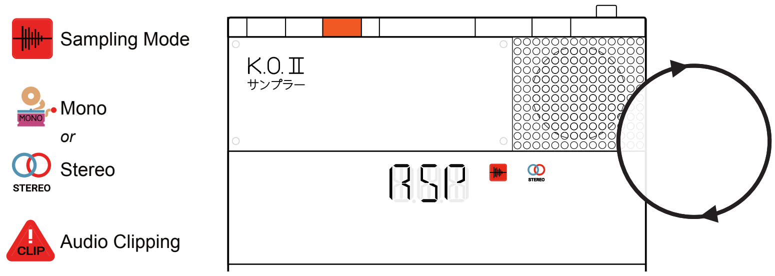
   Example: Slot 32 - Named **Sample #032.PCM**
   Download: Name: **032 sample.wav** Mono or Stereo, 46.9kHz, 16bit

9. The pad can now be played by pressing it. The pad can be overwritten and re-recorded by repeating steps 2 & 5.

10. Press [Main] to exit sampling mode and to revert to the main page.

5.5 Resampling

Resampling records from the internal audio of KOII in either stereo or mono. Patterns or specific bar lengths can be recorded. Resampling is performed when sampling in hands free mode.



RESAMPLING A PATTERN

1. Ensure the desired pattern is selected to record.

2. Press [Sample] to enter sampling mode. The Pad LED's will flash and the sampling icon is displayed along with the sample mono or stereo icon.

3. Press [+] or [-] to select the audio option required.

    - RSP - Mono resampling. Samples internal audio. Mono icon is illuminated on the display.

    - RSP - Stereo resampling. Samples internal audio. Stereo icon is illuminated on the display.

3. Select hands free recording. Hold [Shift] + [Pad] to select a pad to record the pattern into. Sample time is displayed.

4. Press [Play] to start the pattern playing and also start recording. The pattern duration will be recorded.

5. To exit recording earlier than the pattern end, Press [Sampling].

6. When the complete, the Pad LED will be lit solid orange. The sample will be saved into the first available slot in the sample library and named by the slot number:-

   Example: Slot 32 - Named **Sample #032.PCM**
   Download: Name: **032 sample.wav** Mono or Stereo, 46.9kHz, 16bit

9. The pad can now be played by pressing it. The pad can be overwritten and re-recorded by repeating steps 2 & 5.

10. Press [Main] to exit sampling mode and to revert to the main page.

4.

NOTES

RESAMPLING A SPECIFIC BAR LENGTH

1. Press [Sample] to enter sampling mode. The Pad LED's will flash and the sampling icon is displayed along with the sample mono or stereo icon.

2. Press [+] or [-] to select the audio option required.

   - RSP - Mono resampling. Samples internal audio. Mono icon is illuminated on the display.

   - RSP - Stereo resampling. Samples internal audio. Stereo icon is illuminated on the display.

3. Select hands free recording. Hold [Shift] + [Pad] to select a pad to record the pattern into. Sample time is displayed.

4. Tap [-] or [+] to select the specific length to record.

5. Press [Play] to start the pattern playing and also start recording. The pattern duration will be recorded.

6. To exit recording earlier than the pattern end, Press [Sampling].

7. When the complete, the Pad LED will be lit solid orange. The sample will be saved into the first available slot in the sample library and named by the slot number:-

   Example: Slot 32 - Named **Sample #032.PCM**
   Download: Name: **032 sample.wav** Mono or Stereo, 46.9kHz, 16bit

9. The pad can now be played by pressing it. The pad can be overwritten and re-recorded by repeating steps 2 & 5.

10. Press [Main] to exit sampling mode and to revert to the main page.

The option to resample by holding the pad instead of hands free recording is also possible. The pad would be held for the duration of the recording required. When the pad is held recording is armed awaiting the press of play. Recording is stopped when the pad is lifted.

NOTES

5.6 Input Level and Threshold

The input level can be adjusted to ensure a good audio level is recorded. Ideally this should be loud enough to be clear while avoiding clipping and distortion from this being too high. Note that in gain staging the levels may have multiple locations at which to adjust the level including on an external device itself.

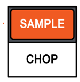
For sampling a useful level is the threshold. This allows recording to commence only when the threshold level is exceeded and after the pad has been held to initiate sampling. This is helpful to capture a recording at the very start of the external audio playback.

Both these parameters are available as recording options and are only available in sampling [Sample] Mode.

Press [Sample]

Screen Indicators

Sampling Mode
Mode Selection



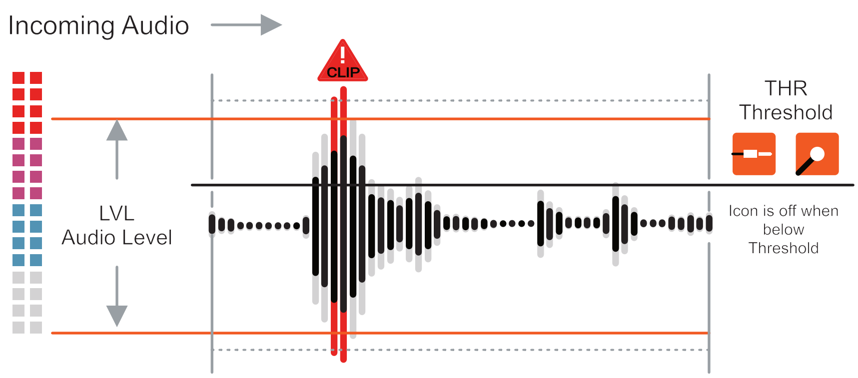
Icons for sampling option

Level
Adjust Input Level

X Quadrant displays value

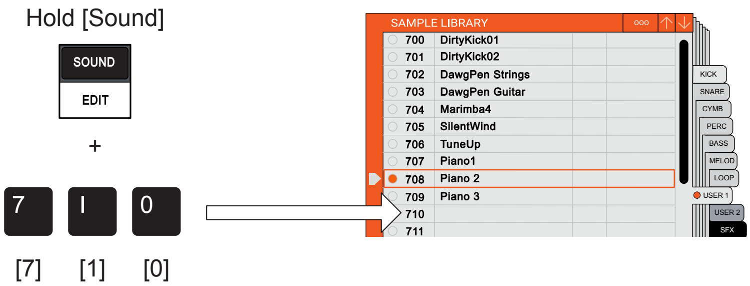
Threshold
Adjust Threshold Level

Y Quadrant displays value



5.7 Sampling to a Selected Library Slot

The default and quick mode when sampling is to store the recorded sample to the first available library slot. There is also the option to record to a specifically selected slot. The general sampling process follows the standard workflow but the sample slot will be selected first. Then press [Sample] and start the sampling process.



SAMPLING TO A SELECTED LIBRARY SLOT

1. Hold [Sound] + Press [Pads] for the sample slot number to record into. For example Hold [Sound] + [7], [0], [9].

2. The number will be displayed and will flash if the slot is empty. If the slot already contains a sample the number will be static and not flash.

3. Press [Sample] to enter sampling mode.

4. Follow the normal steps for sampling as described earlier from here on. Start by pressing [Pad] to trigger sampling or [Shift] + [Pad] for hands free sampling.

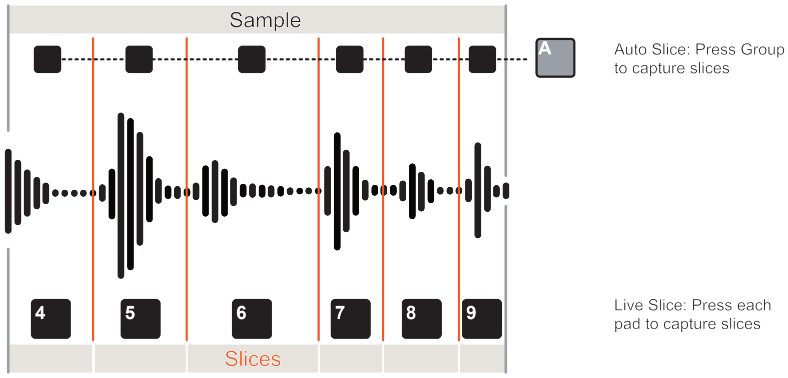
5. The sample will be stored in the selected slot in the library. Any subsequent sample recordings will iterate to the next available position from the originally selected slot.

NOTES

NOTES

5.8 Chopping a Sample

Once a sample is recorded or loaded from the library it can be chopped into individual parts. These slices can then be distributed across the pads. Chopping samples can be performed when any sample is selected but must be performed in sample mode.



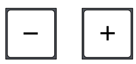
Hold [Shift] + Press [Chop]

Screen Indicators

Chop Mode
Mode Selection

Press [Group]   Press [+] [-]

Auto Chop
Automatically Slice



Change Number of Slices   *Number of Slices*

Press [Pad]

Live Chop
Manually Slice

Tap each pad in turn. The sample will play after pressing a pad. The slice duration will be set based on time between tapping pads.

Pad contains a slice - Solid

AUTOMATICALLY SLICING A SAMPLE TO A PAD GROUP

1. Select a pad with the sample to be chopped, Press [Pad].

2. Hold [Shift] + Press [Chop] to enter Sample 'Chop' mode. The Chop icon will be displayed as will 'Cho' on the alpha numeric display.

3. Automatic slicing will be triggered when a group is selected. Press Group [A], [B], [C] or [D]. Whichever is chosen will be the pad group destination for the slices. Existing group samples will be overwritten.

4. Once the group is selected the sample slices are distributed across a number of pads in the group. The initial number of slices is determined by the KOII beat tracking algorithm and this number is shown on the display. Each pad LED containing a slice will be lit, solid orange, the first slice starting at [.], followed by [0] and so forth.

5. To change the number of slices, Press [+] or [-]. The sample slices are redistributed to match the available pad slots.

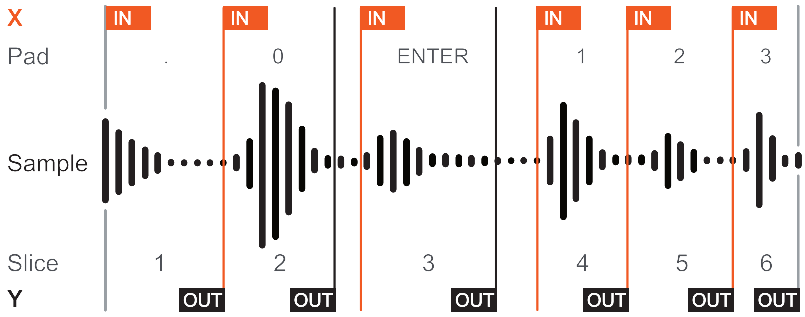
6. Play the slices. Press [Pad] to play its assigned sample slice. Play the pads to audition the sound while editing the In/Out positions. The pad will automatically audition the slice while turning the respective knob.

7. The slice 'In' Start and 'Out' End points can now be edited if required to get a more precise slice. Times are displayed in seconds.

   - Turn Orange (X) Knob to change the Start / In position.
     Use [Shift] + (X) for a more precise resolution.

   - Turn Black (Y) Knob to change the End / Out position.
     Use [Shift] + (Y) for a more precise resolution.

8. Remember, that the undo option is available in chop when the umbrella icon is shown lit. Hold [Shift] + Press Group [B] to undo.



NOTES

MANUALLY LIVE SLICING A SAMPLE TO PADS

1. Select a pad with the sample which will be chopped, Press [Pad].

2. Hold [Shift] + Press [Chop] to enter Sample ‘Chop’ mode. The Chop icon will be displayed as will ‘Cho’ on the alpha numeric display.

3. Live slicing will be triggered when the first pad is tapped and will start to play the entire sample. Each subsequent press of a pad will capture the audio into the pad slot starting at that point in time until the next pad is pressed. Example, Press each Pad [1], [2], [3] or [4] in sequence, at equally timed divisions to capture four slices of the sample to four pads.

4. Once complete, the pads with samples will be recognised by the LED of the respective pad being lit, solid orange.

5. Play the slices. Press [Pad] to play its assigned sample slice. Play the pads to audition the sound while editing.

6. The slice ‘In’ Start and ‘Out’ End points can now be edited if required to get a more precise slice. The pad will automatically audition the slice while turning the respective knob. Times are displayed in seconds.

    - Turn Orange (X) Knob to change the Start / In position.
      Use [Shift] + (X) for a more precise resolution.

    - Turn Black (Y) Knob to change the End / Out position.
      Use [Shift] + (Y) for a more precise resolution.

7. Remember, that the undo option is available in chop when the umbrella icon is shown lit. Hold [Shift] + Press Group [B] to undo.

6

Pattern Basics

The KOII is described as a 'Sampler Composer' and the previous sections have so far covered mainly the sample side of the device, concentrating mainly on sounds. This section turns the attention to the basic principles and functions used to create patterns. The process of creating a pattern is called sequencing. There are two methods of creating a pattern. Firstly live recording allows a quick and simple way to lay down and sequence patterns, recording the performance as its played live. Step sequencing is a more structured method, offering a more precise, step by step 'programming' approach. Both have a place in the composing of a more complete track but the option chosen will depend on the context and the preferred workflow. The general principles are to consider the groups as instrument groups and pads as the specific instrument tracks. Pads contain the sounds used as building blocks for elements such as drums, percussion, bass, melody, effects and so forth. The overall structure would then be created by the assembly of patterns and scenes to form a complete arrangement consisting of sections for introductions, chorus, verses, drop, outro etc. A key part of arranging is the use of scenes. A scene is a set of patterns that can be combined to create a full composition. Not only do scenes help to structure an arrangement but offer the opportunity for experimentation and use of on-the-fly variations. At this point it is worth stating that there is no chaining of scenes to play in series and they can only be changed manually. As a reminder, the KOII has a total of 9 Projects to structure an overall arrangement. For sounds there are 4 Groups, each with 12 Pad slots where samples can be played and sequenced. There are 99 patterns per group and 99 scenes per project.

6.1 Basic Sequencing Modes

NOTES

Sequencing is the process of creating and recording patterns. A pattern is a collection of note steps and settings that are created to build a beat or melody. Whether leads, pads, percussion, bass lines or effect sounds, all can be sequenced to a pattern. Assembling these together is what then creates a full track arrangement.

RECORDING MODES

Two recording modes for creating and programming patterns are available in the KOII. Live recording mode enables tracks to be played and recorded live and the natural note positioning is recorded. Step recording is more precise and allows adding steps into a static sequence, advancing the position and setting duration and parameters precisely.

LIVE RECORDING

- Records while sequencer plays.
- Arm by pressing [Record] then press [Play] or [Pad] to record.
- Record icon is lit red, play white.
- Recording position is displayed.
- 4 beat count in is audible.
- Notes can be erased while playing.
- Play live / real time while the sequencer plays using the pads to record steps.
- Notes can also be played and recorded in key mode.
- Ideal for natural rhythms such as bass lines, hats and shakers.
- Press [Record] to stop recording but continue playing. Press [Play] to stop recording and playing.

STEP SEQUENCING

- Records with sequencer stopped.
- Hold [Record] + [Pad] to record a step note.
- Record icon is lit red.
- Recording position is displayed.
- Use [+] & [-] to increment or decrement the step position.
- Recording a step can be undone or steps can be erased
- Record while the sequencer is <u>not</u> playing using the pads to record the step notes.
- Notes can also be played and recorded in key mode.
- Ideal for precise leads or beats and exact melodic progressions.
- To exit step recording mode, release [Record].

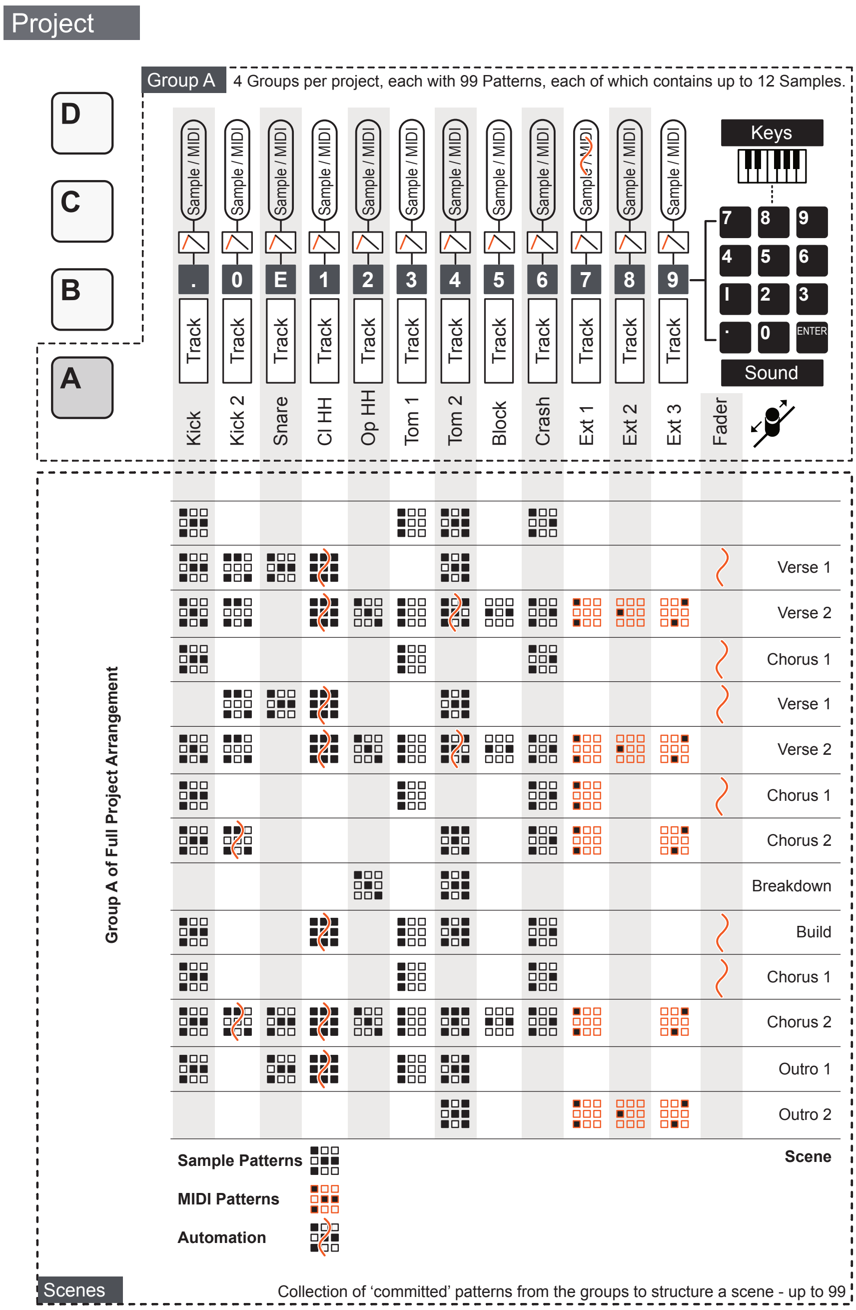
Typically use live recording to capture patterns quickly. Ideal for longer pattern lengths and melodies where a repetitive pattern can be played live. Examples include kick drums, hi hat patterns and melodies. With practice a good technique can be developed to make this method effective.

Step recording is a slower and more precise process. Ideal for specific notes and hits where exact placement and a note settings are important. Can be slow for long sequences. Often its useful to record in live recording and edit using step recording.

NOTES

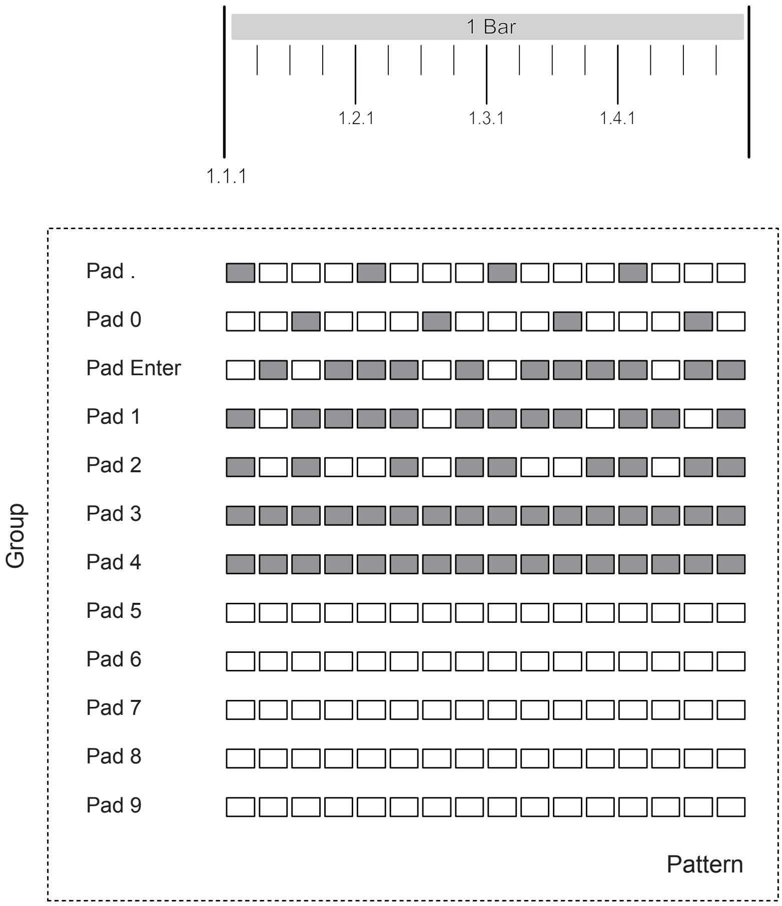
6.2 Overview of an Arrangement

Each group would cluster and instrument section. In this example group A represents the Drums as per the KOII convention. Consider each track to be represented by a Pad containing the individual instruments ie Kick, Snare etc. Patterns can be created for the tracks and then assembled together with others to form a scene. Patterns can be formed from triggering sounds, MIDI Out to other gear and / or automation of parameter values. The structure of these scenes forms the full manually playable arrangement. This is repeated for the other groups in the project.



The number of steps in a pattern is determined by the pattern length and the timing. Timing sets the resolution of the grid while length sets how many bars are covered. Each pad in the group can then be recorded at this resolution into the pattern grid.

Example Pattern: 1 Bar @ 16 Steps / Bar. This will give 1/16th Notes



Other attributes can also be set for example, the note duration and swing. Recording can be made in quantization mode which automatically positions the recorded step notes to the grid by bar or beat etc. Free mode will record notes at the exact point played even if these are recorded off grid and are positioned by tick locations.

A maximum of 99 patterns are available per group.

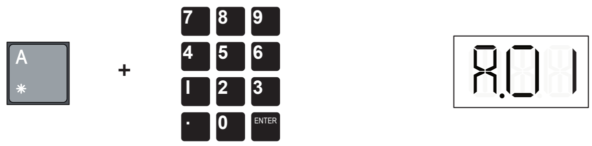
NOTES

6.3 Managing Patterns

Patterns are generally managed in Main Mode. A pattern will always be active to edit. Other patterns can be selected and new ones created. A number of functions such as erasing a pattern can also be performed. Groups A, B, C or D are used to select and erase a pattern within the selected group. Examples shown are for Group A.

Hold [Group] + Press [Pads] — *Screen Indicators*

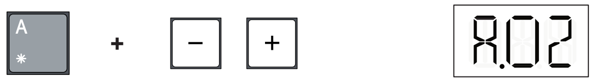
Select Pattern
By Group + Number



Press Number 0-9 — *Group & Pattern No*

Hold [Group] + Press [+] [-] — *Screen Indicators*

Select Next Pattern
Increment Up/Down



Group & Pattern No

Multiple group buttons can be held simultaneously to change the patterns across the held groups

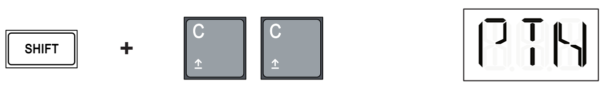
The following are performed in [Main] Mode.

Hold [Erase] + Hold [Group] — *Screen Indicators*

Erase Pattern
Currently selected pattern

Hold [Shift] + Press [Copy] — *Screen Indicators*

Copy Pattern
Currently selected pattern



Hold [Shift] + Press [Copy] — *Screen Indicators*

Copy Bar
Currently selected bar

Hold [Shift] + Press [Paste] — *Screen Indicators*

Paste Pattern / Bar
Currently selected location

Paste Pattern or Bar from clipboard

NOTES

CHANGING PATTERNS

1. There are up to 99 patterns per group. The patterns are accessed using the group button for the respective pattern.

2. Hold Group [A] - [D] + Press Pad [0] - [9] to enter the pattern number using the numeric pad. Example, Hold [A] + Press [2], Press [1] to select A.21.

3. To increment to the prior or next pattern, Hold Group [A] - [D] + Press [+] or [-]. The pattern will be changed to the prior or next one. Example, Hold [A] + Press [+] to change from A.21 to A.22 or Hold [A] + Press [-] to change from A.21 to A.20 Pattern.

4. The group letter and pattern number are displayed when selecting.

ERASING A PATTERN

1. Press [Main] to select Main Mode.

2. Select a pattern. This will be the target pattern to erase.

3. Hold [Erase] + Hold Group [A] - [D] to clear the current pattern within the selected group. Example, Hold [Erase] + Hold [A] to erase the current pattern, A.21 in Group A.

4. The ‘PTN’ and ‘DEL’ messages are displayed confirming pattern delete.

COPYING / PASTING A PATTERN OR BAR

1. Press [Main] to select Main Mode.

2. Select a pattern. This will be the target source for the bar or pattern.

3. To Copy:-

    - Copy Bar. Hold [Shift] + Press Group [C]. The secondary function of Group C is Copy. The current bar is copied to the clipboard.

    - Copy Pattern. Hold [Shift] + Press Group [C], Press [C] i.e. twice. The secondary function of Group C is Copy. The current pattern is copied to the clipboard.

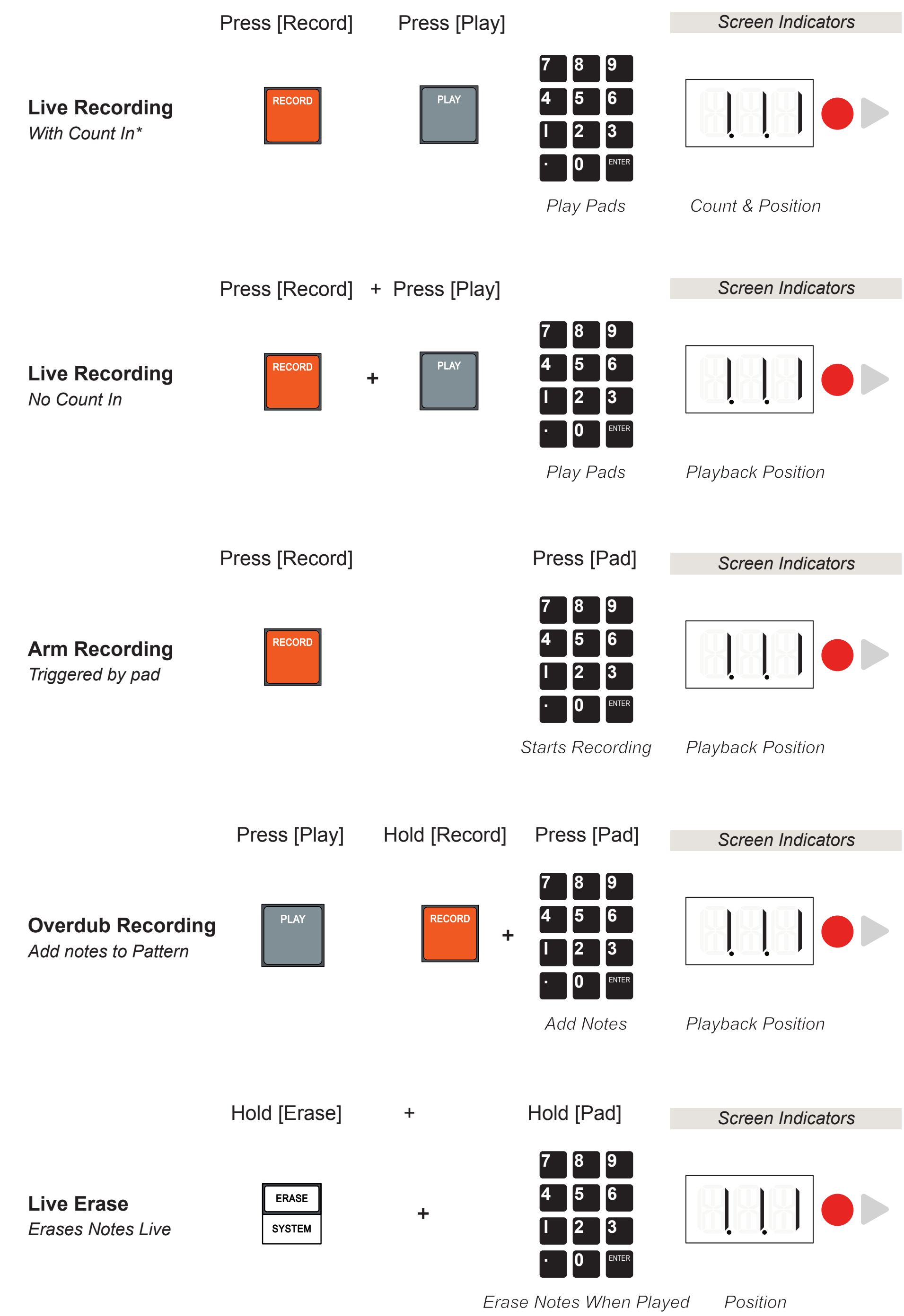
3. Navigate to a destination location, i.e. another group and/or pattern slot.

4. To Paste. Hold [Shift] + Press Group [D]. The secondary function of Group D is Paste. The current clipboard contents, either the previously copied bar or pattern is pasted to this destination.

NOTES

6.4 Live Recording Patterns

Live recording refers to the technique of recording notes using the pads as discrete sounds or a chromatic keyboard and also recording parameter automation into a pattern on the fly while the sequencer keeps playing. Great for improvisations, repetitive sequences and adding a human feel to a pattern.



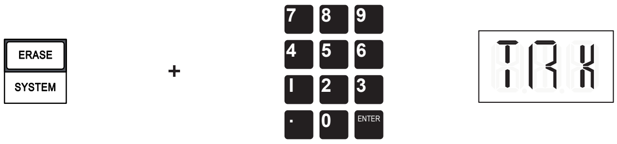
* Ensure the metronome is audible when using count in. This will give a visual and audible count in.

NOTES

Hold [Erase] + Hold [Pad]

Screen Indicators

Track Erase
All Notes When Stopped



Erase Track When Stopped

Track Erase

Hold [Record] + Adjust {Fader}

Screen Indicators

Automation
Live Record

Records Fader Assigned Parameter Changes

Parameter Value

Press [Play]

Screen Indicators

Stop Recording
Stop Rec & Playback

Current Stopped Position

Press [Rec]

Screen Indicators

Stop Recording
Stop Rec & Keep Playing

Current Active Position

The fader should be configured to control defined parameters before recording automation. Live recording will record a momentary value, real time as it changes across the recorded pattern length.

A combination of live recording and step recording techniques can be used in your own workflow. For example live erase of notes can be performed on a sequenced pattern created with step recording or hats recorded live and kick recorded by step recording.

NOTES

LIVE RECORDING A PATTERN

1. Select a group and a pattern to record a sequence into.

2. Ensure the pads have sounds loaded and you are familiar with the notes to play. Each pad will play its loaded sound. You can also record a sound chromatically by selecting a pad then switching to [Keys] mode.

3. Set the pattern length, Press [Record], then [-] or [+] to change number of bars in the pattern. Press [Timing] to set note interval with (X) Knob, and set Quantize / Free mode with [-] or [+]. Quantize records note positions to the grid. See details in this book elsewhere for other pattern settings.

4. Select the live recording process to use based on personal preference and workflow:-

    - To start recording after a count of 4, Press [Record], then Press [Play].

    - To start recording immediately, Hold [Record] + Press [Play].

    - To start recording when the first note is played, Press [Record] to arm, then the first [Pad] played will trigger the start of recording.

5. The record and play icons as well as the current dynamic pattern position in bars, beats and steps are displayed.

6. Any notes played using the pads will be recorded. The pattern will loop allowing elements to added to the recording on each cycle, perhaps using other sounds to expand the patterns.

7. Press [Stop] to stop the pattern recording and playing. Alternatively, to keep the pattern playing but stop recording, Press [Record] while playing.

8. To overdub and add further notes to the current pattern, Press [Play], then Hold [Record] + Press [Pads] to punch in notes when needed. This is called overdub recording. Live recording can also be restarted again.

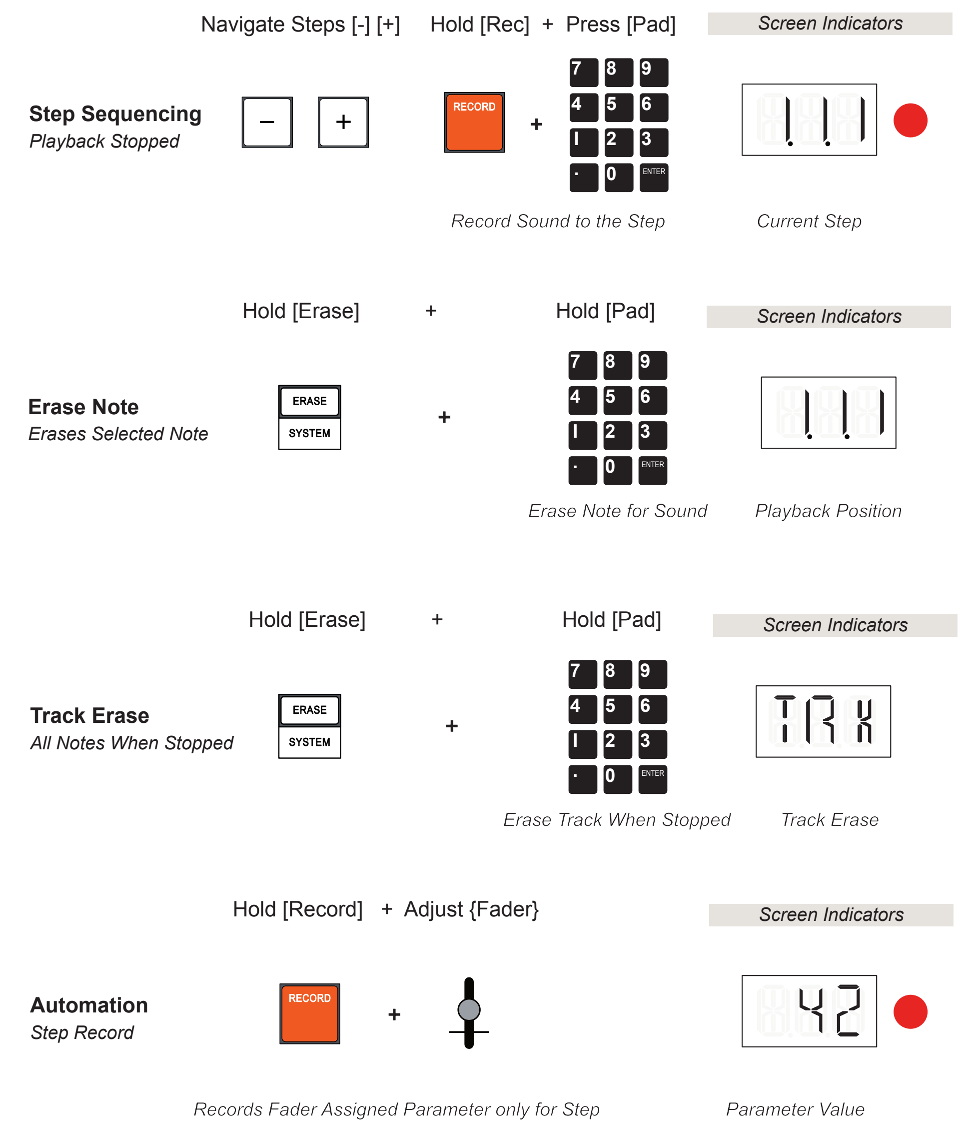
9. To erase all or part of a pattern while playing, Hold [Erase] + Press [Pad] at the pattern location and for the duration to clear. The existing notes for the pad held during this time will be erased. When holding erase and pad while the sequencer is stopped will erase the entire pad track of notes.

10. Parameter changes can also be recorded. This is called automation. While the pattern is playing, Hold [Record] + Adjust {Fader}. In live recording, the continuous value changes will be recorded across the sequence of steps played. The parameter values recorded will be based on the parameters assigned to the fader control.

> A tip for people less familiar with live recording is to slow the project tempo when recording then return to the faster, original project tempo when complete.

6.5 Step Recording Patterns

Step recording is a slower process than live recording. However the advantage is that it offers a more precise level of editing, one step at a time. This is great for exact note positioning on or off grid and to set the note parameters accurately.



As a tip, record a short pattern, say 1 bar and then use the duplicate or copy / past options to extend the bar length and copy over the recorded patterns. This will quickly build longer patterns which can be edited further.

NOTES

NOTES

STEP RECORDING A PATTERN

1. Select a group and a pattern to record a sequence into.

2. Ensure the pads have sounds loaded and you are familiar with the notes to play. Each pad will play its loaded sound. You can also record a sound chromatically by selecting a pad then switching to [Keys] mode.

3. The note interval is particularly important here as it sets the grid resolution of note placement, navigation and selection. Press [Timing] then set note interval by Turning (X) Knob. Higher resolutions offer more precise editing. See the sections elsewhere in this book for details of other pattern settings and functions.

4. Ensure the sequencer is stopped. It must not be playing to work with step recording. Press [Play] if the sequencer is playing to stop it.

5. Press [Main] to ensure the home location is selected.

6. Navigate through the pattern steps at the set note interval resolution.

    - Press [Shift] + [-] to set the playhead 'cursor' to the absolute first step in the prior bar. Repeat to locate to the absolute pattern start 1.1.1.

    - Press [Shift] + [+] to navigate forwards to the start of each bar.

    - Press [-] or [+] to navigate step by step. Resolution of step is based on note interval and quantization setting.

    - If Quantize / Free mode is set to 'Quantize', the resolution of step is based on the note interval. If set to 'Free' the notes can be nudged off grid and will be selected at their tick points when off the grid. Press [Timing] and [-] or [+] to change between quantize and free mode.

7. When a step with an active note is selected, its pad LED will be lit.

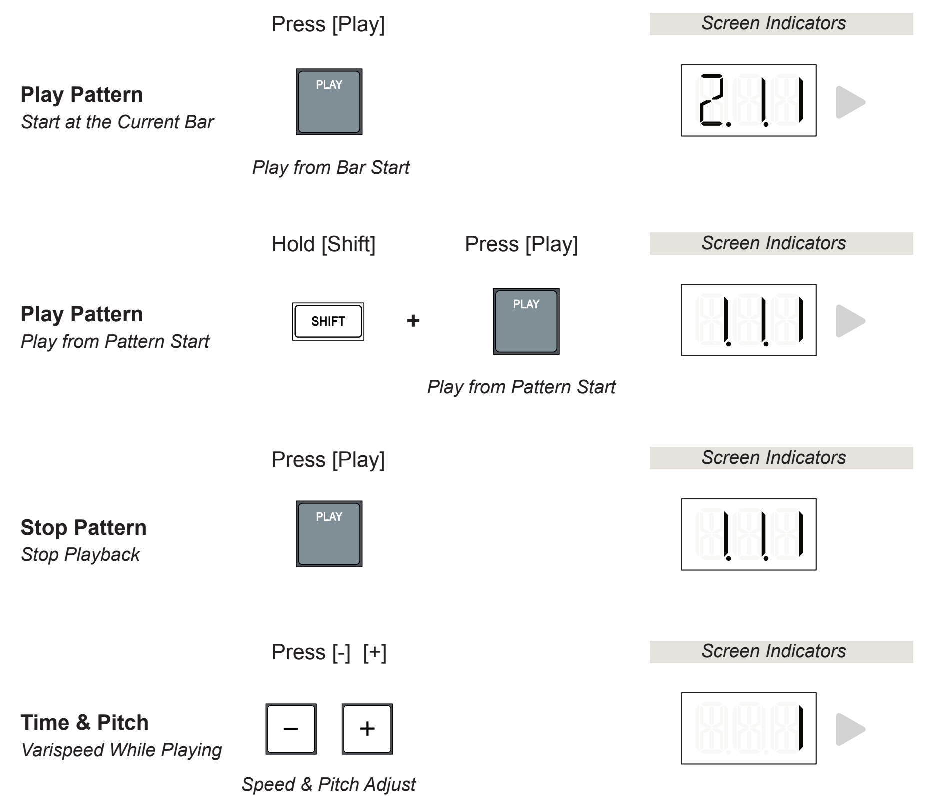
8. To nudge the note for the pad off the grid, select 'Free' mode. The Free icon will be displayed. Hold [Shift] + Hold [Pad] + Press [-] or [+] to move the note in ticks. If 'Quantize' is selected notes are moved across step intervals.

9. To erase a single note, Hold [Erase] + Press [Pad] at the pattern location for the pad and step to clear. When holding erase and pad longer while the sequencer is stopped, the entire pad track of notes is cleared.

10. Parameter changes can also be recorded. This is called automation. While the pattern is stopped, Hold [Record] + Adjust {Fader} on a selected step. In step recording, only the value change for the specific step will be recorded. This is latched meaning that when playing this setting will remain active until a new step parameter value change is recognised. This is different to live recording where changes are recorded momentarily per step and captured across the pattern range recorded.

6.6 Playing Patterns

Once a pattern has been recorded it can be played back in several ways. The normal Play Stop/Start will stop and restart the pattern from the start of the current bar. Using shift will restart from 1.1.1, the beginning of the pattern.



PLAYING A PATTERN

1. Press [Play] to Play a pattern. This will start from the beginning of the current bar. If a pattern is stopped mid way within Bar 2, then pressing play again will restart from the beginning of Bar 2.

2. Press [Shift] + [Play] to Play a pattern from the beginning. This will start from the the first bar at 1.1.1. If a pattern is stopped mid way within Bar 2, then pressing shift & play will restart from the beginning of Bar 1.

3. In all scenarios, pressing [Play] while a pattern is playing will pause playback at the current position. There is no option right now to restart playback from the paused position in the KOII.

4. If required use the [-] and [+] buttons while playing in [Main] mode to introduce a variable speed and pitch effect. The display will animate up / right or down / left. This cannot be recorded.

NOTES

6.7 Solo & Muting Sounds

Soloing and muting is generally performed at group or pad sound level. Group A, B, C and D can be controlled as a collective set of pad sounds. Solo is a temporary option where one or more groups can be soloed while a pattern is playing. Mute operates using the fader so can therefore operate permanently.

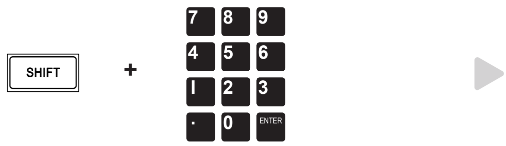
Hold [FX] + Hold [Group] | *Screen Indicators*

Solo Group(s)
Temporary - while holding

Multiple Groups can be held & soloed | *Group(s) displayed while playing*

Hold [Shift] + Hold [Pad] | *Screen Indicators*

Solo Pad(s)
Temporary - while holding



Multiple Pads can be held & soloed

Hold [Group] + Slide [Up] [Down] | *Screen Indicator*

Mute Group(s)
Group Project Volume

Project Vol - Irrespective of assignment | ***Value of Group Volume Level***

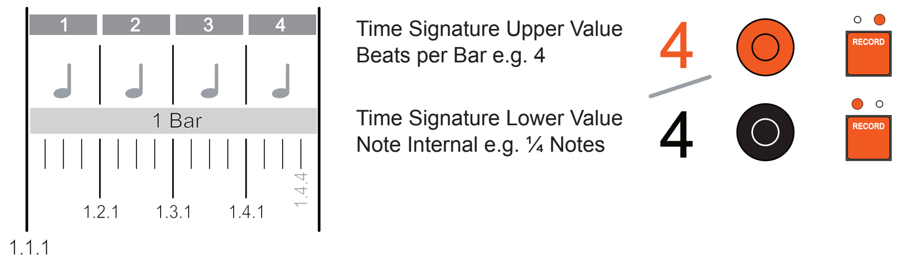
Multiple group and pad buttons can be held simultaneously to mute or solo in these workflows.

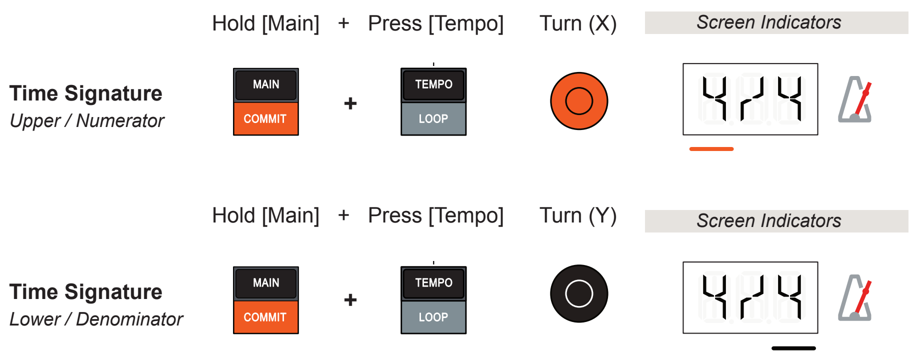
Muting. There are no immediate pad / button orientated muting options available at this OS point in the KOII. The fader is the method used in the currently available functions and workflow.

6.8 Time Signature

The time signature refers to the overall musical timing of patterns based on the relationship between beats in a bar and the actual note values and set the overall timing of the patterns. The metronome will reflect the current time signature in its audible clicks and the record button in the cycle of its 2 LEDs.

Time signatures with uncommon values and time signature variations between scenes can offer some interesting creative possibilities and help create polyrhythmic sequences. Use scenes for mixing time signatures in a project.





SETTING TIME SIGNATURE

1. Hold [Main] + Press [Tempo] to select the time signature mode.

2. Turn (X) Orange knob to adjust the beats per bar i.e. the upper value.

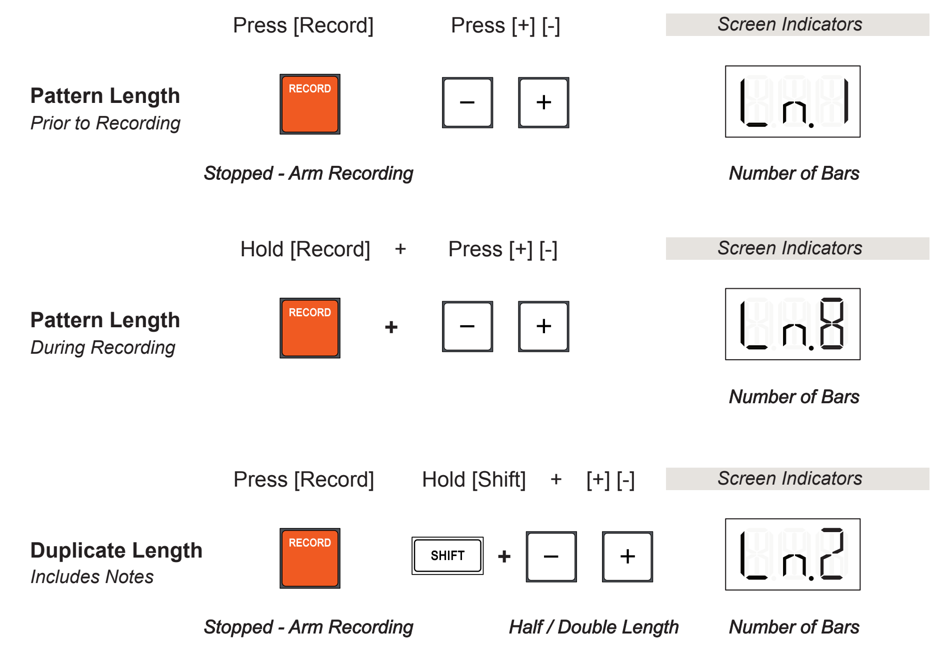
3. Turn (Y) Black knob to adjust the note value in the bar i.e. the lower value.

4. Press [Main] to return to the default ‘home’ position.

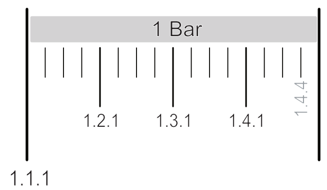
1/2	1/4	1/8	2/2	2/4	2/8	3/2	3/4	3/8
4/4	4/8	5/4	5/8	6/4	6/8	7/8	8/8	9/8

6.9 Pattern Length

The length of the pattern can be set in bars before or during recording in 1 Bar increments up to a maximum of 99 Bars. The default is 1 Bar. It is also possible to use the ‘Duplicate’ function to extend by doubling or retract by halving the bar length. Duplicate also copies over any existing notes.



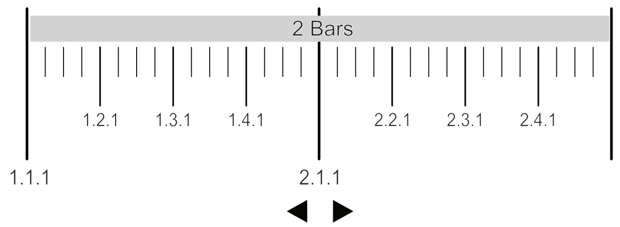
Duplicate Examples



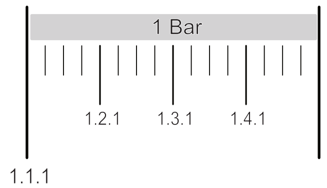
Default is 1 Bar recording length

Can be adjusted before or during recording in 1 Bar increments up to a maximum of 99.

Resolution illustrated is 0.0.1 Tick.



Each time the length is duplicated with [Shift] + [+], it will double the number of bars from the current state and duplicate notes in the pattern too.



Each time the length is reduced by [Shift] + [-], it will half the number of bars from the current state and also reduce the notes accordingly.

NOTES

ADJUSTING THE PATTERN LENGTH

1. With playback and recording stopped.
    - Press [Record] to arm recording. The red record icon will flash.
    - Press [+] or [-] select the number of bars in the length.
    - The alpha numeric display will indicate length a Ln.1, Ln.2, Ln.3 etc.
2. While pattern is actively recording.
    - The record / play icons lit.
    - Hold [Record] + Press [+] or [-].
    - The alpha numeric display will indicate length a Ln.1, Ln.2, Ln.3 etc.

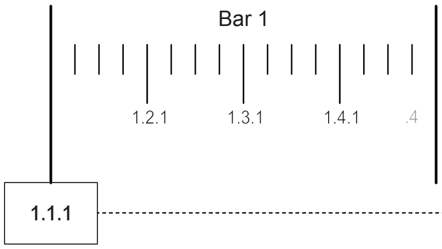
DUPLICATING PATTERN LENGTH

1. Duplication will double the existing pattern length. For example a 2 Bar Pattern will be increased to 4 Bars. Equally the length can be halved.
2. To extend and double the length.
    - Press [Record] to arm recording. The red record icon will flash.
    - Hold [Shift] + Press [+] to duplicate the bars and double in length.
    - Existing notes in the copied bars will also be duplicated.
    - The alpha numeric display will indicate length, Ln.1, Ln.2, Ln.4 etc.
3. To reduce and half the length.
    - Press [Record] to arm recording. The red record icon will flash.
    - Hold [Shift] + Press [-] to reduce bars and half the length.
    - Existing notes in a pattern will also be reduced accordingly.
    - The alpha numeric display will indicate length eg Ln.4, Ln.2, Ln.1.

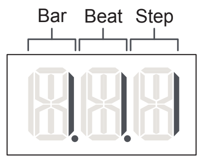
NOTES

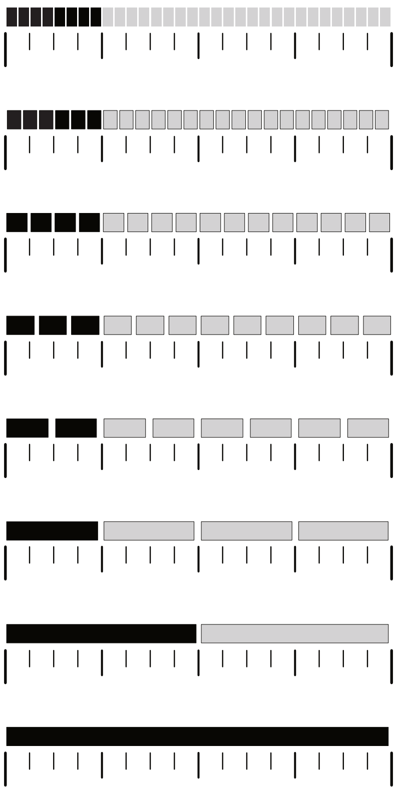
6.10 Note Timing & Swing

The timing interval will determine the note resolution of steps in the sequencer. Swing is also controlled from within this function but only applied to 1/8th and 1/16th Note Intervals. Note timing and swing are set pre-recording to quantize and shuffle notes during recording but can also can be corrected afterwards.



Pattern Length is set in 1 Bar increments.
1 Bar = 4 Beats. 96 Ticks per Beat / 24 Ticks per Step.
[Main] Mode displays the dynamic playhead position.





1/32nd

1/16th Triplets

1/16th
Swing can be applied

1/8th Triplets

1/8th
Swing can be applied

1/4 Notes

1/2 Notes
Half bar

1/1 Notes
1 bar

Timing Options Available

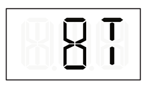
1/32	1/16T	1/16	1/8T	1/8	1/4	1/2	1/1

Press [Timing]

Turn (X)

Screen Indicators

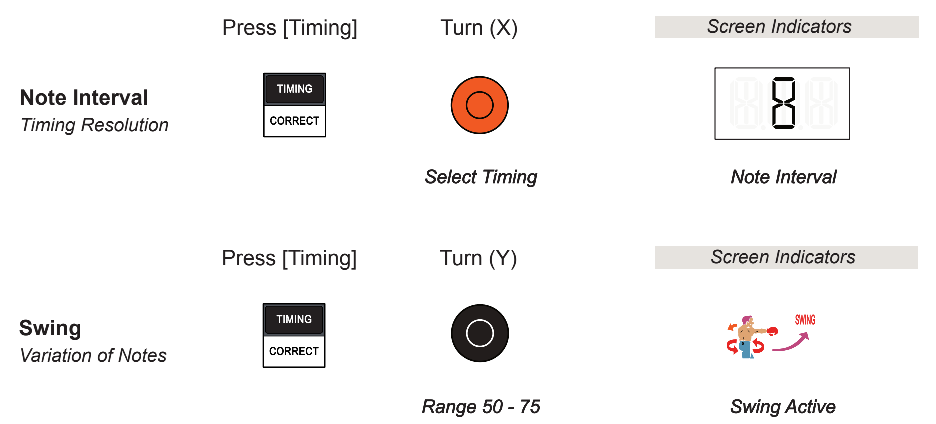
Note Interval
Timing Grid Resolution



Note Interval

Swing

Swing introduces an amount of variation to some of the notes when recording into the sequencer. Only 1/8th and 1/16th notes are affected by swing. The swing setting of 50-75 using the black (Y) control are only adjustable when 8 or 16 is selected as the note timing. The feature is more noticeable in higher resolutions. Swing can give a pattern some variation with a humanised feel or a rhythmic shuffle added into the sequence Swing creates a more natural and less rigid pattern. Swing is a project based function.



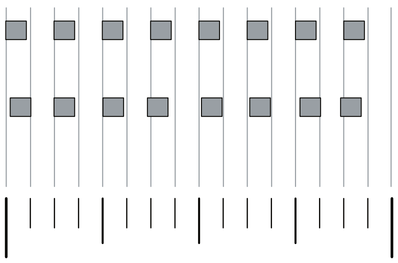
NOTES

NOTES

6.11 Quantize

Quantization is an aid used in order to auto-adjust played notes to align to the nearest interval of a defined grid when recording in the sequencer. This ensures the timing of notes are recorded to a rigid structure and any human errors and timing deviations are corrected automatically. The KOII can record using quantize or free running mode. Free mode will place notes exactly as they are played which may be on or off grid depending on the skill of recording.

Example: Quantize on 1/16th Timing Resolution



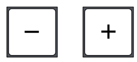
Quantize:
Notes recorded and adjusted on grid.

Free:
Notes recorded exactly as played including deviations from the grid.

1/16th Note Interval for the Grid

Press [Timing] | Press [-] [+] | *Screen Indicators*

Quantize / Free
Recording Quantize



Toggle Free / Quantize

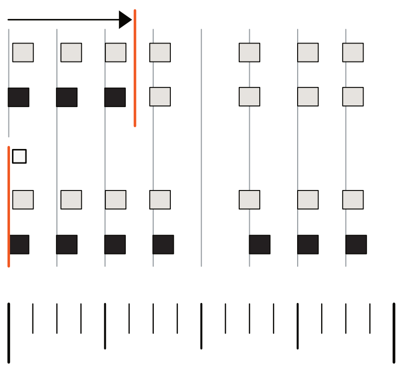
The quantize setting also governs how steps are navigated and nudged in step recording mode. Off grid selections for notes nudged off grid can be selected in free mode and adjusted in tick values. Quantize mode will navigate through in step intervals and be locked to the note interval timing.

6.12 Timing Correct

While the settings such as quantize and swing aim to set the pre-recording options and help manage recording on grid, there may be errors from free recording or manual sequencing that are not as precise as needed.

Timing Correct offers a post-recording option to correct off grid or mistimed notes individually or all together. Live quantize will correct each note as played. The display will count the notes each time one is corrected. Quantize pad will correct all notes in the pad at the same time. In both cases the currently selected note interval is used.

Example: Unquantized notes, 1/16th Timing Resolution



Live Quantize - While Playing

Original: Notes as recorded, some off grid.

Timing Correction: Each note quantized as played.

Quantize Pad - When Stopped

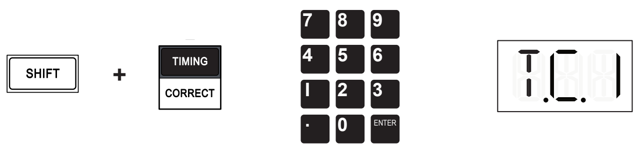
Original: Notes Recorded, some off grid.

Timing Correction: All pad notes quantized.

1/16th   Note Interval

Hold [Shift] + Press [Correct]   Hold [Pad]   *Screen Indicators*

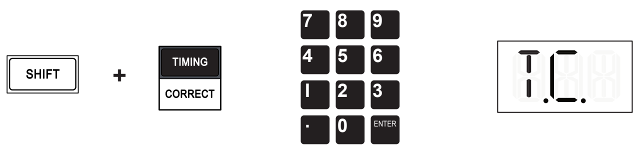
Live Quantize
While Playing



Quantize to selected Note Interval   *Fix played notes*   *Number of Corrections*

Hold [Shift] + Press [Correct]   Press [Pad]   *Screen Indicators*

Quantize Pad
While Stopped



Quantize to selected Note Interval   *Fix all pad notes*

NOTES

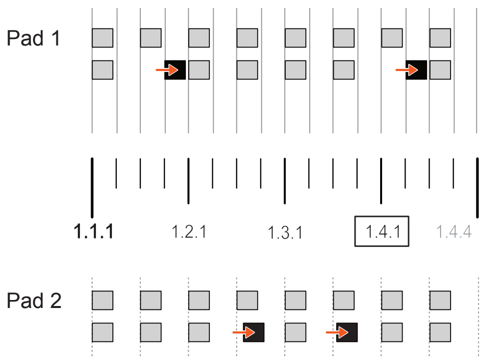
6.13 Offset & Nudge Notes

While quantization and timing correct aim to align notes tightly to a grid, offset notes is all about moving and nudging notes within the grid, perhaps to generate more interest or expression in a pattern. The amount of movement is set by the note interval when in quantize mode and will move freely, outside of the grid in tick intervals when free mode is set.

Firstly select the note in the sequence using [+] & [-] to locate the position. The display will show the position in Bars i.e. note interval setting resolution or ticks depending on the selected note and the selected quantize or free mode.

The sequencer should be stopped to apply a nudge or offset to a note. Also set the Quantize or Free Mode as this determines the nudge interval.

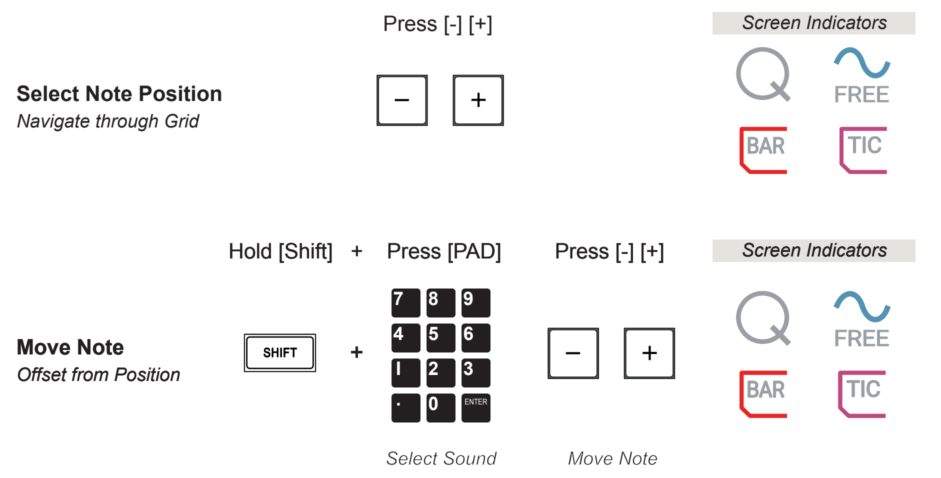
Example: 1/16th Timing Resolution



Quantized:
Notes, selected by pad. Position moved by the note interval i.e. 1/16th, on grid.

1/16th
Note Interval

Free: Notes, selected by pad. Position moved freely i.e. Increment in ticks, off grid.



Quantize On. The notes are nudged by the timing interval set. Free Mode on, the notes are nudged in Ticks. There are 96 Ticks per Beat / 24 per Step. The Alpha-Numeric display will indicate the selected / active note based on the quantized position in the bar or the tic offset position from the grid.

NOTES

6.14 Note Velocity

Velocity refers to how hard a pad or a key is pressed. For example the natural velocity of a piano when playing keys hard or soft will affect the note loudness. The KOII has a system setting that can apply a velocity sensitivity profile to the pads when played and recorded. Also velocity for all notes in a selected step can be adjusted.

SETTING THE VELOCITY SENSITIVITY PROFILE

1. This setting is applied when playing and recording and is therefore a setting that is applied during recording.

2. Select the system settings, [Shift] + [System]. A flashing cursor will be displayed along with the ‘System’ icon. Also the flashing hand icon indicates that the pads can be used to enter selections. The [Enter] Pad will flash when the option to confirm a change is available. Use [Shift] + [Enter] to backup in the menu navigation.

3. Press [-] or [+] buttons to navigate the settings menu. Scroll through the options until ‘PAD’ is displayed. Then Press [Enter] Pad.

4. Press [-] or [+] buttons to navigate the PAD settings sub-menu. Scroll through the options until ‘VEL’ is displayed. Press [Enter] Pad.

5. The 3 Options for the velocity sensitivity are available, navigate using the [-] or [+] buttons. When the selected option is highlighted, Press [Enter] Pad to select and confirm.

    - OFF - No Velocity Curve is applied. Fixed velocity recorded. This is the Default.

    - HI - If your style uses lower pressure to play a pad with less force, this profile will compensate to play or record more impactful sounds.

    - LO - If your style uses a higher pressure to play a pad with more force the profile will compensate to play or record gentler sounds.

6. Press [Main] to exit back to the normal operating mode.

Settings - [Shift] + [System]

Display and Enter LED Flashes when setting has been edited.
Display and LED are static when setting is confirmed with [Enter].

Note Velocity Editing

Velocity can also be altered on some sounds post recording. This can be adjusted for a selected step and its incumbent pad sound. This is independent to the velocity profile set pre-recording in the system menu. In Main Mode, a step is selected from within the pattern by using the [-] & [+] buttons and press [Pad] to trigger it. A step selected with a note will illuminate that note's pad LED.

Press [-] [+] Hold [Shift] + Turn [X] *Screen Indicators*

Adjust Velocity
Selected Step & Pad

Select Step, Pad LED lit *Adjust Velocity* *Sound will play while editing*

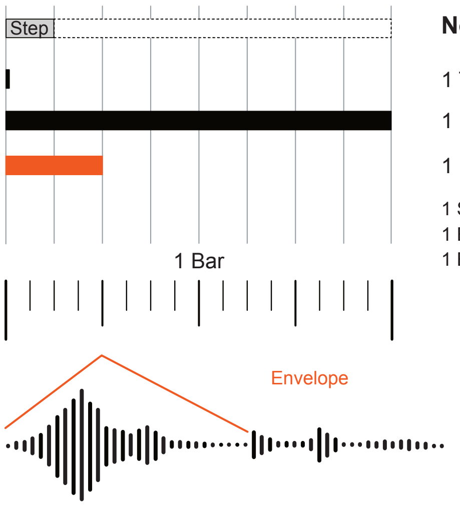
Velocity sensitivity is applied when playing and recording the pads. This is physically assigned to the lower half of each of the Pad buttons. Therefore velocity pressure is applied only when pressing the lower half of each of the pad buttons.

In post recording the ability to change velocity will depend on the sound and sample.

6.15 Note Duration

The duration of all of the notes selected in a specific step can be adjusted. This can be changed between 1 tick and 1 bar. Note that the sample length and envelope also determines how the duration is played. This is a performed in an existing pattern recording.

Example: Note Duration.



Note On 1/8th Step

1 Tick - Minimum Duration

1 Bar - Maximum Duration

1 Beat - *See envelope release**

1 Step = 24 Ticks
1 Beat = 96 Ticks
1 Bar = 384 Ticks

Sample duration and envelope determine sound shape over time. Note duration longer than a sample will be governed by the sample itself and the envelope release time*, which remember is triggered at 'Note Off'.

Press [-] [+] Hold [Shift] + Turn [Y] *Screen Indicators*

Adjust Duration
Selected Step & Pad

Select Step, Pad LED lit *Adjust Duration* *Sound will play while editing*

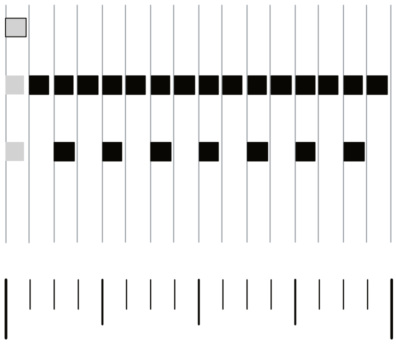
In post recording the ability to change duration will depend on the sound and sample.

NOTES

6.16 Note Repeat

The ability to automatically repeat a played note is possible. This is performed using the timing function which also sets the note interval of the repeated notes. Ideal for recording trills and fills. Repeats can also be held. If velocity is enabled, the lower half of the played pad will affect the velocity based on the amount of physical pressure applied to the pad button.

Example: Quantized notes, 1/8th & 1/16th Timing Resolution



Note On *No Swing*

1/16th Note Repeats

1/8th Note Repeats

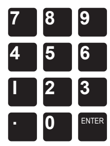
Timing Sets the Note Interval

1/16th Note Interval

Hold [Timing] + Hold [Pad]

Note Repeat
Playing Pad

+

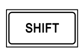


Note repeats are based on the current timing setting. If a note interval of 16 is set, the repeat interval will be 1/16th notes

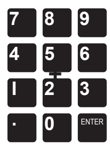
Hold [Timing] + Hold [Shift] + Tap [Pad]

Hold Note Repeat
Lock Pad Repeat

+



+



Repeat the same button combo to unlock the repeats.

The KOI has the ability to play pads with a variable pressure level, normally translated as velocity value for notes played. Velocity would have to be active. Pressure is applied only in the lower half of each of the pad buttons.

Velocity can add even more variation across a repeated set of notes played or recorded. Vary the lower pad pressure while recording to capture natural changes.

7

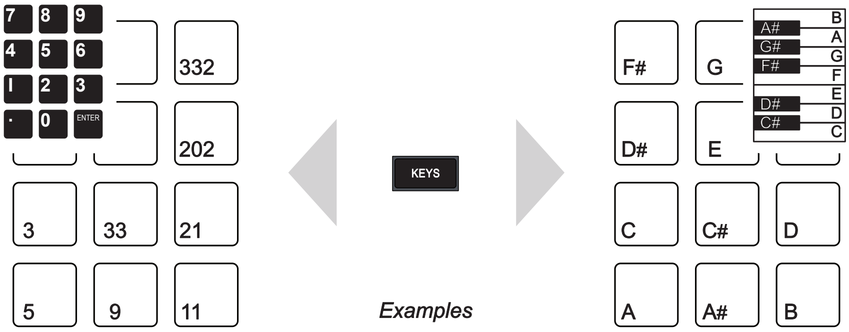
Patterns & Scenes

The basic principles and functions used when creating patterns have been covered. This will serve as a good base reference when building bigger more extended compositions. This section takes this one stage further. A more advanced coverage of the application and elements to building patterns is covered here. More details on editing and developing patterns is covered along with a deeper dive into the 'keys' mode which offers a different dimension to play notes when sequencing a pattern. In addition scenes are covered in detail. A scene is a collection of patterns and is used to structure a more comprehensive arrangement and to build a complete song. Many people will be familiar with 'song mode', a feature in some devices to construct full arrangements. A limited song mode was introduced into the KOII feature set in the Champions 2.0 OS update. One song can be configured per project, with up to 99 scene positions stored. Patterns can be assembled into multiple scenes. The workflow of scene creation is heavily reliant on the 'commit' function used to iterate and progress the composition. This allows the current group of patterns to be captured and copied across to a new set of patterns called a new scene. Scenes therefore build and allow editing from the previously developed scene and to progressively create new variations and arrangement sections. A project can hold up to 99 scenes. Also remember that a project can hold up to 99 patterns.

7.1 Keys Mode

Keys mode switches the pads to an alternative playback mode. In Keys Mode the pads can play a single sound chromatically across the 12 note scale as opposed to playing individual sounds per pad. Keys mode is also useful when connecting an external MIDI Keyboard controller. Select a Pad before selecting [Keys] to choose the sound to map chromatically.

NOTES

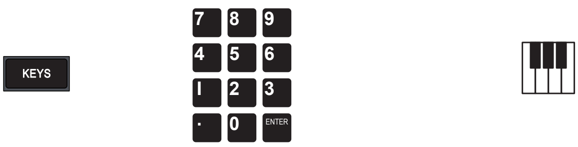


Default. Plays the sound assigned to each pad independently.

Keys Mode. Plays a selected sound chromatically across each of the 12 pads.

Press [Keys] | Press [Pad] | *Screen Indicators*

Keys Mode
Selected Pad



Pads play selected sound chromatically

Hold [Keys] + Press [Pad] | *Screen Indicators*

Root Note
Keys Mode

Select the Root Note - Pad LED will flash

Hold [Keys] + Press [-] [+] | *Screen Indicators*

Octave
Keys Mode

Adjust the Octave Range Up / Down

NOTES

Keys mode is ideal for playing and recording lead or bass lines, chord progressions or any melodic pattern. The sample selected will form the core of the sound. The workflow given below is an example and may differ depending on the sample pitch tuning used.

EXAMPLE IN USING KEYS MODE FOR A SOUND

1. Select a sound for a Pad. For example load a piano sample tuned to note ‘D’ on the [.] Pad. Sample 501 in the factory pack may be helpful as an example. Select the sound by pressing Pad [.].

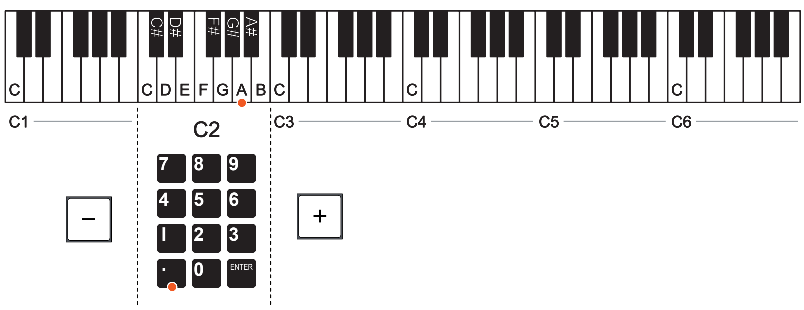
2. Press [Keys] to switch to Keys Mode. The Keyboard icon will be displayed. The sample selected is now mapped across all pads chromatically.

3. The note on Pad [.] has now noticeably dropped by a few semitones in this example. Pad [.] has been assigned to Note ‘A’. In Keys Mode, Pad [3] will now play the note at the same ‘D’ pitch as the original sample sound, assuming the 12T scale is set as default.

4. If required, the root note can be adjusted by transposing the scale. This may be especially useful to align with an external MIDI keyboard to the pads and samples. Hold [Keys] + [Pad] to assign the root note displayed by default as note C2, but this will also depend on the octave selected. In this case example the original root note is retained.

5. If required, the octave range can be adjusted. The default is in the ‘C2’ range. This can be adjusted down to C1 or up to C6. This may be especially useful to align with an external MIDI keyboard to the pads and samples. Hold [Keys] + [-] adjust the range down or [+] to adjust the range up. The range changes with each press of the - / + button.

6. The pads can now be used for recording but in this mode will record the melodic activity played on the pads, or external keyboard.



Keys Mode Scales

In Keys Mode the pads can play a single sound chromatically across the 12 note scale. The scale that is applied in keys mode can be set in the system settings. The 12T Note scale is default.

SETTING THE SCALE FOR KEYS MODE

1. Select the system settings, [Shift] + [System]. A flashing cursor will be displayed along with the 'System' icon. Also the flashing hand icon indicates that the pads can be used to enter selections. The [Enter] button will flash when the option to confirm a change is available. Use [Shift] + [Enter] to back up in the menu if required.

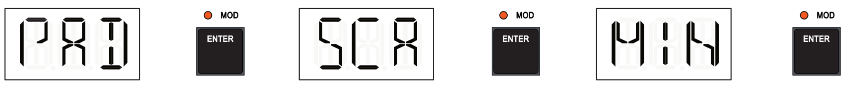
2. Press [-] or [+] buttons to navigate the settings menu. Scroll through the options until 'PAD' is displayed. Press [Enter] Pad.

3. Press [-] or [+] buttons to navigate the PAD settings sub-menu. Scroll through the options until 'SCA' is displayed. Press [Enter] Pad.

4. There are a number of scales to choose from. Navigate using the [-] or [+] buttons. When the selected option is highlighted, Press [Enter] Pad to select and confirm.

5. Press [Main] to exit back to the normal operating 'home' mode.

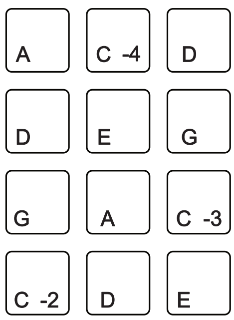
Scale Options



Display and Enter LED Flashes when setting has been edited.
Solid Display and LED when setting is confirmed with [Enter].

etc

Example: MA.P Major Pentatonic



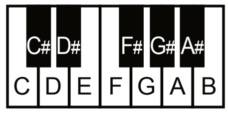
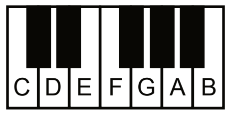
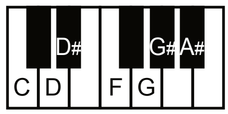
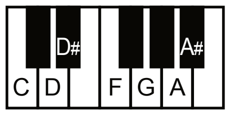
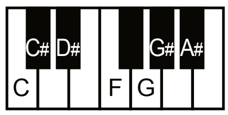
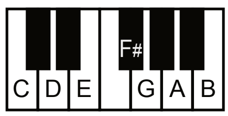
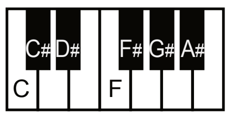
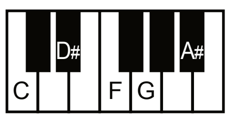
Root Note, C-2

Major Pentatonic uses 5 notes; C, D, E, G & A.

All notes are transposed and mapped across ALL of the KOII Pads in transcending order from the defined root note.

NOTES

Scale Options - Keys Mode

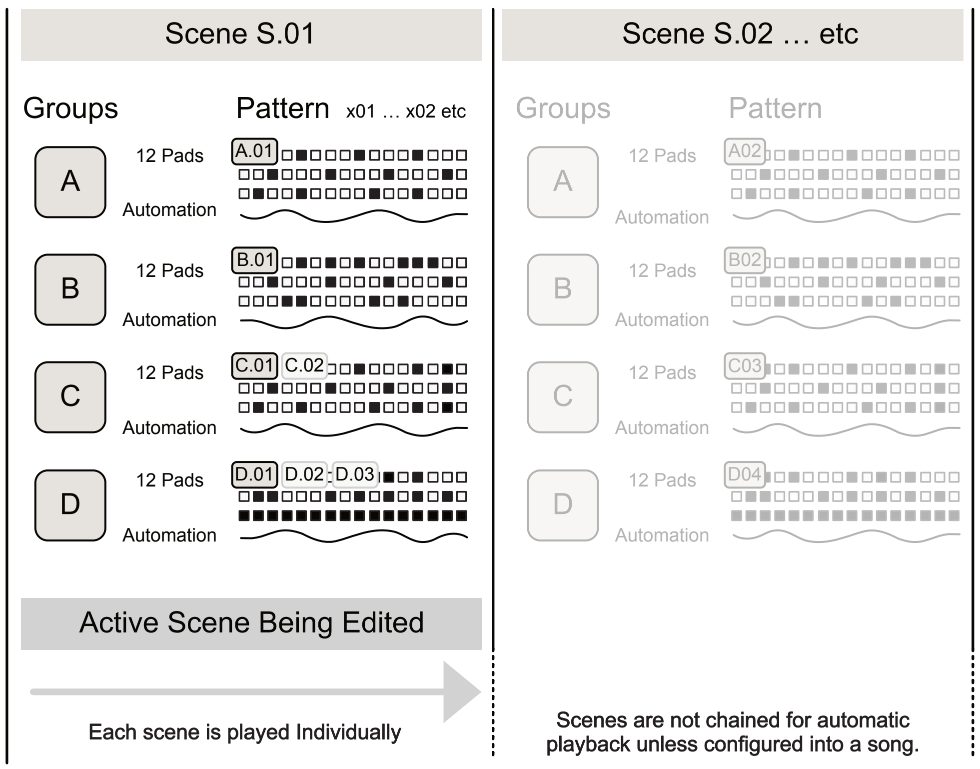
Scale Name	Scale	Notes - Root Note can be changed	
12T	12 Tone Equal Temperament. *Default.*		A# B C G G# A D# E F C C# D
MAJ	Major - Ionian Mode		E F G B C D F G A C D E
MIN	Minor - Aeolian Mode		D# F G A# C D F G G# C D D#
DOR	Dorian Mode		D# F G A# C D F G A C D D#
PHR	Phrygian Mode		D# F G A# C C# F G G# C C# D#
LYD	Lydian Mode		E F# G B C D F# G A C D E
MIX	Mixolydian Mode		E F G A# C D F G A C D E
LOC	Locrian Mode		D# F G# A# C C# F F# G# C C# D#
MA.P	Major Pentatonic		A C D D E G G A C C D E
MI.P	Minor Pentatonic		A# C D# D# F G G A# C C D# F

All KOII pads are populated with a note in the sequence, laid out in the selected scale.

7.2 Scene Overview

A Scene is a workflow and structural feature which helps to construct groups and their patterns into an extended arrangement. Each scene is played individually and cannot be chained to play automatically in series. A scene, by default S-01, is always active and will be the working environment when developing and recording patterns. 99 Scenes are available per Project.

Overview of Scene Structure



- The default scene, out of the box and on a new project is Scene 1, named S.01. This is the only scene available unless 'Commit' has been initiated to create a new one. Hold [Main] to view the active scene.

- Each scene will consist of the 4 Groups and the selected pattern for each group. A pattern will represent the notes or beats created from each pad sound in the group and any parameter automation recorded.

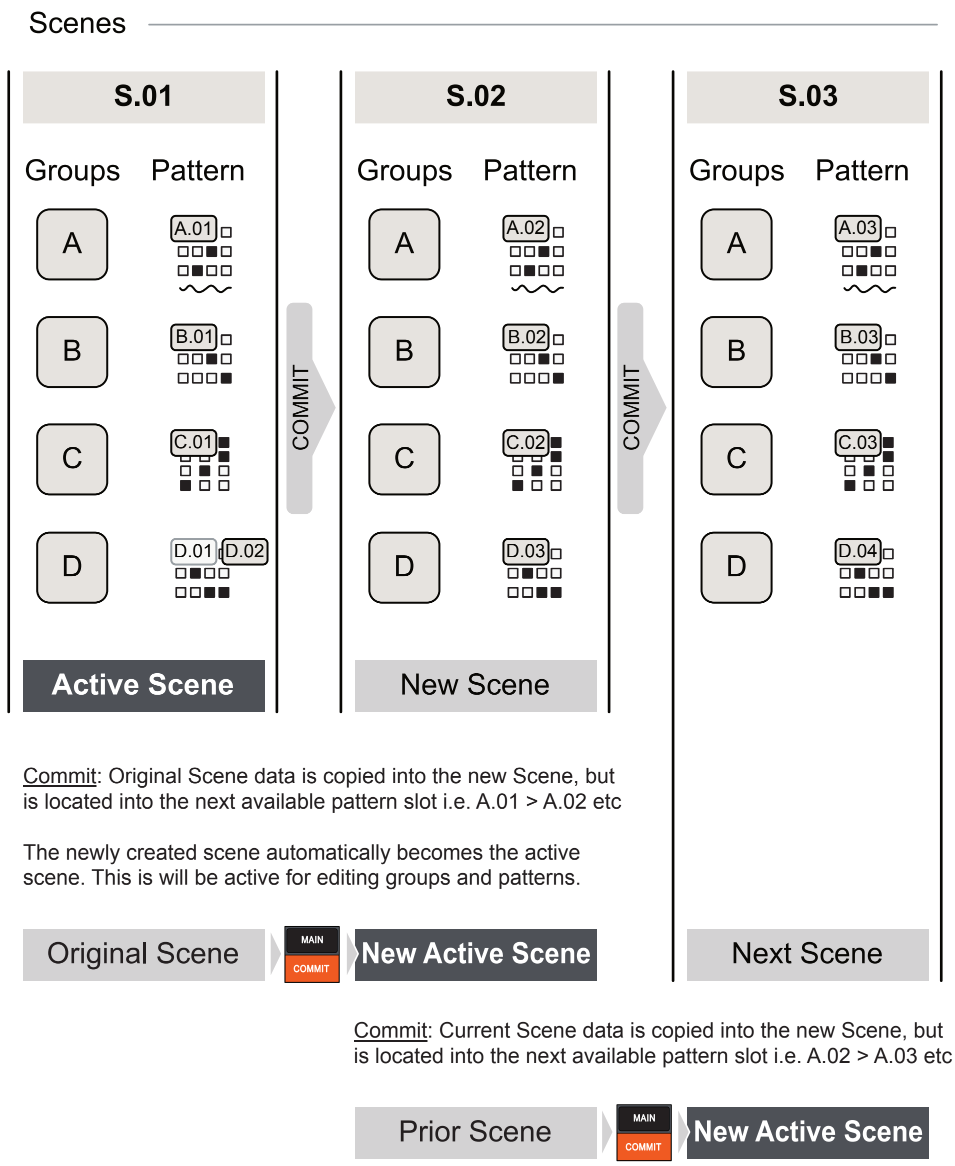
- The active pattern is named by the group and a sequential number, up to 99 per group. So the first Pattern for each group is A.01, B.01, C.01 and D.01. Hold a [*Group*] A-D button to view it's active pattern.

NOTES

NOTES

Scene & Commit

A scene is a structural element which is used to construct extended arrangements. The process used to generate a new scene in called 'Commit'. This is manually requested using the [Commit] button and when doing so will replicate the currently active scene into the next scene slot. The original scene patterns and data are retained but are also copied into the new scene and the next incremental pattern slots.



The commit process is progressive and incremental in order to add speed and efficiency to the song development workflow. The current pattern state is the starting point for a new scene, which can then be edited for that section I.e intro, verse, chorus, build up etc. Think of each scene being a song section. The example here shows incremental pattern slots but in reality any pattern can be assigned to a scene. The commit process increments through the next available pattern slot.

Scenes are a series of ‘containers’ that hold variations of group A, B, C and D Patterns. There are some limitations in the functionality and features. Expectations should be set to how scenes work and how they contribute to ‘song mode’ functionality. Maybe a future OS will address this!

Important notes regarding scenes

- Scenes are a structural elements which organise a series of patterns. Scenes are created as replicas of the current scene and are organised in a fixed numerical order, when created S.01, S.02, S.03 etc.
- Scene S.01 always exists. This will be the default operating environment out of the box.
- Scenes can be played individually, changing the scene number to play manually. Scenes cannot be chained to play in sequence, one after the other, automatically.
- Scenes do not allow the repositioning, copying and re-arranging of existing scenes from their location other than the commit function duplicating the scene to a new one. Scenes however can be deleted.
- Scenes iterate the pattern numbers when replicating through ‘Commit’. It is good practice to try to keep the pattern numbering convention aligned to avoid confusion later. At least make written notes, maybe here in this notebook!!!
- Patterns assigned to scenes can be edited. When doing so this will update in every scene it is assigned. So if Pattern A.05 is used in 3 scenes, editing the pattern A.05 will be updated in every one of the 3 scenes that it is present.
- The general premise is to commit a project early when developing the song. Then build the production iteratively on the new scene and then continue building and committing.
- Take note of the ‘empty pattern’ display icon. This will help identify when a pattern in a scene is empty. Also it may be a good tip to hold an empty pattern intentionally as a ‘mute’ pattern to use in a scene.
- Some elements operate at different levels and therefore will affect scenes differently. Faders operate at group level. The Tempo and FX operate at project level. Time signature for example can be set per scene.
- Synchronization of scene changes is set in the system settings. Default is set to ‘Tic’ so the scene changes immediately it is requested to do so. Other options available are bar and pattern end.

NOTES

7.3 Using Scenes

The general workflow of using scenes is to develop a set of group patterns. Then commit to these and create a duplicate scene to allow the patterns to be edited for the new scene. The original, prior scene state is retained. This creates variations and progressions played manually as a performance.

View Scene #
Check active scene

Hold [Main]

Press then Hold

Screen Indicators

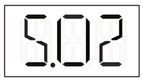
Active Scene

Commit
Replicate to New Scene

Hold [Shift] + Press [Commit]

New scene becomes the active scene

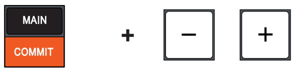
Screen Indicators



Active Scene

Change Scene
Step through Scenes

Hold [Main] + Press [-] [+]



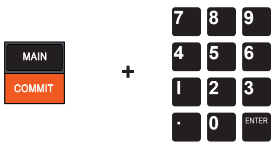
New scene becomes the active scene

Screen Indicators

Active Scene

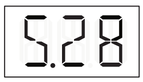
Scene Select
by Number

Hold [Main] + Press [Pads]



Select specific scene by number

Screen Indicators



Active Scene

Delete Scene
Delete active scenes

Hold [Main] + Hold [Erase]

Deleting a scene will move any others down

Screen Indicators

Flashes 'SCN' & 'DEL' to confirm

CREATING A SCENE

1. The prerequisites for creating a scene are that the patterns have been created for one or more groups. Certainly a base set of patterns are required. The options will be to start with a light version and build on it through other scenes. Alternatively creating a more complete scene and then editing out elements is possible across a scene range. The initial working scene by default is Scene 1, S.01.

2. Hold [Shift] + Hold [Commit]. A new, additional scene will be created. This will be a replica of the previous original scene but will iterate to the next available pattern slot for each group. Some examples are:-

   - If only one pattern exists as A.01, then new scene will create an A.02 version and copy the pattern from A.01.

   - If 3 patterns exist in the original scene group, i.e. C.01, C.02 and Active Pattern C.03, the new Scene will copy C.03 to a new pattern C.04. Always try to keep pattern numbers aligned where possible.

   - Scenes are created to the next available slot. So if S.01 and S.02 already exist, using commit when scene S.01 is active will commit and copy it to S.03.

3. The new scene ie S.02 will also become the active scene. This can now be edited to develop a variation from the original. For example the original pattern may only include a kick and now a snare can be added. Add Hi Hats on S.03 etc.

4. Iterate these steps to build more scenes and variations.

SELECTING AN EXISTING SCENE

1. To view the currently active scene in main mode, Hold [Main]. The scene number will be displayed in the format, S.01.

2. To change to another scene, Hold [Main] + Press [-] or [+] to select one scene at a time. The scene number will be displayed.

3. To select to another scene by its number, Hold [Main] + Press [Pad]'s where the number typed on the pads will be the selected scene. The scene number will be displayed. A maximum of 99 scenes can be used.

NOTES

DELETING A SCENE

1. Select a scene to delete. The active scene will be the target.
2. Hold [Erase] + Hold [Main] to delete the scene.
3. The display will flash ‘SCN’ to signify a scene is being deleted.
4. Once deletion is completed, the display will show ‘DEL’. If deleting between scenes, the higher scenes will be shifted down to close the gap and renumbered to the new scene location. For example, if scene S.02 is erased and S.03 exists, then S.03 will be updated as scene S.02.
5. While scene numbers are updated when erasing scenes, the patterns will be retained as per the original scene. i.e. if Scene S.03 , with patterns A.03, B.03, C.03, D.03, is re-ordered as S.02, the patterns will remain a A.03 etc.
6. This process cannot be undone.

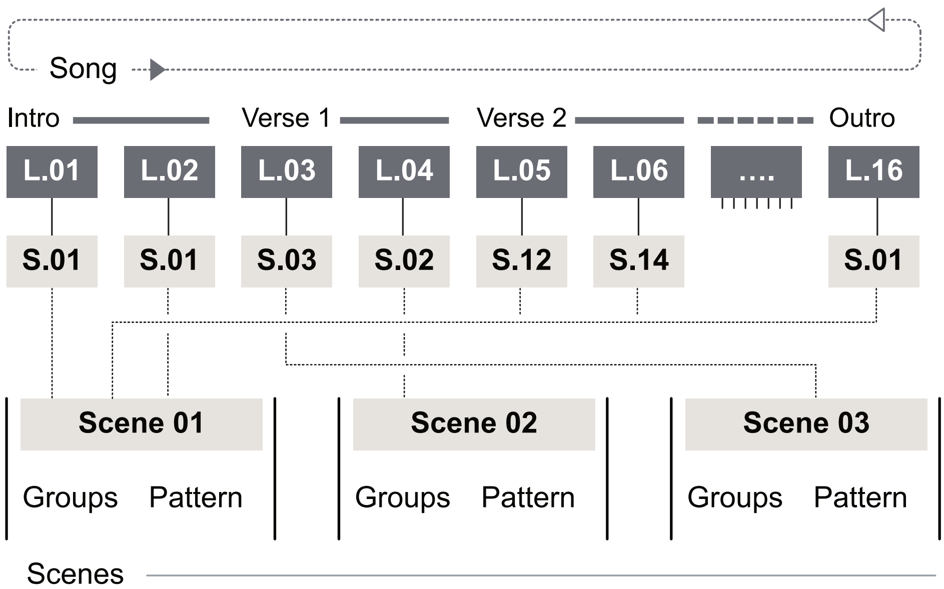
A word of warning when working with scenes and patterns. It is quite easy to get the pattern order out of sync with the Scene + Pattern. Take care when committing scenes and which pattern number is used. Sometimes useful to take written notes to avoid confusion later. Always try to keep the numbers in sequence for patterns in a scene.

NOTES

7.4 Song Mode

Each project has a song where scenes can be organized according to their 'song position' in a sequential arrangement. This chains the scenes to play in a complete end-to-end song composition. Up to 99 scenes can be captured into the song to play in order. Each L.01 - L.99 song position slot (The 'L' represents what… Layer? Line? - go figure!) can hold 1 scene.

Example Song Structure

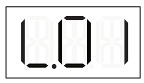


A song line slot can be assigned once scene. Multiple scenes can be applied across the duration of the song and a scene can be repeated into multiple song slots.

Hold [Main] + Press [Enter]

Screen Indicators

Song Mode
Open song editor



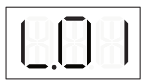
Opens song editor

Song Position - Slot 1

Hold [Main] + Press [Play]

Screen Indicators

Play Song
From any mode

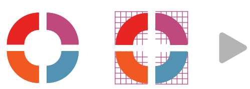


Plays the song from the start

Play Song

Song Playback Position

While playing a song, the 'Y Level' screen indicator will represent the progress of each line scene in real time as its played. The 'X Level' indicator will represent the progress of the entire song.



Full Song *Song Line*
Progress / Position Status

NOTES

The following commands operate in song mode.

Add Song Slots
Extend number of slots

Hold [Shift] + Press [A]

SHIFT + A

In song mode.

Screen Indicators

L.03

Song Position - Slot 3

Assign a Scene
Select scene for the song slot

Hold [Shift] + Press [-] or [+]

SHIFT + − +

Navigate to a scene to assign to the song

S.03

Assign to current slot

Cut Song Slots
Delete slots

Hold [Shift] + Press [C]

SHIFT + C

Delete current slot

L.03

Song Position - Slot 3

Insert Song Slots
Extend number of slots

Hold [Shift] + Press [D]

SHIFT + D

Insert from current position

L.03

Song Position - Slot 3

Navigate Song Slots
Cycle through existing slots

Tap [-] or [+]

− +

L.01

Song Position - Slot 1

Play / Pause Song
From current slot - in editor

Press [Play] Press [Play] again

PLAY PLAY

Play *Pause at current slot start*

L.04

Play Song from Location

Play Song
From start - in editor

Hold [Shift] + Press [Play]

SHIFT + PLAY

Play when in the song editor

L.01

Play Song from Start

NOTES

The song mode workflow follows the commands stated. It is important to have the scenes created and recognised before building the song. Here is where taking mental or written notes helps when creating scenes. Also its good practice to have a clear view or notes options of the intended song arrangement such as Intro, Verses, Chorus, Drops, Build ups etc. It is highly recommended to create a song as the last part of the production process.

CREATING A SONG

1. Select song mode, Hold [Main] + Press [Enter] pad. The display will show, L.01 if this is the first song built. Note that each project has one song. The ‘L’ in the display signifies song mode.

2. Add a series of song lanes ready for assigning scenes. Hold [Shift] + Tap [A] for each slot to add. The display will update to show the added song slot, for example L.02, L.03 etc.

3. Navigate to the slot in the song to assign. Normal workflow would start at L.01. Use [-] and [+] buttons to navigate back and forth in the song editor.

4. Assign a scene to the selected slot / line. Hold [Shift] + [-] + [+] to navigate through the available scenes. Whichever is selected will be assigned to the currently selected song slot. A scene can be assigned to multiple slots. For example, assign Scene 01 to Song slot L.01 and L.02.

5. Continue to iterate through the song to assign more scenes and build a full composition. There are editing options when working in the editor.

    - Hold [Shift] when a song slot is selected to view which scene is assigned.

    - To cut or delete the currently selected song slot line, Hold [Shift] + Press [C]. The slot will be deleted and others moved up.

    - To insert a song slot line at the currently selected position, Hold [Shift] + Press [D].

NOTES

PLAYING A SONG

1. To play the song for the current project from any mode in the KOI, Hold [Main] + [Play]. The Song will play from the start.
    - Playback will start from the first slot and the play symbol flashes.
    - The X Level icon represents the playback duration of the song. The icon will advance clockwise in time with the playback position.
    - The currently played song slot is displayed. In addition the Y Level icon represents the slot playback scene duration and will advance in time with each scene playback position.
2. To play a song from within the song editor, Press [Play]. The song will start playback from the currently selected song slot. To play from the start Hold [Shift] + Press [Play]. The song position will be displayed.
3. Press [Play] while playing to pause playback of the song at the current slot. Position. Restarting play by pressing [Play] will continue from the start of the paused song slot.

8

Effects & Performance

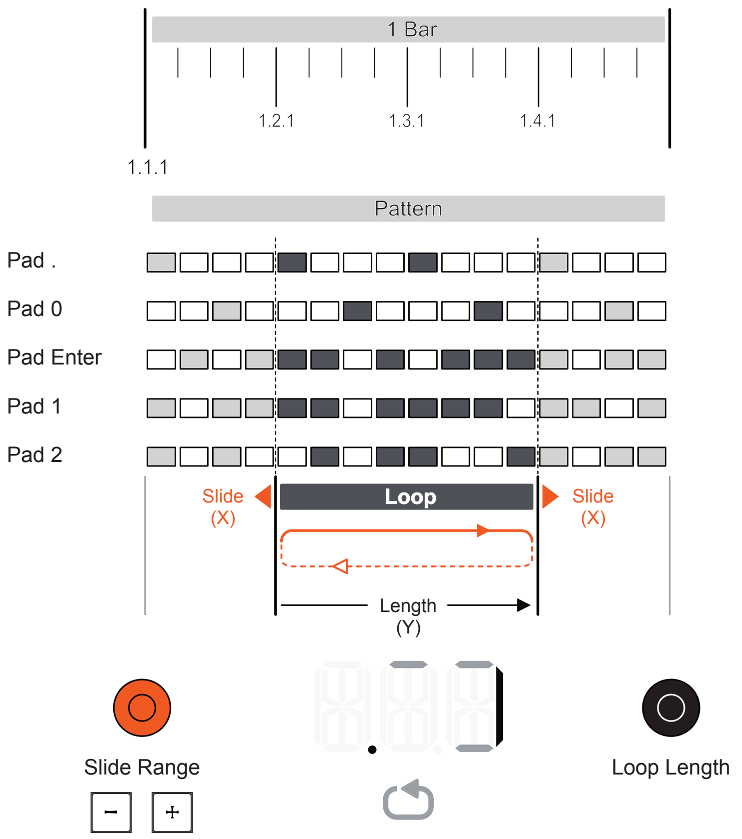
The KOII is a great device for playing out live, delivering dynamic performances and improvisations. There are many features that contribute to the live performing such as the loop function, the time / pitch variation and punch in effects. In addition a number of effects are available which can be used in the production of tracks. Each group can send audio to one of the master effects. These are configured in the FX section. Also the line in audio can be routed to the master effect section. The master effects can be selected from the delay, reverb, distortion, chorus, filter and a compressor. This compressor is independent to the master output compressor which combines all group audio and glues the entire mix together. The output compressor should be the last element to set up when finalising an arrangement and production. A MIDI Sidechain option is also available through the output section. Only one effect can be selected in the master FX section. Send effects allow control of the amount of audio from the group while the main, unaffected audio also remains routed separately. The master effects are ideal for developing the sound as part of the production process, adding character and tone. Consider automation on the sends to apply effects at various points to add emphasis and accents. In addition a set of on-the-fly effects also exist triggered from the pads. These are used more for live performances and improvisations to add variety and interest. Punch-in effects are triggered manually and can be combined with other punch-in effects and functions such as the loop feature to expand the audio performance options even more.

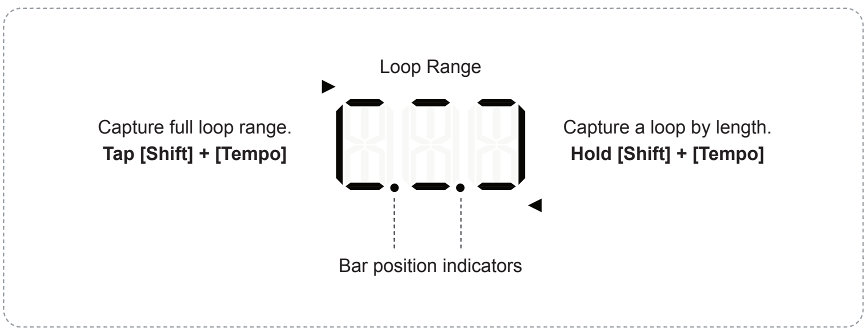
NOTES

8.1 Looping Patterns

The KOII Loop function allows capturing and replaying cyclic pattern sections. These are pattern elements that repeat and loop on playback. Loops help to create variations and allow on the fly ad lib performances and improvisations.

Loops are captured and played as a snapshot from the pattern when playing and are tempo locked including when sliding the loop. Loops are a maximum of 1 bar in length. Loops cannot be recorded into a pattern but can be used with punch-in effects.





NOTES

Loops should be considered more of a performance effect and with practice can form part of a live performance and ad-libs. They are used when playing an existing pattern.

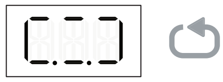
Capture Loop
Full Loop - 1st Selection

Press [Shift] + Tap [Loop]

SHIFT + TEMPO / LOOP

Pattern playing or stopped

Screen Indicators



Entire Loop

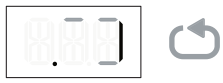
Capture Loop
Loop Length

Hold [Shift] + Hold [Loop]

SHIFT + TEMPO / LOOP

Capture length based on time buttons held

Screen Indicators



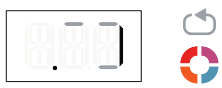
Loop Range & Current Position

Adjust Length
Loop Length Change

Turn (Y)

While Loop is Active

Screen Indicators



Loop Range & Current Position

Slide Loop
Shift Pattern Loop Section

Turn (X)

While Loop is Active

Screen Indicators

Loop Range & Current Position

Exit Loop
Exit loop immediately

Press [Tempo]

TEMPO / LOOP

Exit Loop
Exit loop on Next Bar

Press [Main]

MAIN / COMMIT

NOTES

CAPTURING AND PLAYING A PATTERN LOOP

1. Ensure a Project is selected with patterns to play. Loop will act on all groups, capturing a snapshot of all patterns to repeatedly play back. Maximum loop length is 1 bar.

2. To capture a Loop, options are:-

    - While the playback is stopped, Press [Shift] + Tap [Loop]. The full loop length is captured from the pattern start when first applied.

    - While playing, Press [Shift] + Tap [Loop]. The full loop length is captured and playback remains in time with the pattern and the point of capture.

    - While playing, Hold [Shift] + Hold [Loop] for the duration of the loop capture. The loop length captured is based on the duration that the buttons are held up to the maximum length.

3. Once a loop is captured playback will continue playing in time, but will only play the loop captured. The display will show the loop icon and also will display the captured range and the current playback position. The lower left dot in the loop display indicates the start of the bar.

4. The [Play] button can stop or start playback of the loop.

5. To change the length of the loop, Turn (Y) Black knob. This will be reflected in the display and while playing will offer some interesting characteristics to the audio played.

6. To change the section of patterns contained in the loop, Turn (X) Orange knob or Press [+] [-]. This will be reflected in the display and while playing. The looped section will slide back or forth from the original position within the pattern. The tempo is synchronised while doing so.

7. To exit loop mode:-

    - Press [Tempo]. This will exit immediately and continue playing the original pattern.

    - Press [Main]. This will exit the loop at the end of the bar and then continue playing the original pattern.

8. Once exited, the captured loop can be restored once again. Press [Shift] + Tap [Loop] to switch back to the captured loop playback.

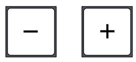
8.2 Variable Speed & Pitch

This is a feature normally associated with DAW's, for example Varispeed is a feature associated with the Apple Logic DAW where the speed and pitch can be changed live. The KOII has a similar feature used to add interest to performances that is often overlooked. This can add some fun, interest and variation to a live performance.

USING VARIABLE SPEED/PITCH LIVE

1. Select [Main] to ensure Main mode is selected. The display will show 'Main'.
2. Play the patterns in the scene to apply the effect. Press [Play] or [Shift] + [Play] to play from the start.
3. While playing, Press [-] to lower the speed and pitch for the project. This is fairly subtle but can be used punch-in style. The effect is removed when the button is lifted.
4. While playing, Press [+] to increase the speed and pitch for the project. This is fairly subtle but can be used punch-in style. The effect is removed when the button is lifted.
5. Playback remains in sync.

When in [Main] Mode, Press [-] or [+] to punch in the speed and pitch effect.



8.3 Punch-In Effects Overview

Punch-in effects refer to the effects that can be triggered temporarily on and off while a track is playing. These are normally used as performance features. Lower part of each pad is pressure sensitive, affecting the punch-in effect sound. Therefore not only the effects selected have an impact but also how the pad pressure is applied too.

The KOII has 12 punch-in effects. There is no official TE reference for the punch-in effects, so the headline names and descriptions are based on user observed behaviour. Use [FX] + [Pad] to activate one or more punch-in effects.

Summary of Punch-In Effects

Headline	Pad	Description
Random Pitch	.	Stepped pitch effect. Pressure affects the randomisation of the steps.
Sample Swap	0	Swaps the pad sound sample within the group for the punch-in duration. Triggered pads will be lit.
Ratchet	Enter	Granular ratchets of the audio buffer. Pressure of pad affects the size and therefore ratchet length.
Beat Repeat	1	Beat repeat type effect. Latches and repeats a short audio buffer snapshot at the point pad is pressed.
Tape Stop	2	Tape stop slowdown style effect. Speed determined by pressure the pad is held. Also affects MIDI clock out.
Modulated Filter	3	A filter style effect with LFO style cutoff modulation. Modulation speed by pad pressure.
LPF	4	Low pass filter. Cutoff is based on the pressure applied to the lower pad button.
HPF	5	Low pass filter. Cutoff is based on the pressure applied to the lower pad button.
FX Sender	6	Controls the send amount to the current master effect. Pressure on the lower pad determines the amount.
Tremolo / Gate	7	Introduces a gated style effect, dropping audio at sequenced beat points. Frequency set by pressure applied to lower pad button.
Pitch Bender	8	Pitched effect that systemically varies down and up the audio pitch based on pressure applied to lower pad.
LoFi	9	Bit crusher / sample rate reduction. Amount applied is based on the pressure applied to the lower pad

With many Teenage Engineering devices, a lot of the algorithms and configurations for the effects are often kept under the hood. Little official information is provided by TE in order to keep that edge of fun, surprise and mystery. The KOII is no different.

NOTES

8.4 Punch-In Application

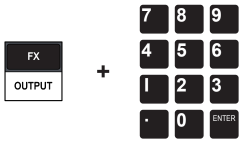
Punch-in effects are normally used for live performances, triggered manually and especially polarised to certain track points or beats. They are used less so on sound productions and when developing fixed song arrangements. Pressing and applying variable pressure to the lower part of the pads controls elements within each of the triggered punch-in effect.

Punch in effects are applied to all groups, manually triggered while playing live and cannot be recorded.

Hold [FX] + Hold [Pad]

Screen Indicators

Trigger Effect
Punch-In



FX

Hold [Pad] for duration of effect

Animated Display

TRIGGERING PUNCH IN EFFECTS

1. Ensure the tracks are playing, press [Play] or [Shift] + [Play].

2. While playing, Hold [FX] + Hold [Pad] for the respective punch in effect to apply. Note that multiple effects can be applied simultaneously by holding several pads.

3. The effect is applied and the display will animate. The [FX] button can be lifted and the effect will remain as long as the [Pad] buttons are held. The animation will not be displayed in this state.

4. Applying variable amounts of pressure to the lower part of the pad button when activating a punch-in will also vary the effect. The application outcome is dependant on the punch-in effect triggered. Use the pad pressure sensitivity as a modulator. Pads are only pressure sensitive on the lower half of the pad button.

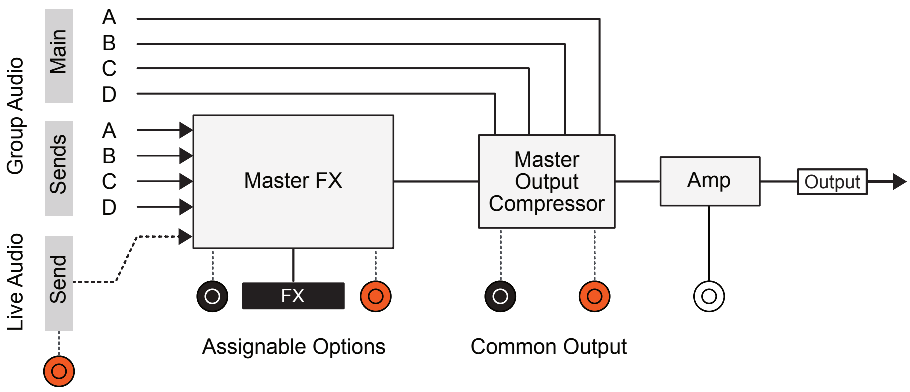
5. The effect will continue until all of the buttons are lifted.

NOTES

8.5 Master Effects Overview

Master effects refer to the effects that can be configured into the main output slot and groups can then ‘send’ audio to these effects. Also live audio can be directly routed to the master effects. The Main mix then is a combination of the affected and original audio. One Master effect can be configured from the available options. In addition a common Master Compressor will operate at the final master output stage to sum and glue all the audio before the final output.

The KOII has 6 selectable master effects.



Assignable Master Effects - 1 to be assigned to the FX Slot

Effect	Orange (X)	Black (Y)	General Description
DLY Delay	Length. Time between repetitions.	Feedback. Affects the number of repetitions.	Classic delay, repeating echo style effect.
RVB Reverb	Length. Replicates room size.	Colour. Filters high/low frequencies.	Emulates room space and its reflections and ambience.
DST Distortion	Drive. Overdrive the audio to distort.	Colour. Filters of high/low frequencies.	Generates audio distortion and resonant filtering to create grunge and dirt.
CHO Chorus	Modulation. Rate of modulation applied.	Feedback. Audio fed back and ring mod.	Chorus modulates the audio to spread and thicken the overall sound
FLT Filter	Cutoff. High or Low cutoff frequency point.	Resonance. Boosts the freq at cutoff point.	Carves out frequencies in the low / high range to affect tone.
CMP Compressor	Drive. Controls loudness in.	Speed. How fast the compressor reacts.	Controls and tames audio dynamics to create punch, loudness and impact.
OFF			Effects OFF

Master Output Compressor - Common effect on output

OUT Compressor	Drive. Pronounced and impactful compression.	Speed. How fast the compressor reacts.	Glues and sums the entire audio output. Controls dynamics and loudness.

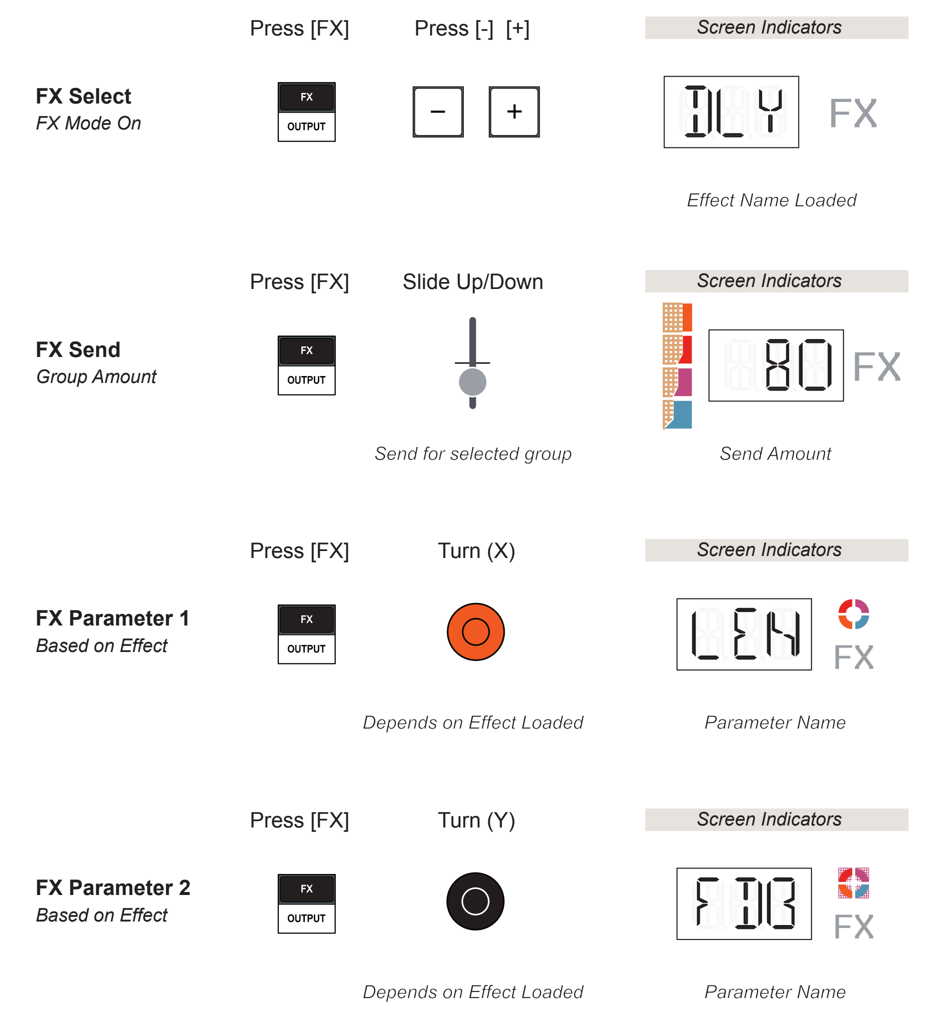
NOTES

8.6 Master Effect Application

Master effects are normally used sound design and to develop full productions and arrangements. There are 6 master effect models. One can be assigned to the master effect slot. Additionally a common master compressor then follows in the audio chain.

Master Effects

The 6 master effects concept in the KOII is based on the Send/Return model where each group sends an amount of audio to a single master effect. The effect output is returned and combined with the main audio.



The 'FX' Mode should be active to edit Master Effect information. The send amount is also available directly using [Fader] + Pad [6] and then send by adjusting the fader. The active effect is also used for the Punch-In 'FX' Effect, Pad 6.

NOTES

Master Output Compressor

A single, common master output compressor will combine the full mix at the end of the chain, prior to the amp. This sums and combines all audio from the effects and main group outputs. The compressor is used to glue together and control the dynamics of the entire mix.

Press [Shift] + Press [Output]

Screen Indicators

Compressor Select
Master Output

Output Compressor

Turn (X)

Screen Indicators

Drive
Compressor

Parameter Name

Turn (Y)

Screen Indicators

Speed
Compressor

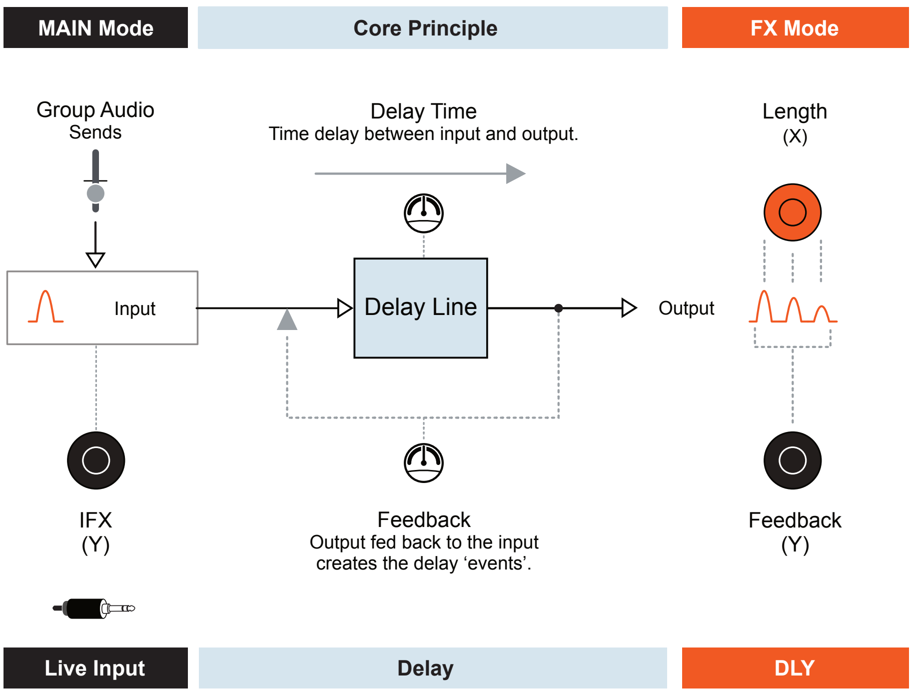
Parameter Name

NOTES

8.7 Delay

A delay is a classic effect used to create echoes and repeats of the original sound. This adds sonic interest but can also thicken sounds and be used to fill gaps in a mix. Basically a delay takes an audio input and delivers it to the output in a timed fashion. Feedback introduces more of the audio from the output and into the input creating repetitions.

The KOII delay has two controllable elements. The length which affects the time between repetitions and feedback amount which controls how many repetitions are generated.



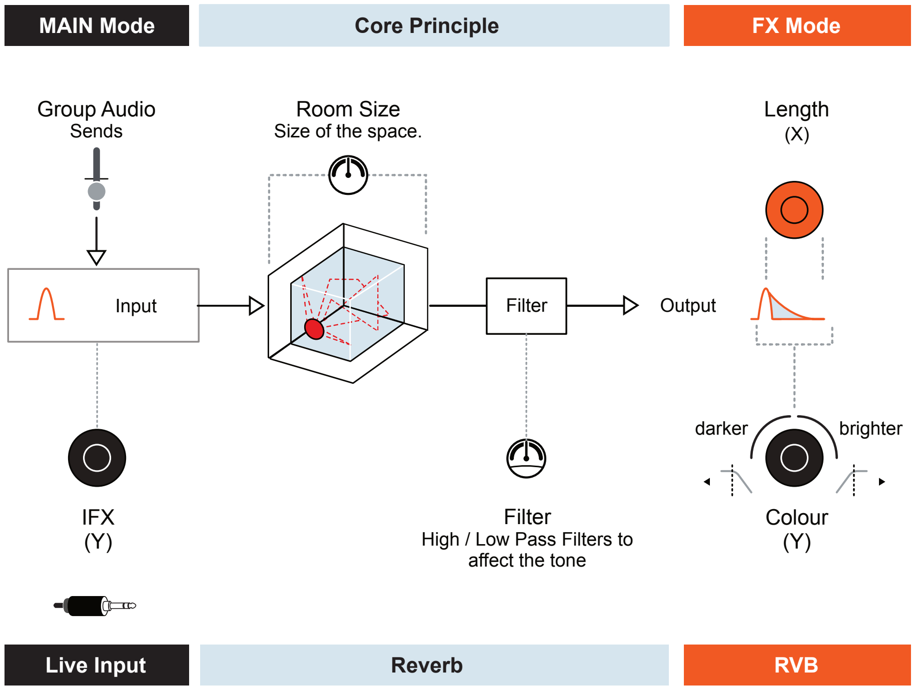
The live input, if connected to an external audio device can also be routed to send audio directly to the master effect section.

NOTES

8.8 Reverb

A reverb is another common effect which represents the characteristics of sound behaviour within a room. The sound will behave differently in a large room versus a small room, whether elements absorb sound or reflect sound. This reverberative nature can be used to create ambience and intimacy or expand and extend when designing sounds.

The KOII delay has two controllable elements. The length which emulates the room size and expansiveness of the reverberations. Colour adjusts the high and low pass filtering to affect the tone to be brighter or darker.



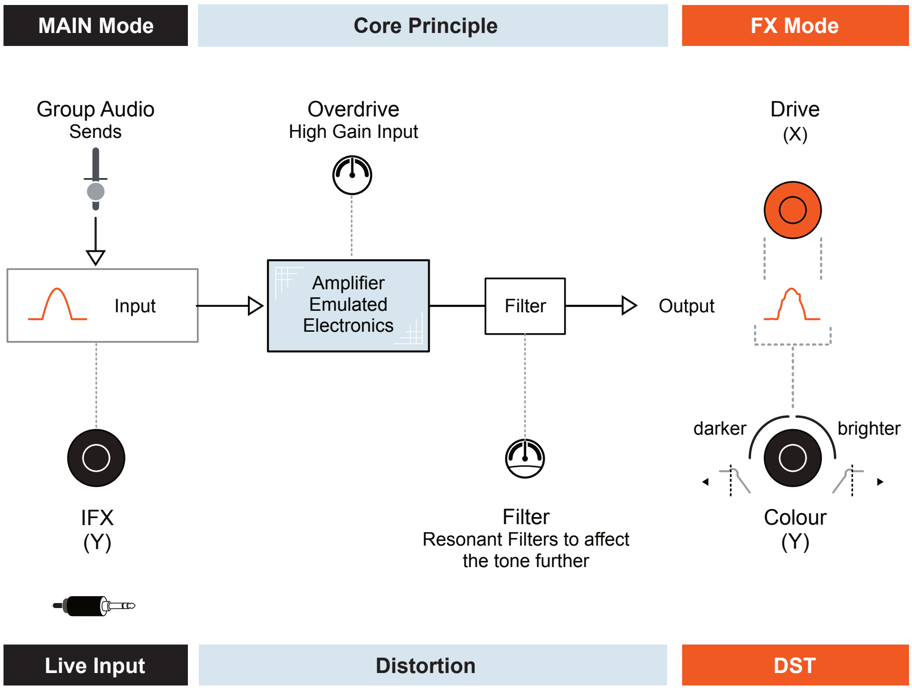
The live input, if connected to an external audio device can also be routed to send audio directly to the master effect section.

NOTES

8.9 Distortion

There are many effects in the general distortion family. The KOII distortion operates based on the classic model of driving the electronics of an amplifier's audio input hard, distorting the speaker audio output. Use heavy distortion to add a grungy feel to the mix or subtly to give a gentle yet slightly rough texture to a sound.

Distortion has two controllable elements. Drive introduces the overdrive, an emulation of an amplifiers overloaded input. Colour adjusts the high and low frequencies and adds resonance for an even more aggressive tone and feel.

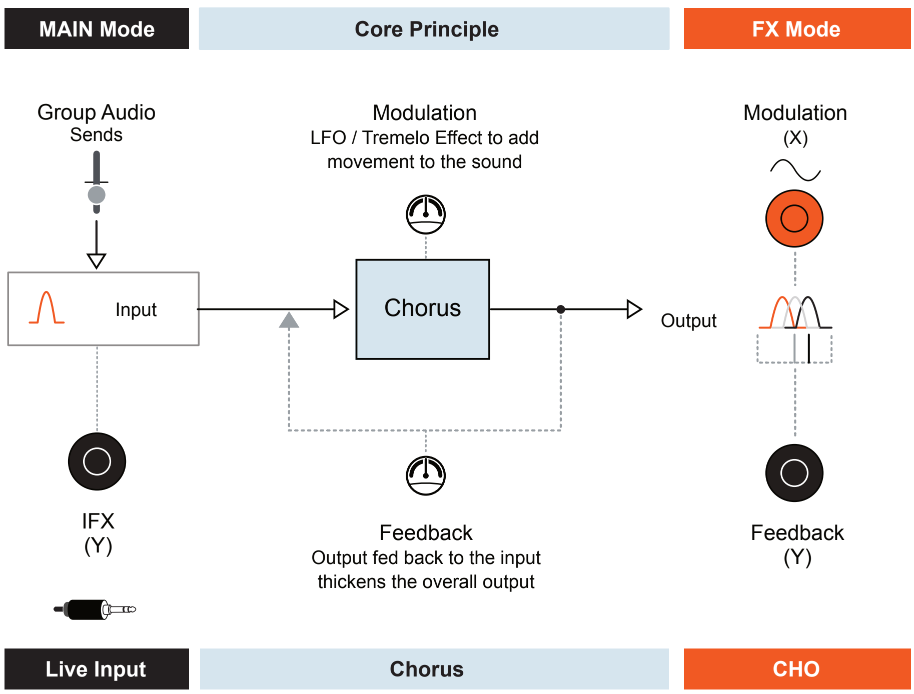


The live input, if connected to an external audio device can also be routed to send audio directly to the master effect section.

8.10 Chorus

A chorus has similar characteristics to both a delay and reverb. The key difference is that a chorus will introduce a spread and thicken a sound through duplicating, delaying and modulating the audio. Think of a choir with multiple singers all slightly different in pitch and time.

The KOII chorus has two controllable elements. Modulation controls the rate when adding movement to the audio. Feedback sets how much output audio is channeled back into the input in order to thicken and spread the sound.



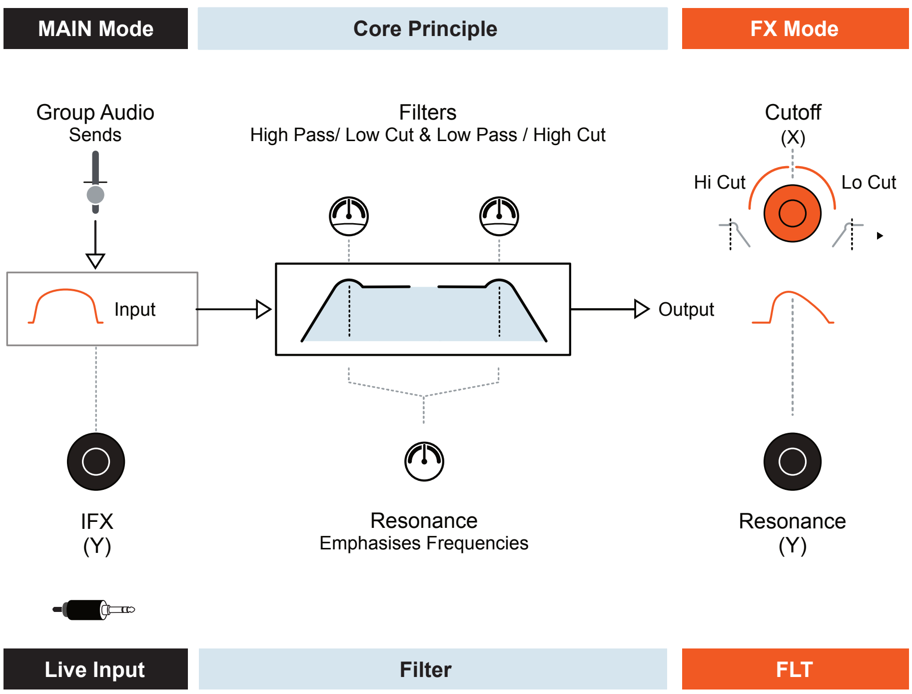
The live input, if connected to an external audio device can also be routed to send audio directly to the master effect section.

NOTES

8.11 Filter

A filter is designed to attenuate frequencies above or below a defined cutoff point. This helps to shape the sound, carving out frequencies to set the tone. Resonance boosts and emphasises the frequency at the cutoff point. Filters are useful in developing the character of a sound.

The two controllable elements in the KOII are the cutoff frequency and resonance. A high cut is performed when rotating counter clockwise, left and low cut when rotating clockwise - right. The 12’O’Clock central point is neutral. Resonance will boost the frequencies at the cutoff point.

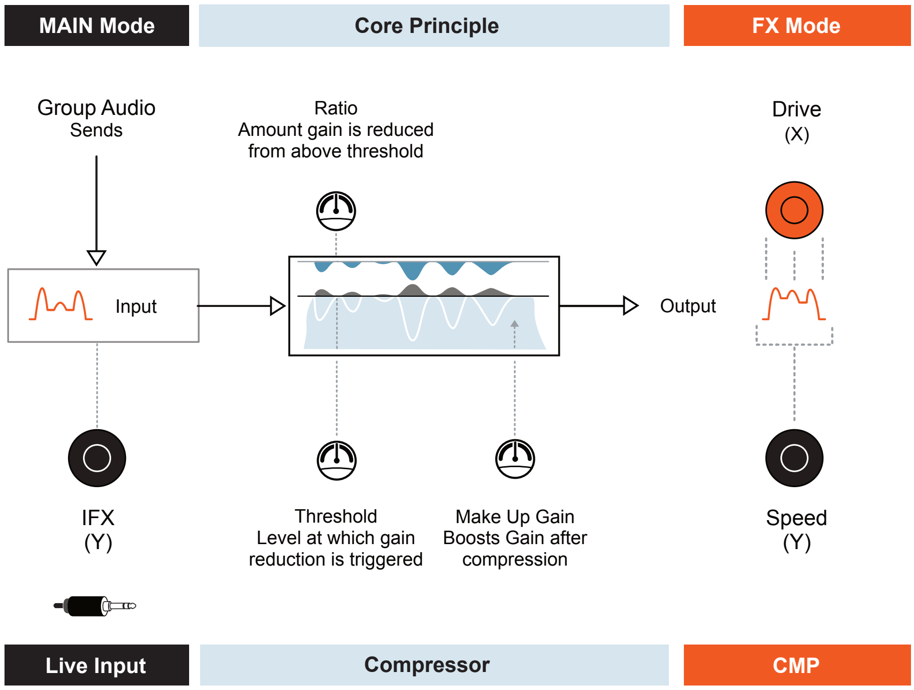


The live input, if connected to an external audio device can also be routed to send audio directly to the master effect section.

8.12 Compressor - Master Effect

A compressor is designed to control peaks and tame errant audio dynamics. This is performed by attenuating peaks above a threshold by an amount governed by a ratio. The overall gain can then be increased to add loudness and punch.

The KOII has several compressor parameters integrated into the algorithm which are presented by two user controls. Drive controls how loud the input audio is while Speed controls the reaction of the compressor. Adjust these as a pair to affect the impact and to nurture the loudness, punch or gentle dynamics.



The live input, if connected to an external audio device can also be routed to send audio directly to the master effect section.

Note that this refers to the Master Compressor, accessible as a send effect from each group. There is another common compressor that follows the master effects to control the entire mix called the Master Output Compressor.

A compressor is used less frequently as a send effect. However the technique of compressing audio in parallel to the original audio is a recognised technique in audio production. This is often called Parallel or New York Compression.

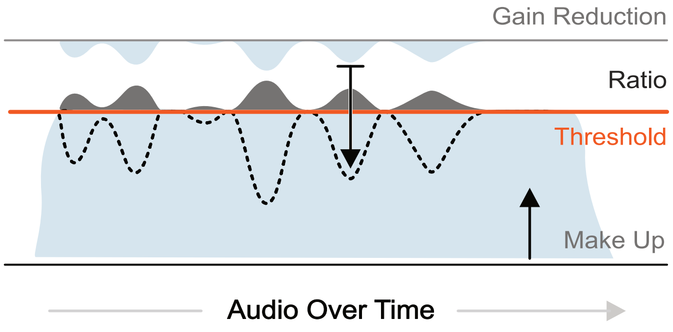
NOTES

NOTES

8.13 Compressor - Master Output

The compressors in KOII have simplified parameter set with many generic compressor elements built into the algorithm. Ratio, Threshold, Envelope, Make Up Gain etc are typical generic parameters. These are integrated into a drive and speed controls.

Generic Compressor Concepts



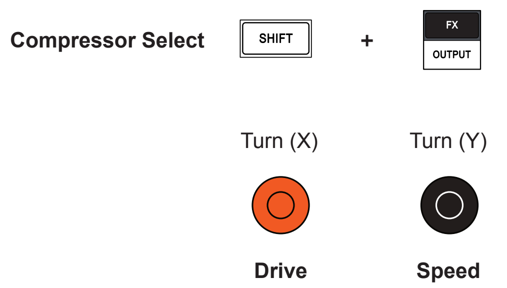
KOII Master Output Compressor

Drive controls loudness and emphasises and amount of the compression applied. Speed controls the reaction of the compressor and can control the punchiness and impact or tame things to a more subtle level of control.

The main purpose of the output compressor is to glue together all of the main audio along with the effect output audio, bringing the dynamics into a smoother plane and controlling perceived loudness and punch.

The group sound levels and final mix should be completed first. Adjusting the output compressor being the last step in developing a production.

It is advised to iterate between adjusting the drive and speed controls in small incremental steps while listening to the overall sound by ear. There are no precise metering functions to measure or record the compressors behaviour, but after all its how it sounds that truly counts.



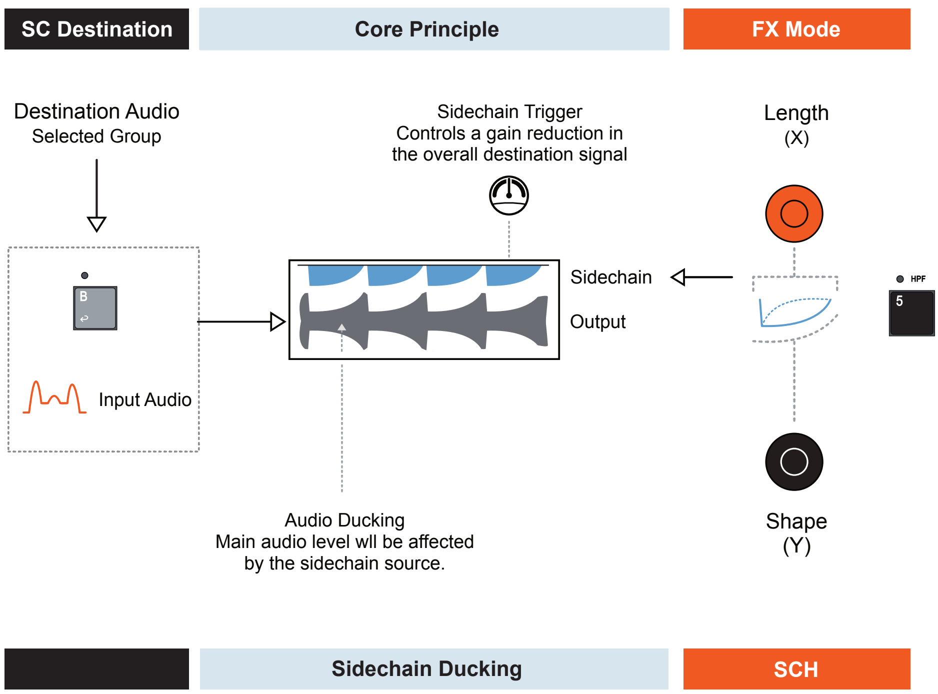
Set the output compressor as the last stage of developing a sound and production.

Adjust the controls incrementally by ear until the output sounds dynamic, loud and good.

8.14 MIDI Sidechain Compression

Sidechaining allows the automatic adjustment of the compressor volume level of a groups sounds by the control from another group as a source. This is called ducking and can create interesting creative sounds and rhythms. The sidechain option for the KOII operates from MIDI note messages as opposed to the sound itself as in traditional methods.

The KOII sidechain option is found in the main output settings along with the main compressor output.

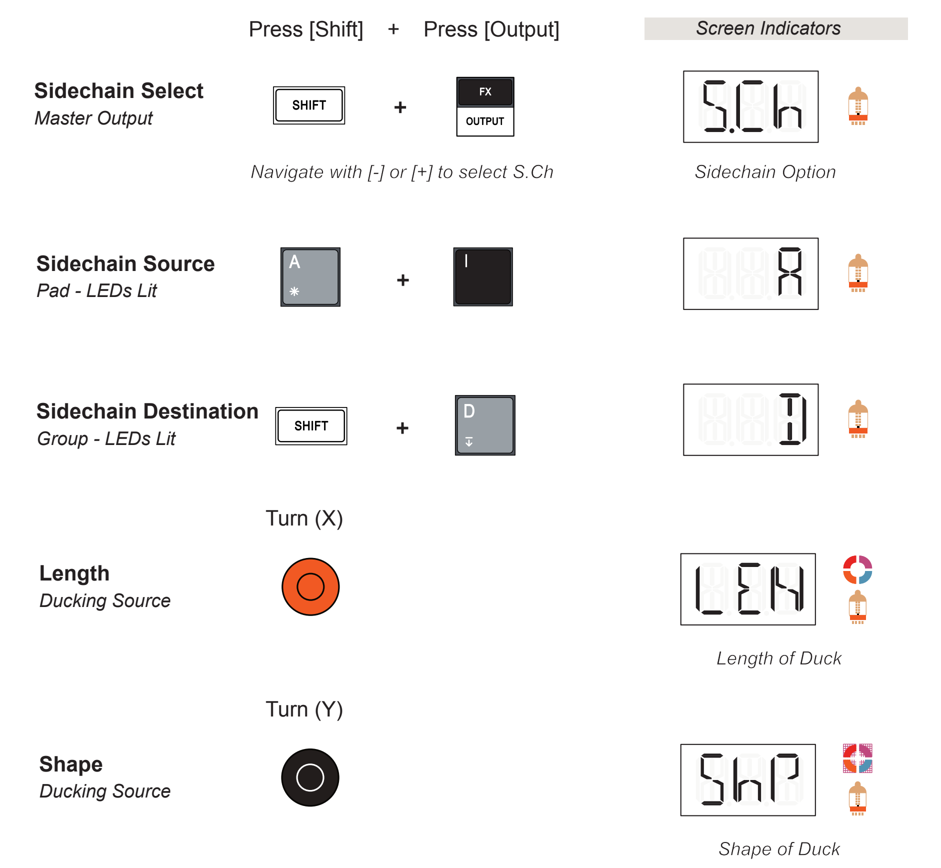


The source pad sound will act as the MIDI trigger to initiate a gain reduction. The note message is what triggers the sidechain not the audio sound itself. This is a benefit from using a MIDI sidechain model. This is selected as a pad sound. The destination will be based on one or more pads groups A,B,C, D.

NOTES

APPLYING A SIDECHAIN DUCK

1. Identify the sound source - the trigger for the ducking, normally a drum trigger. This is selected from pad sounds. Identify the destination - normally a more sustained audio passage, identified as a group.
2. Select sidechain mode. Hold [Shift] + Press [Output]. Navigate using [-] or [+] to select the sidechain function, S.Ch.
3. Assign the trigger source. Hold Group [A] - [D] + [Pad] for the selected sound to act as trigger. The LED of the pad will be lit.
4. Set the destination. Hold [Shift] + Group [A] - [D] for the selected sound groups to be affected by the trigger. The LED of the group pad will be lit.
5. Play the scene, pattern or song. Iterate between steps 6-7 to get the right balance of punch and release.
6. Turn (X) Orange knob to adjust the sidechain source length. The value is shown with the X-Level icon.
7. Turn (Y) Black knob to adjust the sidechain source shape. The value is shown with the Y-Level icon.



9

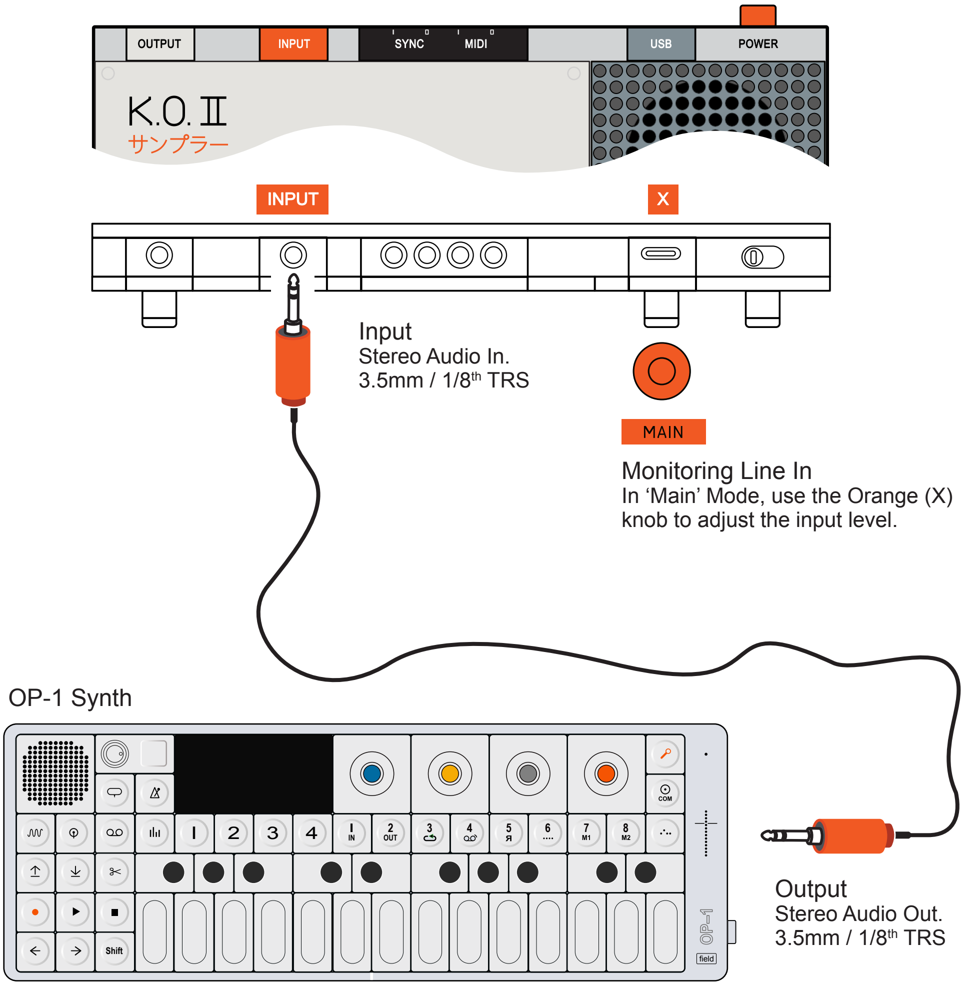
Connecting Gear

The KOII has the both audio, MIDI and Sync input and outputs. MIDI short for Musical Instrument Digital Interface is available across the USB or TRS connections based on the 3.5mm Stereo Mini jacks located at the top of the device. Also located at the top are 3.5mm Sync and Audio in and outs. There are no cables for these connections supplied in the box with the KOII device as it will also depend on what you are connecting to. The audio input and outputs are both stereo TRS, the output doubling as the headphone connection. MIDI In and Out are also 3.5mm TRS and would need to connected with a compatible configuration to other gear based on the Type ‘A’ TRS MIDI standard. Often this is by using a 5 Pin to 3.5mm adapter but also can work with a straight stereo TRS cable if the sending or receiving gear allows. Note that Type ‘A’ MIDI is the adopted standard and most common format but there are also other types i.e. Type ‘B’ used on some gear which are not directly compatible. This is a source for frustration and confusion in the desktop MIDI gear community. In addition, the USB connection can also transmit and receive MIDI. Finally the Sync options are based on the PPQN - Pulses Per Quarter Note Standard with configurable options. This is based on a TRS 3.5mm connection and is ideal for synchronisation between other KOII devices, TE Pocket Operators, Eurorack gear or any device which uses this format for its clock or sync. This section covers the typical configurations and connections. The KOII can receive mod wheel and pitch bend controls as well as external notes. Of course it takes two to tango and the send or receiving device also will needs some careful consideration regarding connectivity. Every eventuality and scenario cannot be covered but a guide to get started with expanding the set-up beyond the KOII alone is provided here.

NOTES

9.1 Audio In

Audio in is connected via the 3.5mm / 1/8th TRS Mini Jack. This is a stereo input but mono can also be used. A standard mono patch cable would be input on the left channel. The L/R can be changed when sampling. The KOII will automatically switch from Mic to Audio In when a physical input is connected on the 'Input' socket.



MONITORING INCOMING AUDIO

1. Press [Main]. The Main indicator is displayed.

2. Set the Line 'IN' level. Adjust the Orange (X) Knob. The display will show 'IN' on the alpha numeric display and the level is shown on the left quadrant level indicator. The actual combined pattern and input audio level is also shown real time.

3. Turn (X) fully counter clockwise to mute the external audio in.

4. Note that any other audio, i.e. from a pattern will also be heard if playing.

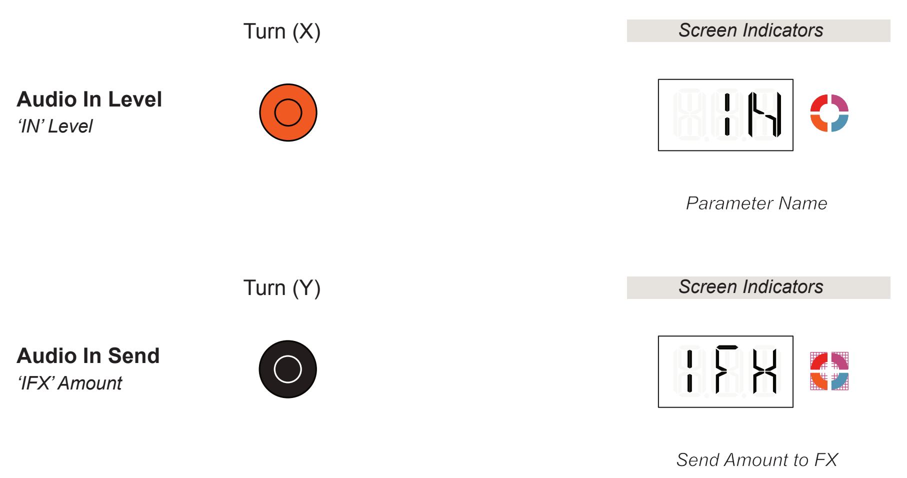
NOTES

Live In FX

A selected amount of the incoming audio input can also be sent, live to the FX section. The currently selected master effect will also process the audio. For example is the Delay - DLY is assigned, the live, incoming audio can be processed by this delay effect. This is performed in Main mode.

SENDING INCOMING AUDIO TO MASTER FX

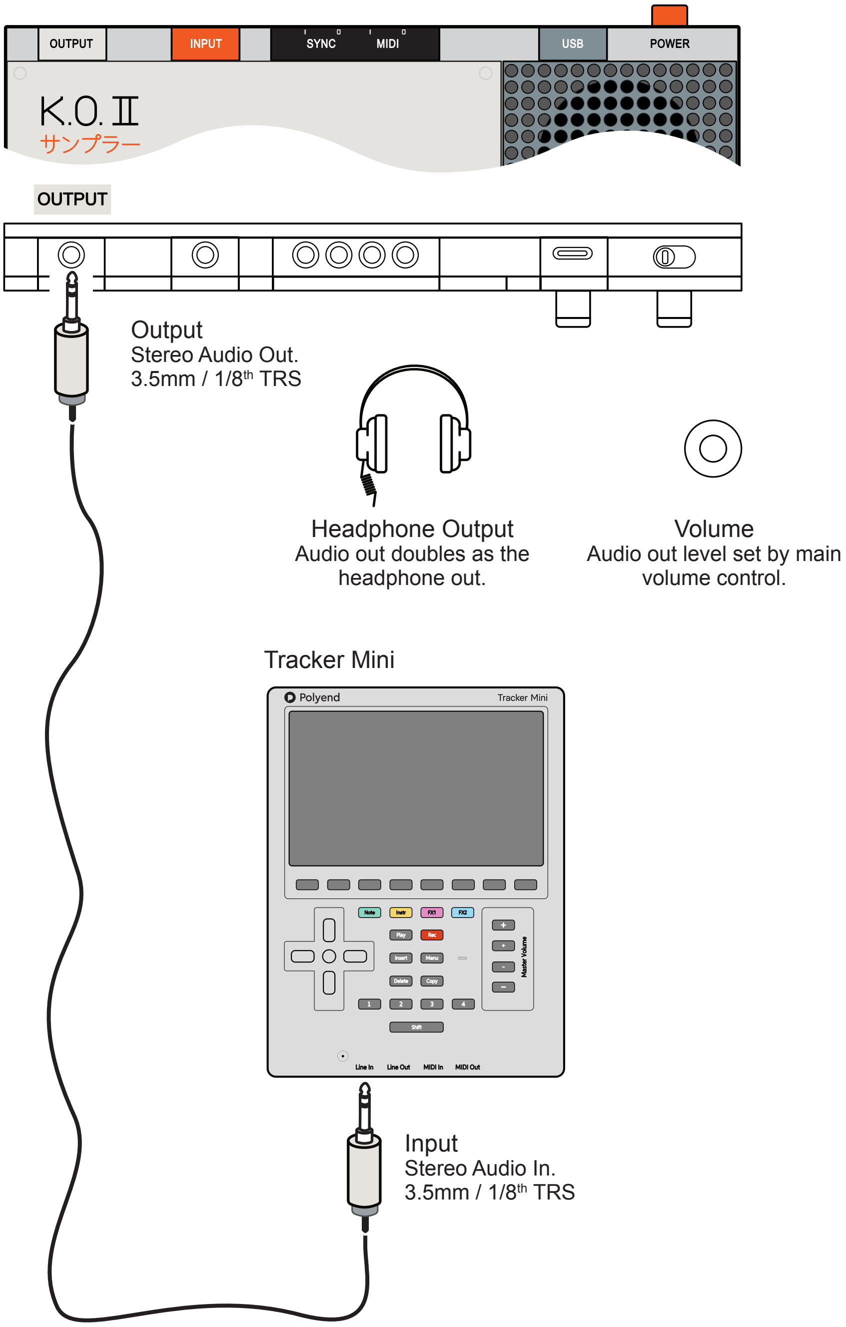
1. Select the desired effect for the master slot.
2. Press [Main]. The Main indicator is displayed.
3. Set the Line 'IN' level. Adjust the Orange (X) Knob.
4. To adjust the amount of audio sent to the FX, Turn Black (Y) Knob. The display will show 'IFX' on the alpha numeric display while adjusting. Also the amount is shown on the right quadrant level indicator.
   - Fully counter clockwise, not audio sent to the effect.
   - Fully clockwise, 100% of the audio is sent to the effect.
   - Maybe 25% is a good starting point.
3. Note that any other audio, i.e. from a pattern will also be heard if playing.



Line Input Specification: 3.5mm / 1/8th Mini Jack, Stereo, 24 Bit, Signal to Noise Ration 96 dBA, Impedance 6.5 kOhm, Analog Gain 0-12 dB, Max Level 8 dBU, 2 V RMS.

9.2 Audio Out

Audio out is connected via the 3.5mm / 1/8th TRS Mini Jack. This is a stereo output but mono can also be used. This can be connected to a variety of destinations including headphones, mixer, amp, recorder, effect processor or to another sampling or audio device. The output is the master out consisting of all groups, routed incoming audio and the master effects.



Line Output Specification: 3.5mm / 1/8th Mini Jack, Stereo, 24 Bit, Signal to Noise Ration 98 dBA, Max Level 5 dBU, 1.4 V RMS. Headphone compatible.

NOTES

NOTES

9.3 MIDI Implementation

MIDI implementation in the KOII may seem a little complex, especially based on the Teenage Engineering documentation. Here we will concentrate on the elements that are relevant to practical use rather than dive deep into the technical MIDI standard definitions.

The KOII functions which are relevant for MIDI are:-

MIDI Reference - Send / Receive

Function	MIDI Send	MIDI Receive	Comments. vs *MIDI Standards*
Clock	YES	YES	Synchronisation of clocks including the tempo BPM. KOII can send or receive a clock sync.
MIDI Channels	YES	YES	Channels 1-16 can be set in the system MIDI settings globally or assigned per pad which then takes priority.
MIDI Mode	Mode 1	Mode 1	Operates as Omni on, polyphonic. The setting to respond to all MIDI input channels is configurable.
Start	YES	YES	Transport Play. Manually triggered by Play button on KOII. Plays sequencer.
Stop	YES	YES	Transport Stop. Manually triggered by Play button on KOII.Stops sequencer.
Continue	YES	YES	Transport Continue. Continues sequencer from current position.
Position	YES	YES	Playhead position from start. KOII measures in Bars, Beats and Steps / Ticks.
Note	YES	YES	MIDI note information for playing chords and melodies.
CC	YES	YES	Continuous Control Change messages used to send parameter value changes.
Velocity	YES	YES	Transmitted with note on message.
Pitch Bend		YES	Responds to the standard pitch bend input. Does not send pitch bend.
Mod Wheel CC 1		YES	Responds to the standard modwheel on CC1 input. Does not send modwheel.
Bank Select CC 0	YES	YES	Control change message is only used for bank change using number 32.
Program Change PC		YES	PC messages will change the sample for the selected pad on the device. Use bank change to access all samples.

NOTES

9.3 MIDI Basic Settings

The KOII can transmit and receive MIDI with a number of system commands, notes and clock settings. For clock synchronisation over MIDI, the KOII can act as the primary device sending clock, transport, tempo or a secondary follower taking its sync from another device.

Global MIDI Channels

The KOII can operate with a global MIDI channel or pads can be assigned to a channel individually. Global channel assignment applies to the entire device in order to send or receive MIDI messages to / from another device. The pad channel assignment setting takes precedence.

Global MIDI Channels - System Settings [Shift] + [System] > MIDI > CHN

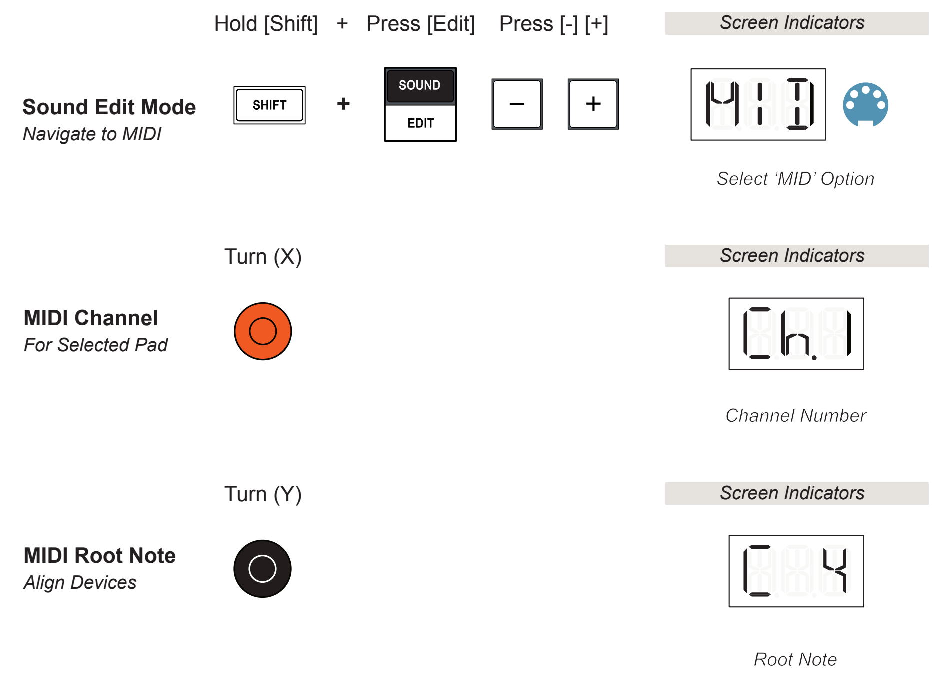
Option	Comments.
MIDI Channel All	Receive on all MIDI Channels and send on MIDI Channel 1 *
MIDI Channel 1	Receive & send only on MIDI Channel 1
MIDI Channel 2	Receive & send only on MIDI Channel 2
MIDI Channel 3	Receive & send only on MIDI Channel 3
MIDI Channel 4	Receive & send only on MIDI Channel 4
MIDI Channel 5	Receive & send only on MIDI Channel 5
MIDI Channel 6	Receive & send only on MIDI Channel 6
MIDI Channel 7	Receive & send only on MIDI Channel 7
MIDI Channel 8	Receive & send only on MIDI Channel 8
MIDI Channel 9	Receive & send only on MIDI Channel 9
MIDI Channel 10	Receive & send only on MIDI Channel 10
MIDI Channel 11	Receive & send only on MIDI Channel 11
MIDI Channel 12	Receive & send only on MIDI Channel 12
MIDI Channel 13	Receive & send only on MIDI Channel 13
MIDI Channel 14	Receive & send only on MIDI Channel 14
MIDI Channel 15	Receive & send only on MIDI Channel 15
MIDI Channel 16	Receive & send only on MIDI Channel 16

** Default*

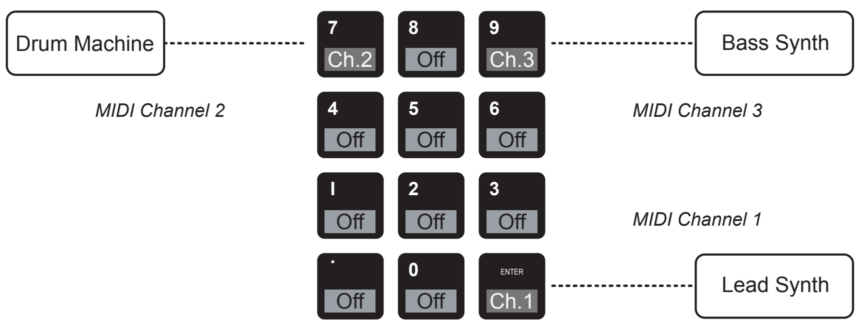
NOTES

Pad MIDI Channel & MIDI Root Note

The KOII can operate with a MIDI channel specifically assigned at pad level. Also the MIDI root note can also be set. The pad MIDI assignment takes priority when set.

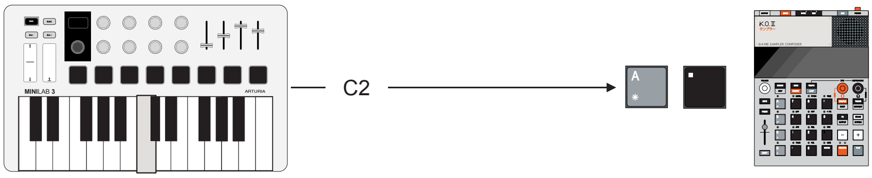


Example Pad Channel Configuration



MIDI Note Map Assignments

When using an external MIDI controller to control the KOII, a set of predefined note assignments can be used as a reference. This will always depend on the configuration of the connected device for perfect alignment. It is not uncommon to see C2 represented as C1 or C3 for example in some devices. The table below will act as a reference and starting guide. If notes don't match with your connected device, try experimenting by moving an octave range.



KOII Group	KOII Pad	Note	MIDI Std Note #
A	.	C2	36
A	0	C#2	37
A	Enter	D2	38
A	1	D#2	39
A	2	E2	40
A	3	F2	41
A	4	F#2	42
A	5	G2	43
A	6	G#2	44
A	7	A2	45
A	8	A#2	46
A	9	B2	47

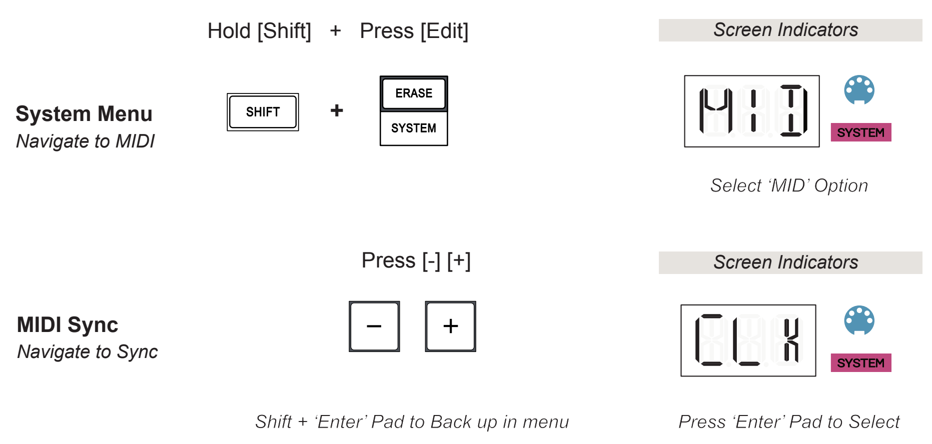
KOII Group	KOII Pad	Note	MIDI Std Note #
B	.	C3	48
B	0	C#3	49
B	Enter	D3	50
B	1	D#3	51
B	2	E3	52
B	3	F3	53
B	4	F#3	54
B	5	G3	55
B	6	G#3	56
B	7	A3	57
B	8	A#3	58
B	9	B3	59

KOII Group	KOII Pad	Note	MIDI Std Note #
C	.	C4	60
C	0	C#4	61
C	Enter	D4	62
C	1	D#4	63
C	2	E4	64
C	3	F4	65
C	4	F#4	66
C	5	G4	67
C	6	G#4	68
C	7	A4	69
C	8	A#4	70
C	9	B4	71

KOII Group	KOII Pad	Note	MIDI Std Note #
D	.	C5	72
D	0	C#5	73
D	Enter	D5	74
D	1	D#5	75
D	2	E5	76
D	3	F5	77
D	4	F#5	78
D	5	G5	79
D	6	G#5	80
D	7	A5	81
D	8	A#5	82
D	9	B5	83

MIDI Synchronisation

As well as communicating note information, the clock can also be assigned to send or receive control to/from a device. This will allow transport play, stop and tempo to be controlled by a primary lead for other devices to follow. This applies to the generic clock and is therefore set globally in the system menu.



Clock Options

Menu	CLK Option	Comments.
System > MID > CLK	OFF	Clock is not communicated by MIDI. This is the default setting in the KOII.
System > MID > CLK	IN	The KOII will receive a clock in on the USB or TRS MIDI Input. The KOII Clock i.e. for tempo, transport etc will be controlled externally from another device.
System > MID > CLK	OUT	The KOII will send a clock out through the USB or TRS MIDI Output. The KOII Clock i.e. for tempo, transport etc of other connected devices is controlled by KOII.

MIDI Clock synchronisation operates at 24 PPQN. This means that 24 pulses are sent per quarter note. Generally this is a MIDI standard so a clock will be synchronised between MIDI devices without too much setup needed.

The main clock configurations should concentrate on which device is the primary lead. This is typically set for only one device in a network. This will send the clock out. Other devices on the network will follow this lead. Multiple devices would be set as the secondary follower of the lead device.

NOTES

CONFIGURING MIDI

1. It is useful to map out and document or mentally assess the desired configuration of connected devices. Is the KOII controlling or being controlled? What is the clock lead? How many other devices will follow? Which need independent control? This task will always consider for the entire set up plus the other gear and not just the KOII alone.

2. Set the MIDI clock of the KOII based on the setup.

    - Hold [Shift] + [System], then [-] or [+] to navigate to ‘MID’ for MIDI.

    - Press [Enter] to access the sub-menu and the CLK for ‘Clock’ settings.

    - Press [Enter] to access the clock parameter options. Static text shows the current option and flashing text shows the available inactive options.

    - Select either, OFF, IN or OUT. When the desired option is in view press [Enter] to select and confirm as active.

    - Hold [Shift] + [Enter] to back up a level in the config menu if required.

3. The KOII can be configured as an Omni device where the KOII listens to ‘ALL’ 16 MIDI channels from an external device or a single global channel can be set . As an alternative the KOII MIDI Channel can be assigned per pad. This is useful when KOII is the lead, controlling external gear. This allows multiple devices to be controlled from the KOII.

4. Assign each device to a Pad and set its MIDI Channel. For example, Pad 7 will control a drum machine. This means that any pattern note for pad 7 will trigger the note out in or out of Key Mode.

    - Select the [Pad] to assign.

    - Hold [Shift] + [Edit] to open the sound edit menu.

    - Press [-] or [+] to navigate to ‘MID’ for MIDI. The MIDI channel will be displayed which is assigned to the selected pad. For example Ch.5. Generally this is set to ‘Off’. Off is advised for pads not used for MIDI.

    - Turn Orange (X) Knob to set the Pad’s MIDI Output Channel assignment between 1-16 or Off.

    - Turn Black (Y) Knob to set the MIDI root note.

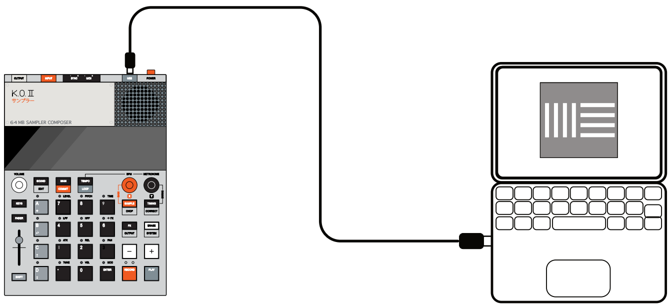
4. The KOII is now set for MIDI communications and can work with other connected MIDI Devices.

NOTES

9.4 USB MIDI In / Out

MIDI can be transmitted and / or received over the USB-C connection using a data compatible USB cable when connected to a USB Host. The KOII will operate as MIDI Device.

USB Device:
A USB device responds to any communication initiated by a host. A device can communicate for example with USB MIDI but may need to be managed from a Host.



USB Host:
A function that initiates, manages and controls communications across the USB network. For example a PC/Mac would act as a host for USB MIDI interfacing with the KOII.

The USB Interface is therefore good when working with a DAW, say Ableton Live or Logic and synchronising clocks and transport. USB would be a preferred connection when working with a PC / Mac connection. To use the KOII as a MIDI controller with another hardware synth, i.e. Play KOII notes to control another synth it may be better to use the 3.5mm MIDI output.

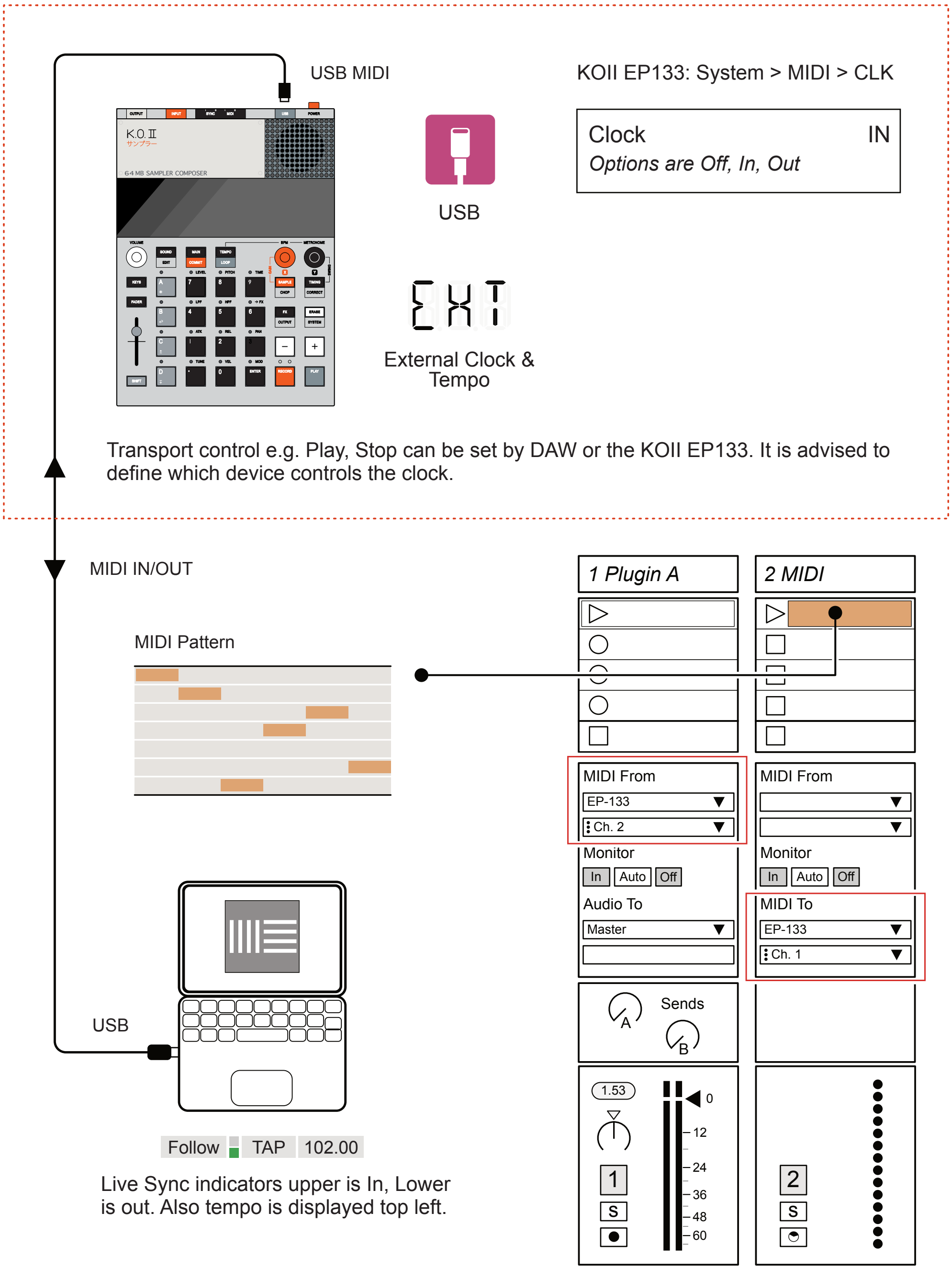
If connecting a MIDI Keyboard / Controller to the KOII using USB then it will need host capabilities. Consider using an intermediate hub that provides this function. This may also be useful to support the power supply to the MIDI controller as many use the USB connection also as it's power supply.

The KOII is a USB Device. Some other Teenage Engineering gear such as the OP-1 Field and the OP-Z have USB Host functionality. This makes interfacing the KOII with these devices easier.

NOTES

Example: KOII + Ableton Live in a MacBook Pro

The MIDI configuration must be set in the Ableton Live ‘Settings’. The KOII will be identified as the EP133. Also the KOII MIDI settings for channel and clock control. The KOII can be the primary controller, Ableton secondary follower or vice versa. Example shows Ableton as the primary clock, control lead.



MIDI settings in the Live preferences page will need to be set MIDI Ports, Input & Output. For this example Sync Out should be ON when controlling the KOII EP133. Ableton transport and clock will control the KOII EP133 device.

NOTES

ABLETON LIVE CONTROL OVER KOII WITH USB MIDI

1. Connect the KOII and the Ableton Live PC/Mac with a USB cable. This must be data compatible cable.

2. In the KOII, Set the Sync option.

    - Hold [Shift] + [System] to Open the settings.

    - Use the [-] & [+] to cycle through options. Select the MIDI Option ‘MID’. Press [Enter] Pad.

    - Use the [-] & [+] to cycle through options. Select the Clock Option ‘CLK’. Press [Enter] Pad.

    - Use the [-] & [+] to cycle through options. Select the In Option ‘IN’. Press [Enter] Pad.

3. In Ableton Live open the Setting menu. In the MIDI Ports, set the output ‘Sync’ on for the EP133. This is by checking the box. MIDI options are found in the sub menus; Live > Settings > Link Tempo MIDI > MIDI Ports.

4. At this point Live’s transport should control the KOII Play, Stop and Set Tempo. The KOII will display ‘EXT’ when an external tempo is recognised. Sync out (and in) and tempo are shown top left in Ableton Live.

5. The KOII is Omni Channel for receiving MIDI Notes. Any channel that Live sends on will be picked up by the KOII. Sounds only in the currently selected KOII group, in or out of Keys mode are triggered. It is good practice to keep a group just for MIDI in sounds with no internal patterns

6. In Ableton Live open the Setting menu. In the MIDI Ports, set the output ‘Track’ on for the EP133. This is by checking the box. MIDI options are found in the sub menus; Live > Settings > Link Tempo MIDI > MIDI Ports.

7. At this point notes will also be sent when a MIDI pattern is configured in an Ableton Live MIDI Track. The track will need the ‘MIDI To’ set to EP133 and to any MIDI Channel.

8. Note that audio sounds triggered by the MIDI i.e. Pad sounds and also from any other pattern configured in the KOII will be heard when playing.

Always use a good quality USB cable when connecting MIDI. A common issue with USB cables is that often many are manufactured only as power charger cables i.e. for phones or devices. These may not handle data communication properly.

NOTES

KOII CONTROL OVER ABLETON LIVE WITH USB MIDI

1. Connect the KOII and the Ableton Live PC/Mac with a USB cable. This must be data compatible cable.

2. In the KOII, Set the Sync option.

    - Hold [Shift] + [System] to Open the settings.

    - Use the [-] & [+] to cycle through options. Select the MIDI Option ‘MID’. Press [Enter] Pad.

    - Use the [-] & [+] to cycle through options. Select the Clock Option ‘CLK’. Press [Enter] Pad.

    - Use the [-] & [+] to cycle through options. Select the In Option ‘OUT’. Press [Enter] Pad.

3. In Ableton Live open the Setting menu. In the MIDI Ports, set the <u>input</u> ‘Sync’ on for the EP133. This is by checking the box. MIDI options are found in the sub menus; Live > Settings > Link Tempo MIDI > MIDI Ports.

4. Click the ‘EXT’ Tag in the upper left corner of Live to activate external sync. The upper orange sync tag will flash when received from the KOII.

5. At this point Live will be controlled by the KOII Play, Stop and Tempo.

6. The KOII MIDI Output Channel for sending MIDI Notes can be set for the selected pad. Hold [Shift] + [Edit], Use [-] or [+] to cycle to the MIDI setting, ‘MID’. Then Turn the Orange (X) Knob to set the channel.

7. Any MIDI channels that KOII sends on should be matched by the Live track MIDI From setting. Also a plug-in synth or sample should be loaded to the MIDI track in Live in order to trigger audio from the KOII.

8. In Ableton Live open the Setting menu. In the MIDI Ports, set the <u>input</u> ‘Track’ on for the EP133. This is by checking the box. MIDI options are found in the sub menus; Live > Settings > Link Tempo MIDI > MIDI Ports.

9. The Live track will need the ‘MIDI From’ set to EP133 and to the KOII MIDI Channel. At this point notes will also be sent when a pattern or pad is played on the KOII. These notes are received by the Ableton Live MIDI Track and its associated sound / plug-in. This MIDI can be recorded.

NOTES

9.5 TRS MIDI In / Out

While USB is useful for DAW interfacing, the traditional 5-Pin DIN MIDI is more common and simpler for connecting to other hardware gear such as a MIDI keyboard or sequencer inputs or another synth out. The KOII uses 2 x TRS, 3.5mm Mini Jacks for the MIDI Inputs and Outputs.

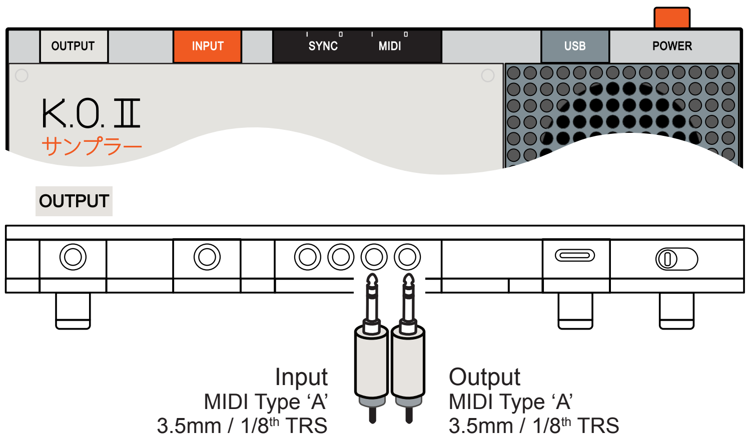
MIDI TRS Connectivity

TRS Stereo for MIDI Type ‘A’ i.e. KOII to Type ‘A’ Connections

The KOII can be connected between MIDI Compatible devices using a Stereo TRS 3.5mm Jack to Jack cable. This will connect only MIDI Type ‘A’ gear. This cannot be used if the devices have mixed types, i.e. KOII is Type ‘A’ and cannot be connected directly to a Type ‘B’ device

MIDI Adapter for MIDI TRS Type ‘A’ i.e. KOII to 5 Pin MIDI Connections

The KOII can be connected between MIDI Compatible devices using the MIDI standard 5-Pin Cable. However a Type ‘A” MIDI Adapter is required. This then allows a 5-Pin Cable to be used

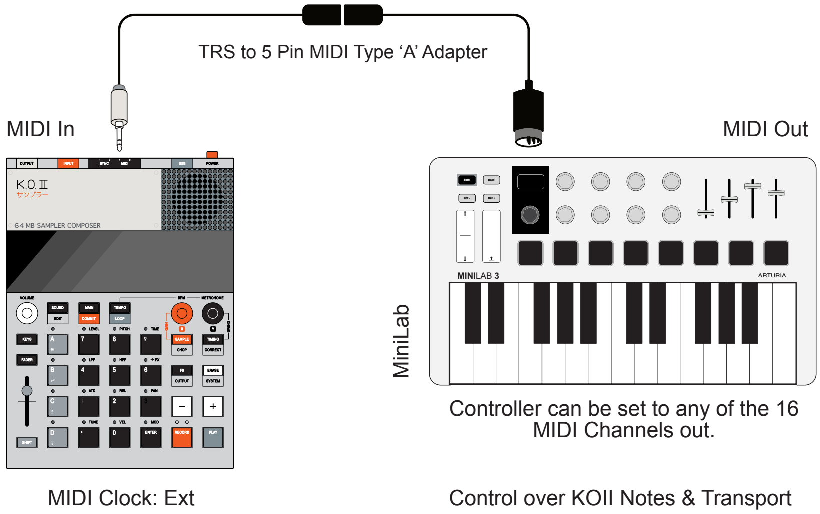


The MIDI Settings in the ‘System’ or ‘Sound Edit’ menus are applied equally to both MIDI for USB and for TRS connections.

TRS Represents, Tip, Ring, Sleeve i.e. the 3 connections available in the physical plug and socket. The sleeve would represent the ground. The Ring and Sleeve handling the stereo or data transmission. TS only contains Tip and Sleeve and would handle Mono or Analog signals.

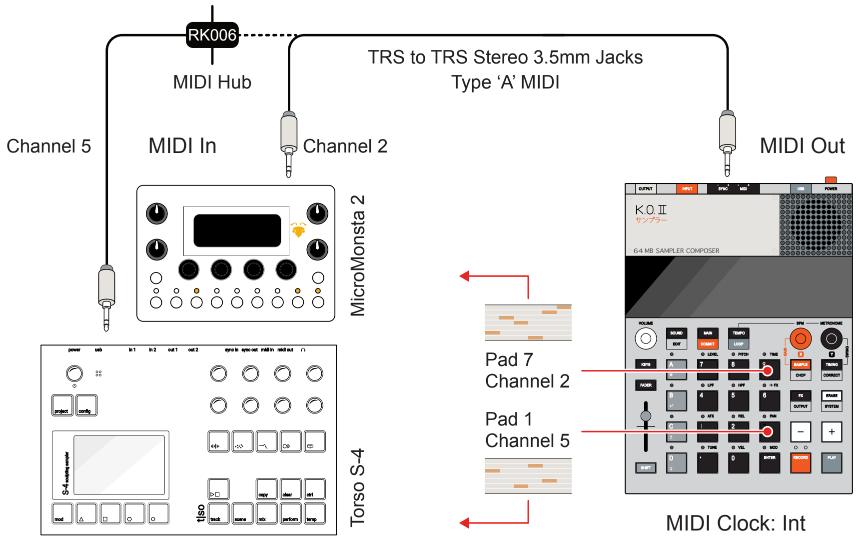
There are numerous connectivity options with other gear. A couple of typical examples are provided here.

External MIDI Controller (MIDI In)



The external controller will play the individual KOII pad sounds for the manually selected group on any MIDI Channel. If Keys Mode is selected on the KOII, the selected pad sound will play chromatically based on the KOII Scale setting (System > PAD > SCA) from the external keyboard. The MIDI controllers transport will also Play, Stop, Continue i.e. Restart the playback on the KOII when MIDI Clock is set to external on the KOII (System > MID > CLK).

External Synth (MIDI Out)



Each pad of the KOII can be assigned a MIDI Output Channel. This allows control of several devices from each pads pattern or by manually playing pads. Pad channels are set in the Sound Edit options (Edit > MID > Ch.X). A single stereo cable can be connected directly between two Type 'A' devices. A MIDI Hub such as the Retrokits RK006 takes 2 incoming MIDI TRS and distributes to 10 TRS Outputs.

NOTES

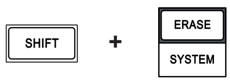
9.6 MIDI Thru

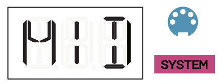
MIDI thru allows any incoming MIDI messages on the USB interface to be transmitted, unaffected, out on the TRS MIDI Output. Equally, any incoming TRS MIDI messages can be passed through to the USB MIDI interface. This is configured in the system settings options.

Hold [Shift] + Press [Edit]

Screen Indicators

System Menu
Navigate to MIDI



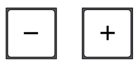


Select 'MID' Option

Press [-] [+]

Screen Indicators

MIDI Thru
Navigate to THR

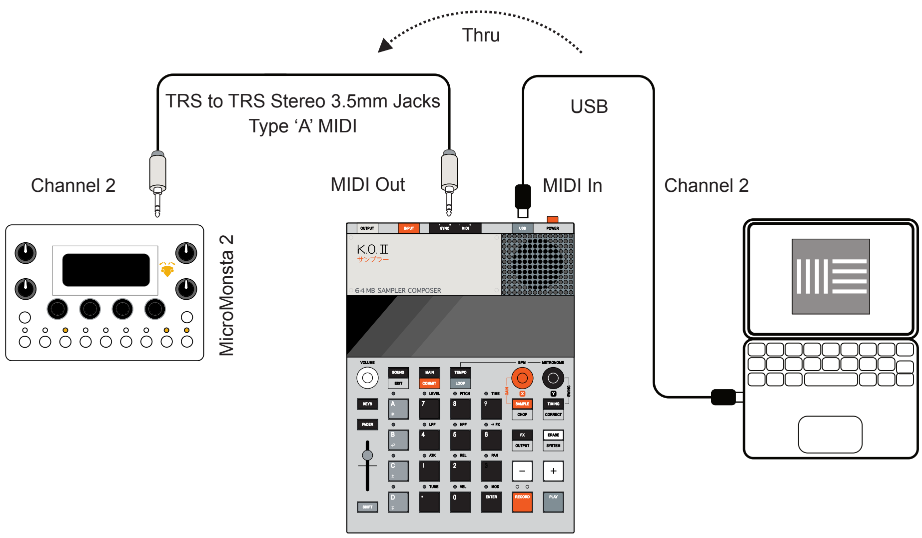


Shift + 'Enter' Pad to Back up in menu

Press 'Enter' Pad to Select

MIDI Thru Options

Menu	THR Option	Comments.
System > MID > THR	OFF	MIDI Thru is disabled. Default.
System > MID > THR	ON	Thru is enabled. The KOII will receive MIDI on the USB MIDI Input and pass out to the TRS output. The KOII will receive MIDI on the TRS MIDI Input and pass out to the USB output.

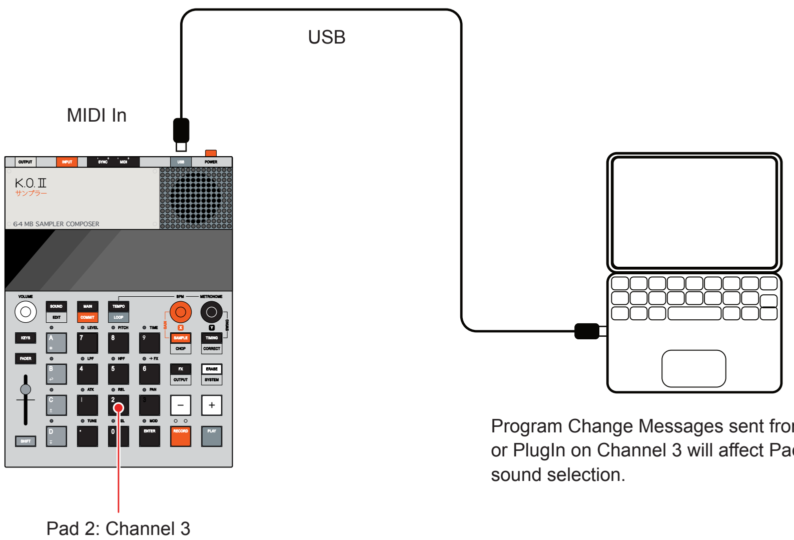


NOTES

9.7 Program Change

Program change or PC messages are used in the KOII to change sounds. PC implementation varies on different devices some starting with a 0 and others with a 1 in the program count. Some testing may be needed to match your devices. However the general principle is that KOII sounds can be changed to a new sound selected using the Pads MIDI channel with an external device. This can choose from the 999 available sounds.

PC messaging is based on a 0-127 range so a 'bank' change will be required using CC#1 - Control Change messaging to step to higher sound slots above the 127 value. Program change needs to be enabled in the system settings under MIDI > P.CH = ON to apply a sound change.



Program Change Messages sent from a DAW or PlugIn on Channel 3 will affect Pad 2 sound selection.

Standard	Sound Library		Program Change
	1. Dirty Kick	◄······	PC 1
	2. Soft Kick		PC 2
	3. Long Kick		PC 3
	4. Snare Loud		PC 4
	5. Shaker		PC 5
	6. Zaps		PC 6

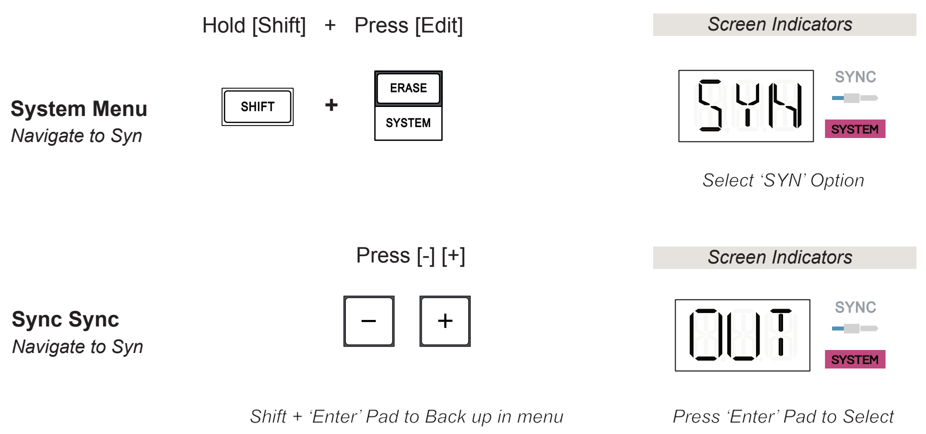
To change a sound beyond the 128 limit of the PC values, use CC#0 Bank Select MSB and CC#32 Bank Select LSB Control Change messages or any bespoke bank change message from the sending device to change banks. As an example with Ableton Live, set the 'Sub' value in the Bank / Prog section to 2 for 129-256, 3 for 257-384, etc to access each set of 128 sounds.

Bank +	Sound Library		Program Change
	129. Hat1	◄······	PC 1
	130. Kick3		PC 2
	131. Snarezz		PC 3
	132. Clickclack		PC 4
	133. Zzzzz		PC 5
	134. Zaps2		PC 6

NOTES

9.8 Sync Settings

Synchronisation of the clock can also be assigned to send or receive control with another device. This will allow the tempo to be controlled by a primary lead for other devices to follow. Or for KOII to be controlled by other gear. This applies to the generic clock and is therefore set globally in the system menu.



Clock Options

Menu	CLK Option	Comments.
System > SYN > OUT	8	Synchronisation output at 1/8th rate, 8 Pulses per Quarter Note. Ideal for Pocket Operators and other compatible devices.
System > SYN > OUT	16	Synchronisation output at 1/16th rate, 16 Pulses per Quarter Note. Ideal for desktop and modular gear.
System > SYN > OUT	24	Synchronisation output at 1/24th rate, 24 Pulses per Quarter Note. This is standard for MIDI Sync.
System > SYN > IN	8	Synchronisation output at 1/8th rate, 8 Pulses per Quarter Note. Ideal for Pocket Operators and other compatible devices.
System > SYN > IN	16	Synchronisation output at 1/16th rate, 16 Pulses per Quarter Note. Ideal for desktop and modular gear.
System > SYN > IN	24	Synchronisation output at 1/24th rate, 24 Pulses per Quarter Note. This is standard for MIDI Sync.

The main clock configurations should concentrate on which device is the primary lead. This is typically set for only one device in a network. This will send the clock out and dictate the primary tempo of the setup. Other devices on the network will follow this lead. Multiple devices would be set as the secondary follower of the lead device.

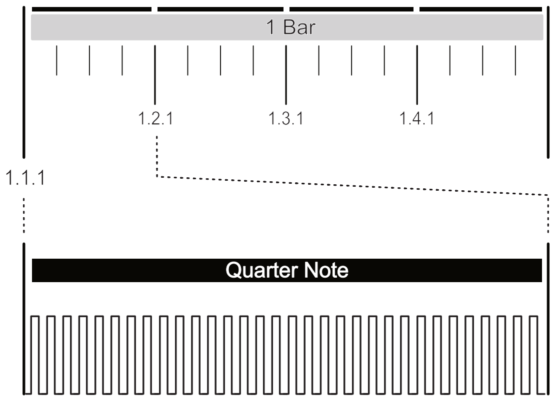
9.9 Sync In/Out

In addition to MIDI, the KOII also has the ability to synchronise with other compatible gear using the 'Sync' features. This aligns the connections typically with modular gear, Eurorack, Pocket Operators and som desktop gear. This feature only synchronises the clock functions and does not communicate note information.

PPQN

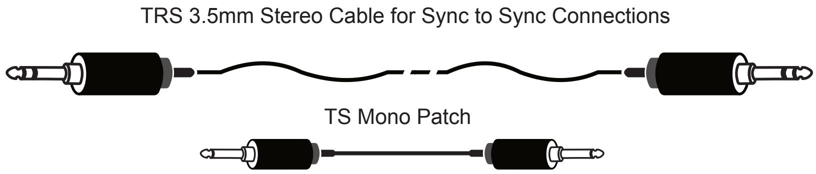
The standard measure of an audio clock both in MIDI and in the Sync connections is based on the PPQN standard. This represents Pulses Per Quarter Note. MIDI operates at 24 PPQN while Sync may operate at a various settings and the connected devices should be set to match.

Example 4/4



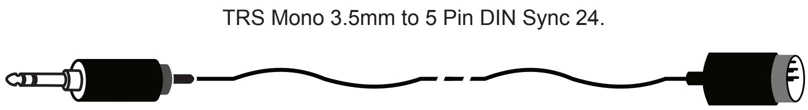
Clock: 24 PPQN

Sync TRS / TS Connectivity



The KOII can be connected between Sync compatible devices e.g. TE gear using a TRS 3.5mm Jack cable. A standard patch or mono cable may be used for some devices i.e. Eurorack Modules.

Sync TS to DIN Connectivity



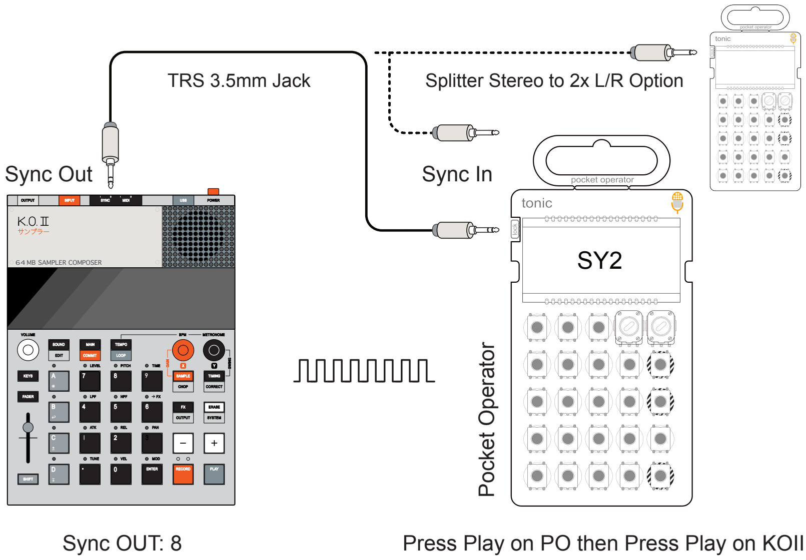
A 3.5mm to DIN cable is required to connect KOII Sync to Sync 24 DIN gear. MIDI Cables and DIN cables are often used interchangeably. However while MIDI needs 3 Pins connected, DIN cables needs all 5 pins to be connected.

NOTES

NOTES

There are numerous connectivity options with other Teenage Engineering and Other gear. Some typical examples are provided here.

External Pocket Operator (Sync Out)



Synchronisation must be set on the KOII. Set the Sync to 8 (System > SYN > OUT) in the KOII system settings. The Sync settings options available are 8, 16 and 24. This will set a 1/8 Sync pulse output to any connected device. The connected PO device is ready to receive 1/8th pulses / 8 PPQN. Press Play on the PO first and its clock tempo and transport will be controlled by the KOII when it is also played.

PO Sync Modes

Sync Mode

Set the PO to SY2 using Hold [Function] + Tap [BPM] to cycle options on the PO. The ‘function’ button is dependant on the PO model but usually is the button located directly under the right knob. Refer to the specific PO documentation for exact details.

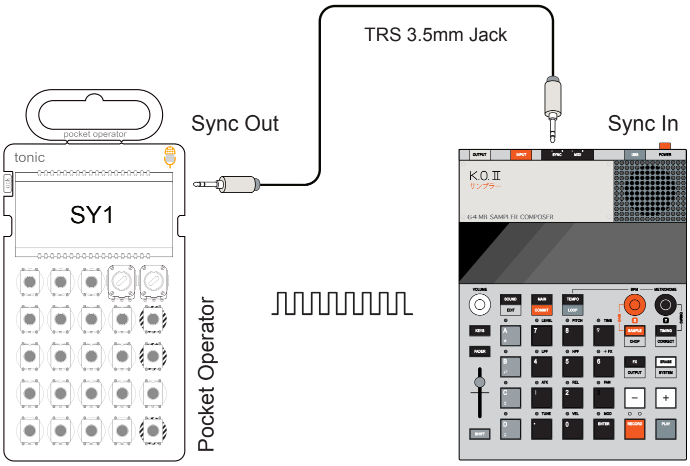
Sync Modes

Mode	Input	Output
SY0	Stereo	Stereo
SY1	Stereo	Mono / Sync
SY2	Sync	Stereo
SY3	Sync	Mono / Sync
SY4	Mono / Sync	Stereo
SY5	Mono / Sync	Mono / Sync

The KOII operates with a TRS 3.5mm Stereo Jack, Sync with Tip, Left Channel and Ring, Right Channel as Sync connections. This allows a splitter cable to sync two PO devices on the left and right stereo channels.

The 1/8th rate is ideal for Pocket Operators but can also be used on other gear which has a 8 PPQN setting.

External Pocket Operator (Sync In)



Press Play on OP then Press Play on KOII

Sync IN: 8

Synchronisation must be set on the KOII. Set the Sync to 8 (System > SYN > IN) in the KOII system settings. The Sync settings options available are 8, 16 and 24. This will set a 1/8 Sync pulse input from any connected device. The connected PO device is ready to send 1/8th pulses / 8 PPQN. Press Play on the KOII first and its clock tempo and transport will be controlled by the PO when it is also played.

PO Sync Modes

Sync Mode

Set the PO to SY1 using Hold [Function] + Tap [BPM] to cycle options on the PO. The 'function' button is dependant on the PO model but usually is the button located directly under the right knob. Refer to the specific PO documentation for exact details.

Sync Modes

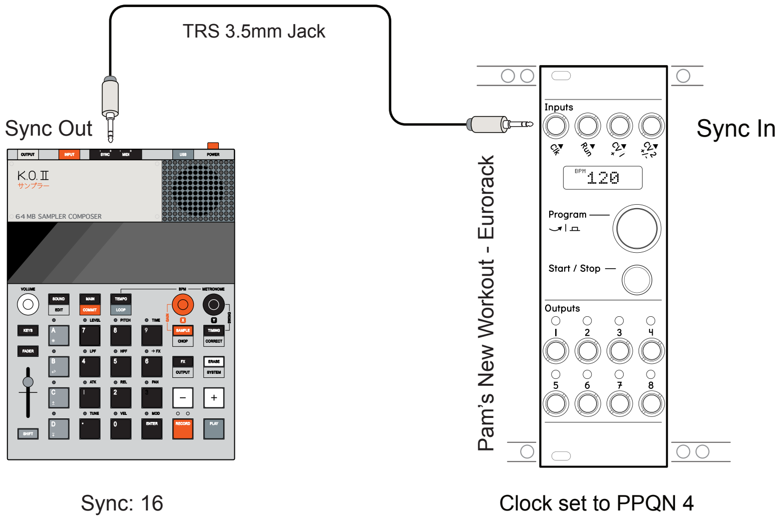
Mode	Input	Output
SY0	Stereo	Stereo
SY1	Stereo	Mono / Sync
SY2	Sync	Stereo
SY3	Sync	Mono / Sync
SY4	Mono / Sync	Stereo
SY5	Mono / Sync	Mono / Sync

The KOII operates with a TRS 3.5mm Stereo Jack, Sync with Tip, Left Channel and Ring, Right Channel as Sync connections. The SY1 Setting on the PO allows a stereo to 2x mono splitter cable from the PO to sync (left channel) to another device and also to send mono audio out (right channel). Maybe connect the PO audio to the KOII Input.

NOTES

Modular gear may operate via a range of clock settings. Sometimes it may need some trial and error settings to match up the device clock tempo's..

External Module - Pamela's New Workout (Sync Out)



Synchronisation must be set on the KOII. Set the Sync to 16 (System > SYN > OUT) in the KOII system settings. The Sync settings options available are 8, 16 and 24. This will set a 1/16 Sync pulse output to any connected device. The connected module will be set to PPQN 4 to match clock tempo.

The KOII operates with a TRS 3.5mm Stereo Jack output. For modular gear it is sometimes ok to use a mono TS cable with 3.5mm jacks. Eurorack modules often operate using mono patch cables..

The principles of connecting to modular gear pretty much also apply with other desktop devices in general. The key element is ensuring the cable has the correct standard for both devices and that the clock configuration is align or set to match the tempo of both devices.

10

System & Reference

This final section is more of a reference than a walkthrough of processes and practices. Whiles some step by step features are includes it is an area that may be visited time to time when needed. However do not overlook this section. There are some important and useful elements contained within. The process of performing a firmware update is the most obvious one.

10.1 System Menu Options

NOTES

The system menu contains the settings for a number of global options. This is an area that could be considered a set and forget function but it is worth getting to know what features it contains and what options can be changed.

CHANGING A SYSTEM OPTION IN THE MENU

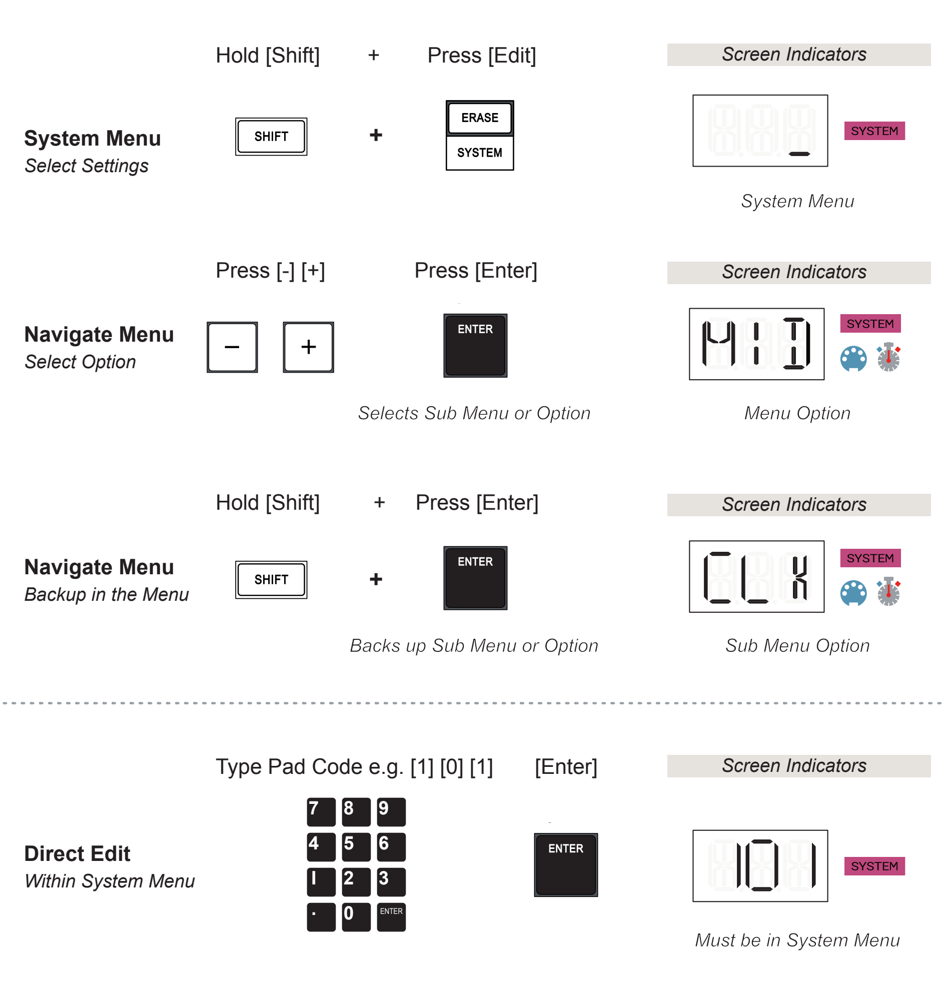
1. To open the ‘System’ menu, Hold [Shift] + Press [System].
2. The display will show the ‘System’ icon along with a flashing cursor to state the top level system menu location and any other associated icon.
3. To navigate the menu, Press [-] or [+]. This will cycle through the available system options. At the top level the available options are:-
   - MID - MIDI Settings for TRS and USB MIDI
   - SYN - Synchronisation settings for Sync In/Out
   - PAD - Pad velocity and scales.
   - SEQ - Sequencer settings for metronome and scenes.
4. When the desired option is in view, Press [Enter]. This button is located as the bottom right Pad.
5. The sub-menu containing the options for each function are presented. These are specific to each function. Note that you can back up a level in the menu structure by Hold [Shift] + Press [Enter].
6. To navigate the options, Press [-] or [+]. This will cycle through the available sub menu options. When the desired option is displayed, Press [Enter] to select it for editing.
7. The next level has the actual settings. To navigate the options, Press [-] or [+]. This will cycle through the available settings. The currently active option will display static text while the other inactive options will flash.
8. Once an option is displayed, Press [Enter] to select it. It will flash quickly then the text will be static. The value has now been changed.
9. Press any other function i.e. ‘Main’ to exit the system menu. Auto save will be performed when making changes in the system.

NOTES

The system menu settings can also be accessed directly using menu option codes. This is a quicker way to make system changes but it relies on you remembering the code numbers. Maybe just remembering the useful and regularly accessed ones will work best.

CHANGING A SYSTEM OPTION DIRECTLY

1. To open the ‘System’ menu, Hold [Shift] + Press [System].
2. The display will show the ‘System’ icon along with a flashing cursor to state the top level system menu location and any other associated icon.
3. Type the direct access codes using the pads, I.e 102 selects the MIDI Clock OUT option.
4. Press [Enter] to change and confirm the selection. The option will flash on the display and will become the active setting.



Direct Access Codes

The system menu options can be directly accessed using the following codes. Select the ‘System’ menu with [Shift] + [System] first and press [Enter] when the option code has been selected to confirm.

System Menu Codes

Option	Code	System Sub-Menu	Comments.
MIDI Clock Off	100	> MID > CLK > OFF	Sets the MIDI clock off. *
MIDI Clock In	101	> MID > CLK > IN	Sets the KOII to receive an external MIDI clock
MIDI Clock Out	102	> MID > CLK > OUT	Sets the KOII to send an MIDI clock out to external gear
MIDI Channel All	110	> MID > CHN > ALL	Receive on all MIDI Channels and send on MIDI Channel 1 *
MIDI Channel 1	111	> MID > CHN > CH.1	Receive & send only on MIDI Channel 1
MIDI Channel 2	112	> MID > CHN > CH.2	Receive & send only on MIDI Channel 2
MIDI Channel 3	113	> MID > CHN > CH.3	Receive & send only on MIDI Channel 3
MIDI Channel 4	114	> MID > CHN > CH.4	Receive & send only on MIDI Channel 4
MIDI Channel 5	115	> MID > CHN > CH.5	Receive & send only on MIDI Channel 5
MIDI Channel 6	116	> MID > CHN > CH.6	Receive & send only on MIDI Channel 6
MIDI Channel 7	117	> MID > CHN > CH.7	Receive & send only on MIDI Channel 7
MIDI Channel 8	118	> MID > CHN > CH.8	Receive & send only on MIDI Channel 8
MIDI Channel 9	119	> MID > CHN > CH.9	Receive & send only on MIDI Channel 9
MIDI Channel 10	120	> MID > CHN > C.10	Receive & send only on MIDI Channel 10
MIDI Channel 11	121	> MID > CHN > C.11	Receive & send only on MIDI Channel 11
MIDI Channel 12	122	> MID > CHN > C.12	Receive & send only on MIDI Channel 12
MIDI Channel 13	123	> MID > CHN > C.13	Receive & send only on MIDI Channel 13
MIDI Channel 14	124	> MID > CHN > C.14	Receive & send only on MIDI Channel 14
MIDI Channel 15	125	> MID > CHN > C.15	Receive & send only on MIDI Channel 15
MIDI Channel 16	126	> MID > CHN > C.16	Receive & send only on MIDI Channel 16

* Default Settings

NOTES

Remembering the codes may not be an easy task. However using the access codes is a quick and easy method. It is advised to focus only on remembering the ones you will access often in your workflow e.g. Metronome or Velocity.

System Menu Codes

Option	Code	System Sub-Menu	Comments.
MIDI Thru	130	> MID > THR > OFF	MIDI Thru disabled *
MIDI Thru	131	> MID > THR > ON	MIDI Thru enabled
MIDI Prog Change	140	> MID > P.CH > OFF	Ignore MIDI PC messages *
MIDI Prog Change	141	> MID > P.CH > ON	MIDI PC messages change sounds
Sync In Rate 8	200	> SYN > IN > 8	Sync In rate 1/8th
Sync In Rate 16	201	> SYN > IN > 16	Sync In rate 1/16th *
Sync In Rate 24	202	> SYN > IN > 24	Sync In rate 24 PPQN
Sync Out Rate 8	210	> SYN > OUT > 8	Sync Out rate 1/8th
Sync Out Rate 16	211	> SYN > OUT > 16	Sync Out rate 1/16th *
Sync Out Rate 24	212	> SYN > OUT > 24	Sync Out rate 24 PPQN
Pad Velocity Off	300	> PAD > VEL > OFF	Pad velocity off *
Pad Velocity High	301	> PAD > VEL > HI	Pad velocity set to play for low pressure / soft play users.
Pad Velocity Low	302	> PAD > VEL > LO	Pad velocity set to play for high pressure / aggressive play users.
12 Tone Scale	310	> PAD > SCA > 12T	12 Tone equal temperament *
Major	311	> PAD > SCA > MAJ	Major Ionian Mode
Minor	312	> PAD > SCA > MIN	Minor Aeolian Mode
Dorian	313	> PAD > SCA > DOR	Dorian Mode
Phrygian	314	> PAD > SCA > PHR	Phrygian Mode
Lydian	315	> PAD > SCA > LYD	Lydian Mode
Mixolydian	316	> PAD > SCA > MIX	Mixolydian Mode

* Default Settings

NOTES

System Menu Codes - Continued

Option	Code	System Sub-Menu	Comments.
Locrian	317	> PAD > SCA > LOC	Locrian Mode
Major Pentatonic	318	> PAD > SCA > MA.P	Major Pentatonic
Minor Pentatonic	319	> PAD > SCA > MI.P	Minor Pentatonic
Scale Key C	320	> PAD > SCA > C	Sets Scale Key *
Scale Key Db	321	> PAD > KEY > Db	Sets Scale Key
Scale Key D	322	> PAD > KEY > D	Sets Scale Key
Scale Key Eb	323	> PAD > KEY > Eb	Sets Scale Key
Scale Key E	324	> PAD > KEY > E	Sets Scale Key
Scale Key F	325	> PAD > KEY > F	Sets Scale Key
Scale Key Gb	326	> PAD > KEY > Gb	Sets Scale Key
Scale Key G	327	> PAD > KEY > G	Sets Scale Key
Scale Key Ab	328	> PAD > KEY > Ab	Sets Scale Key
Scale Key A	329	> PAD > KEY > A	Sets Scale Key
Scale Key Bb	330	> PAD > KEY > Bb	Sets Scale Key
Scale Key B	331	> PAD > KEY > B	Sets Scale Key
Sample Audition	340	> PAD > AUD > ON	Preview audio of samples when tweaking samples
Sample Audition	341	> PAD > AUD > OFF	Preview of audio samples is muted when tweaking samples *
Metronome	400	> SEQ > MET > ON	Enable Metronome when Record + Play is used. *
Metronome	401	> SEQ > MET > REC	Enable Metronome when Record only is used.
Scene Change	410	> SEQ > SCN > TIC	Change scene immediately. *
Scene Change	411	> SEQ > SCN > BAR	Change scene at bar end
Scene Change	412	> SEQ > SCN > PTN	Change scene at pattern end

* Default Settings

PPQN Stands for Pulses Per Quarter Note and is a commonly used standard for clock rates. MIDI clocks operate at 24 PPQN standard but Eurorack and some desktop gear can operate at a variety of clock rate options. The internal KOII clock runs at 96 PPQN.

NOTES

10.2 Back up and Restore (Beta)

The ability to back up and restore the KOII is possible. At the time of writing, this is a beta feature of the TE Online Sample tool and as stated on the tool site, 'Use at your own risk'. A fully tested and officially certified version may be updated in future by TE. This needs a Web MIDI compatible browser such as Chrome or Edge to work.

BACKING UP THE KOII

1. Connect the KOII using the USB connection to a PC or Mac and power on the device.

2. Access the TE EP Sample Tool using a compatible web browser. Note the backup and restore options are beta features right now and the following steps are at the user risk.

3. On the EP Sample Tool page, your samples and KOII will show connected. The KOII OS and Serial Number are shown bottom left.

4. Click the 'BACKUP & RESTORE' button on the lower right of the page. The backup and restore function will be presented.

5. To backup your device data to the connected Mac or PC.

   - Click the 'BACKUP' button. The backup of samples and projects will start immediately.

   - Activity is displayed by the progress bars. This may take several minutes.

   - Once completed, the file will be available as a 'download'. For example in Google Chrome the Download History is presented in the browser bar window with the down arrow symbol and will be located on the PC/Mac in the download files folder.

   - The name format will include the device serial number and date. Example: EP-133_E3xxxxxK_2024-06-09_backup.pak.

   - If required, the file can be moved or copied to a more convenient and safe location on the PC / Mac or an external storage device.

6. Once completed, Click 'BACK' to exit the backup and restore page.

NOTES

RESTORING THE KOII FROM A PREVIOUS BACKUP

1. Connect the KOII using the USB connection to a PC or Mac and power on the device.

2. Access the TE EP Sample Tool using a compatible web browser. Note the backup and restore options are beta features right now and the following steps are at the user risk.

3. On the EP Sample Tool page, your samples and KOII will show connected. The KOII OS and Serial Number are shown bottom left.

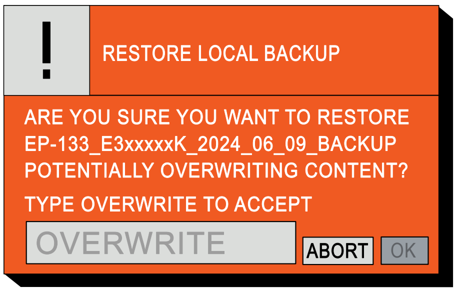
4. Click the ‘BACKUP & RESTORE’ button on the lower right of the page. The backup and restore function will be presented.

5. To restore your device from a saved backup file on the PC/Mac.

    - Click the grey ‘RESTORE’ button to ‘Restore Local Backup’. This will start the process to restore the previously backed up samples and projects.

    - The file browser will open and allow selection of the back up file to restore. Click ‘Open’ when highlighting the file to restore.

    - A security pop-up will appear. This requires the typing of ‘OVERWRITE’ in the prompt to avoid any accidental erasure. This will overwrite your existing data.



    - Once the text has been typed, click ‘OK’. The option to abort is also presented.

    - Restoration will commence. Activity is displayed by the progress bars. This may take several minutes. Successful completion will be reported once complete.

6. Once completed, Click ‘BACK’ to exit the backup and restore page.

NOTES

10.3 Factory Restore (Beta)

The original factory projects and samples can be restored so that the KOII is back to its original out of the box state. At the time of writing, this is a beta feature of the TE Online Sample tool and as stated on the tool site, 'Use at your own risk'. This needs a Web MIDI compatible browser such as Chrome or Edge to work.

RESTORING THE KOII FACTORY PROJECTS & SAMPLES

1. Connect the KOII using the USB connection to a PC or Mac and power on the device.

2. Access the TE EP Sample Tool using a compatible web browser. Note the backup and restore options are beta features right now and the following steps are at the user risk.

3. On the EP Sample Tool page, your samples and KOII will show connected. The KOII OS and Serial Number are shown bottom left.

4. Click the 'BACKUP & RESTORE' button on the lower right of the page. The backup and restore function will be presented.

5. To restore the factory content to your device:-

    - Click the orange 'RESTORE' button to 'Restore Factory Content'. This will start the restore factory content process.

    - A security pop-up will appear. This requires the typing of 'OVERWRITE' in the prompt to avoid any accidental erasure. This will overwrite your existing data.

    - Once the text has been typed, click 'OK'. The option to abort is also presented.

    - Restoration will commence. Activity is displayed by the progress bars. This may take several minutes. Successful completion will be reported once complete.

6. Once completed, Click 'BACK' to exit the backup and restore page.

10.4 Updating the Firmware

Occasionally TE will release firmware updates that fix bugs or add new features and improvements. This will require downloading of the firmware and updating the KOII. The actual instructions applied should follow those supplied by TE for each update. The following example update is provided as a general guide. It is recommended to back up the device first.

The recommended update process for the KOII and perhaps the simplest is available by using the TE online update utility tool. A Web MIDI compatible browser is required such as Edge, Chrome or Brave.

FIRMWARE UPDATE USING THE UPDATE UTILITY

1. Connect the KOII using the USB connection to a PC or Mac and power on the device.

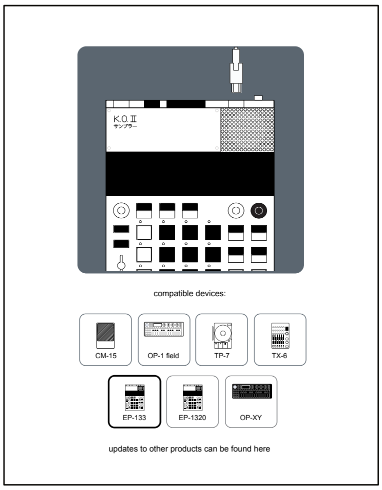
2. Access the TE update utility using a compatible web browser.

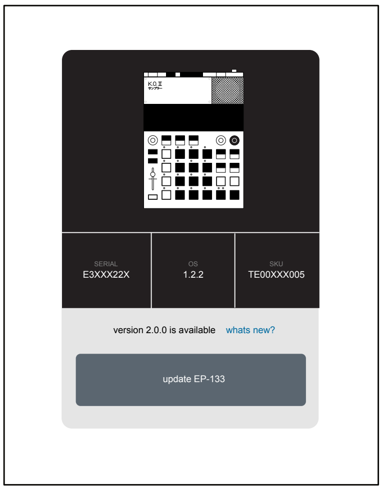
3. On the update utility page select the EP-133.

4. The current installed OS will be presented along with any available OS update. The option to update is directly accessible unless the device is at the current version in which case this will be presented as such.

5. To update the connected KOII, Click ‘update EP-133’. Do not disconnected or power down the device during this process.

6. The update will commence, progress and completion will be indicated





10.5 Error Codes

There are several error codes declared by TE that may be useful if they ever appear (hopefully not). While these may indicate a system error there isn't much a user can do to rectify. Some need device repairs while re-formatting the drive may cure some issues. It is a good reminder to keep a regular backup of the device.

Error codes will appear on the display:

Code	Display	Description
E.01		SRAM (Static Random Access Memory) Error. No user remedy. Device needs to be repaired.
E.02		Flash (Memory) Error. No user remedy. Device needs to be repaired.
E.03		Codec (data compression management) Error. No user remedy. Device needs to be repaired.
E.04		LFS Mount Error (File System). Flash needs to be reformatted. Erase & Format Drive.
E.05		Sample_FS Error (File System). Flash needs to be reformatted. Erase & Format Drive.

It is important that the OS firmware is maintained up to date on the KOII. This not only ensures the most recent features are available but will also reduce any risk of errors as bugs are fixed and performance improvements introduced.

10.6 Erase & Format Drive

This is an activity that should only be performed as a last resort and if an unrecoverable error E.04 or E.05 is displayed. It is always recommended that when any device is faulty or 'bricked' i.e. appears dead that the manufacturers support is sought in the first instance.

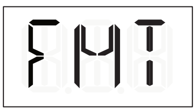
Once a KOII drive has been erased and reformatted a backup or the factory restore can be performed to get back to a prior state.

**** Warning ****

This is a destructive process and will erase all data, samples, projects and factory data etc from the KOII. There is no undo or going back. Only perform this function if you feel competent to do so, advised by a TE professional or in circumstances where the KOII is in an unusable state.

ERASE AND REFORMAT THE KOII DRIVE

1. Power off the KOII.
2. Hold [Shift] + Hold [Erase], while holding the buttons, Power the KOII ON.
3. The KOII will start its erase and reformat process and the display will show 'FMT on the screen. This will be the state for about 10-15 Seconds.
4. Once formatting is complete, the KOII will automatically restart.
5. Once restarted the device will be empty. All content including the factory and user samples will have been erased.
6. If a backup has been created this can be restored using the EP Sample Tool. Also the option to restore factory content is available in the tool. Note that the EP Sample Tool Backup & Restore options are currently only beta versions.



Formatting erases all data content. The process takes approx 10 Seconds to complete.

NOTES

10.7 Frequently Asked Questions

FAQ’s or Frequently Asked Questions enable a number of common questions to be addressed collectively. The intimate details and workflows are covered in the respective section elsewhere in this book, while this section only provides quick answers to the queries often asked.

Global and Architecture

Q. How do I save projects or patterns?

A. Projects and the settings are automatically saved when the sequencer is not playing. This will be performed as changes are made.

Q. Does KOII have a song mode?

A. Yes. Song mode was introduced in version 2.0 of the OS. It allows song positions to be captured into a song and therefore playback a chained composition. One song per project is available in KOII. Scenes also allow variations of an arrangement to be created and manually selected in order to playback in a manual sequence.

Q. What is the memory capacity?

A. 64 MB on board memory.

Q. Are the 12 pads velocity sensitive?

A. The simple answer is yes. However the pressure is only recognised in the lower half of the pad. Also the velocity profile and setting need to be configured in the settings.

Q. Can the KOII be powered by a power bank and USB?

A. Yes a USB power supply will be ok to power the device which is a light energy user.

NOTES

Projects

Q. How many Projects are there?

A. There are 9 Projects in total. Out of the box the first 5 projects have samples loaded to the 12 pads for each group.

Groups & Patterns

Q. Does the groups allow for choke groups?

A. Yes. In the KOII they are called mute groups and are available per group.

Samples & Audio

Q. What is the sampling rate?

A. Sampling is performed at 46.9kHz, 16 Bit, and cannot be changed. Sampling can be in Stereo or Mono.

Q. Can samples be renamed once recorded?

A. Not in the KOII device. Editing sample names is performed in the EP Sample Tool, an online aid to managing the KOII samples via a connected Mac/Pc.

Q. How many voices does the KOII have?

A. The device has 12 Stereo / 16 Mono voices. These are dynamic and will adjust depending on the system loading.

Q. How many samples does the KOII Hold?

A. The number of samples is dependant on the samples audio length and whether stereo or mono. Stereo samples take twice as much memory as mono. The on-board sample library has 999 sample slots.

NOTES

Effects & Performance

Q. How many effects exist?

A. There are 6 effects in the master effect section but only one can be assigned at a time. These operate as send effects. The Master output also has a compressor. A set of punch in effects can also be applied manually for performance purposes.

MIDI & Synchronisation

Q. What type of MIDI TRS connection is used?

A. The KOII uses Type ‘A’ MIDI for the TRS MIDI In and Out.

Q. Can a MIDI Channel be assigned per group?

A. Not directly. MIDI Channel is set per pad so a work around could be to set all pads in a group to the same channel. Then have different channels, set via the pads, per group.

Q. Can the KOII Receive MIDI Control Change Messages?

A. TE state the KOII can send and receive CC however there is no published or defined mapping to support setting this up.

Q. What is the PPQN of the KOII?

A. Internally the KOII operates at 96 PPQN. There are various setting that can be applied to sync to other gear using PPQN.

END

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Disclaimer

This book is for general reference and guidance only when working with the named devices. This book does not replace or negate the manufacturers official manuals. Any actions or risks taken in using the devices are the full responsibility of the user.

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